On The Articulation of English Grammar: Intonation, Prosody, and Related Topics

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Abstract

This paper culminates into a preliminary grammar of English, a phrase structure grammar defined over vectors which combine syntactic categories with tonal and prosodic properties. In the grammar, transformations from one structure to another are assumed to occur pre-performance, so that the result doesn't have to be moved, and can stand as its own independent structure. From prosody and intonation, it can be shown that all the syntactic information of the initial structure is preserved—in particular, a theta role based case system—even when the initial constituents are not articulated. The information about syntax is communicated by computational properties of tones and the face and jaw muscles. Leading up to the grammar includes an overview of the speech act, from the perspective of muscular control. This provides motivation for introducing prosodic and tonal articulation, their spectral corollaries, and computations derived from them. Some non-syntactic observations of these articulators are also discussed and helps provide more familiarity and context for the grammar.

Contents

1 Preliminaries

(1) In Room 402, the doctor cured a patient.

Speaking a sentence like (1) involves much more than just producing the phonemes. It also includes producing the intonation, which is handed down according to syntactic structure: there are intonation properties determined for "in Room 402" and "the doctor cured a patient" once they are determined for one or the other; they are determined for "doctor" and "patient" once it is determined for "cured;" and they are determined for "in" once they're determined for "Room 402." The grammar constrains, but doesn't determine, a small collection of possible tonal dynamics over a syllable. A similar thing happens with prosody, whose values can be identified with properties of the face and jaw muscles. The topic of this paper is how these tonal and prosodic values apply to syllables according to the sentence's syntax. The two main tasks to present this is to describe the articulation in general (§2), and then to specify how the articulation is used in the grammar (§4). What follows in this section is some background information for the former task. How the data was generated is presented, and the general physiological nature of the tongue is discussed to give more context to the particular concerns of this paper. Then in §2, the 3 main modules of articulation are introduced, along with their formal treatment. This involves describing their perceptual and any acoustic properties, their physiology, and the discrete values that are used in the grammar. In §3, applications which don't feature in the grammar are presented, as well as a head start on the grammar. A discussion of clauses, and in particular inputs to the clause, is given. Definitions for prosodic modals, which play a small role in the grammar, are also given. Besides this, topics such as melody, aspect, and differences of speech acts are discussed. Finally, the grammar is pursued. It expands Chomsky's (1965) early framework, allowing it to accommodate the new empirical details. §4.1 Specifies the new framework. §4.2 deals with noun, prepositional, and verb phrases. §4.3 and §4.4 defines multiple clauses. §4.5 specifies the rules for embedding one clause into another. §4.6 then takes on the various cases that form a sentence or valid speech act. Finally, §4.7 provides a summary and addresses some topics not in the grammar.

While this is presented as an English grammar, it's not clear how much of this is a grammar of English in general as opposed to a grammar of a particular accent of English.

1.1 Spectrogram

Syntax is produced by tones, the muscles of the jaw, and the muscles around the lips. Thus, for each sound cue which contributes to the syntax of English, or at least certain accents of English, we have muscles associated with its articulation. These sounds of syntax will be discussed in terms of physiological, auditory, and acoustic effects. Below, a general model of this articulation will be presented, but here we discuss its audition and acoustics. Identifying these elements as auditory properties is a skill, like relative pitch, which has to be developed and practiced. The acoustic signatures show up on spectrograms to different levels of satisfaction. For this study, the spectrogram is used for visual identification and no numerical nor computational properties are identified.

A spectrogram is a graph with the two axes of frequency and time, with colors indicating the amplitude of a given frequency at a given time.[1] In the settings for the examples of this paper, red is the highest amplitudes, followed by yellow and then green corresponding to lower amplitudes. A spectrogram must have various parameters set. Which parameters are valid must be obtained from matching the information to auditory perceptions. For instance, the tonal information can be inaccurate if the temporal scale is too fine or coarse; determining the right temporal scale to analyze data at requires calibration with auditory information

[1]All spectrograms come from Sonic Visualiser 5.0.1, available from `sonicvisualiser.org`. The audio files are available at `drive.google.com/drive/folders/1OPYkRw6dK_m5lA-rv5cVFtE7gpspJzkD`

about articulatory features. But once this is done, the parameter should stay invariant, for the given property, across all speech data.

The proper temporal scale for a tone can be approximated by experiment. There is data that gives distinct tonal results when one changes the number of samples used for the spectrogram. By using auditory judgments with this data, the temporal length that matches auditory judgment can be deduced. Playing around with the length of a sound sample, it's known that short lengths produce a pop (<.005 seconds (approx.)), slightly longer lengths produce static auditory effect, such as with phonographs (Olson (1967), p. 250). Then at ~.01 seconds, one gets a sound whose tonal structure is not entirely clear and its ontology is not determined, but no longer sounds like static or undifferentiated noise. At ~.05 seconds, one can identify male vs. female speaker, though the word(s) and phoneme(s) spoken still can't be processed. Finally, the solving for the ontology (both the source and the behavior) can be heard at a sufficient length. All of this data can be used to deduce different parts of the auditory processing algorithm and the scale they work at. The files were always 44,100 samples per second, and the spectrogram is limited to samples of powers of 2. Given this constrain, the best match between auditory tone and the spectrogram occurred with 8,192 samples, which is a temporal scale or window of ~.19 seconds, or approximately twice the wave length of the lowest tone human perception can hear.

A useful terminology is the harmonic range and the prosodic range. The harmonic range for the male voice studied occurs from approximately 80 Hz to 2,700 Hz. The prosodic range occurs from approximately 3,000 Hz to 18,000 Hz. These areas can be distinguished by how many samples are used to obtain useful imagery. Regardless, one will see three different spectrogram windows: one a close up on the tone, one for the harmonic region, and one for the prosodic region.

1.2 Anatomy and Physiology

The muscles used in speech are difficult to hone in on and presents a challenge. While the task isn't complete, the general layout of what's being implemented can be given at least a first pass. The muscular control works under the tongue's unique anatomical conditions, on some general principles, and with multiple such actions being coordinated to produce longer sequences. The articulation of the grammar is but one part of the entirety of what is articulated for any given syllable. Besides the tone, face, and jaw muscles, this involves phonemes and elements of prosody that are achieved with the tongue. Here, the concern is with the muscular control of the tongue and its more central dependencies, and ignores the air mechanisms and nasalization.

Muscular architecture and principles. The tongue is a muscular hydrostat, which means that it's volume is invariant. It has various intrinsic muscles and extrinsic muscles which allow its shape and position to change. All muscles pull when tightened and produce no push when loosened or relaxed, and this fact along with the volume invariance can be used to reason about muscular effects. The exterior muscles are attached to the hyoid bone, jaw, and skull, which are then the tongue's connection to more central elements. The hyoid bone is not locally connected to other bones and can come closer to or be further away from the mandible or jaw. That is, it can be raised and lowered, and in turn have an effect on the height of the tongue.

Each muscle is made up of many fibers and neither the muscle nor the fibers are the right level of analysis for describing behavior. The more appropriate level of analysis would be motor units, and if our interests are limited to just the effected fibers, we can name this level muscle units to indicate a collection of muscular fibers that belong to both the same motor unit and same muscle. Each fiber is connected to a *unique* lower level motor neuron; more complex motor units are produced by activating multiple lower motor neurons (Gallistel (1980), Marieb and Hoehn (2023)). So fibers are smaller divisions of muscles, and are thin, string like structures. Multiple fibers make up a muscle unit, and multiple muscle units make up a muscle. If the

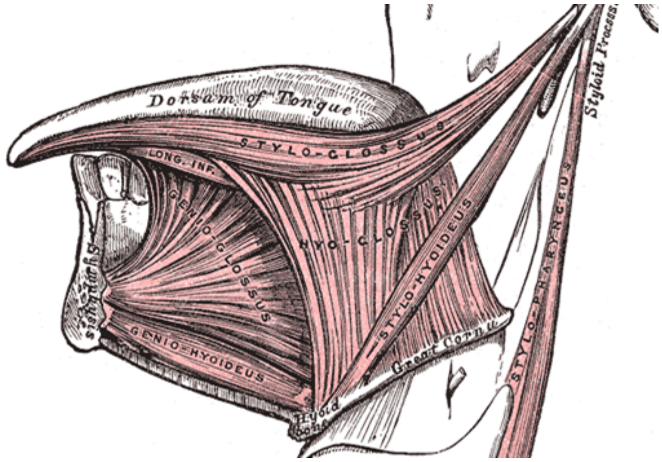

Figure 1: Extrinsic Tongue Muscles. Not pictured is the palatoglossus. Taken from Gray (1918).

depth of our study included phonology, we would be interested in the smallest muscle units which produce an adequate pull e.g. for acoustic change or positional change.

The tongue is usually split into two parts, the body and the root, and unless stated otherwise “tongue” will refer to the body. The body lies parallel to the throat at the back and parallel to the top of the mouth at its front. It starts in the back with attachments to the hyoid bone and ends in the front at the tongue tip. The root lies in front of and parallel to the back part of the body, also connects to the hyoid and to the top of the tongue, and contains the hyoglossus muscle, an extrinsic muscle.

The 4 intrinsic muscles are the inferior and superior longitudinal muscles, the transverse muscle, and the vertical muscle. The longitudinal muscles run along the length from the hyoid to the tip, and their pulls shorten the tongue, and as a consequence widen and give it more height. The superior muscle runs along the top surface, and the inferior muscle runs along the bottom surface. If they are set at distinct lengths, then the tongue will curl or bend in the direction of the tauter muscle. The transverse muscle runs from side to side, and its units can target particular places of articulation. Its pull narrows the tongue, and thus lengthens it and gives it height. The vertical muscles’ fibers run from the upper surface to the bottom surface, and its units can target places of articulation as well as central or side locations. Its pull shrinks the height, and thus widens and lengthens the tongue. The pulls of the intrinsic muscles assist the extrinsic muscles for actions elongating or widening the tongue.

The 4 extrinsic muscles are the genioglossus, hyoglossus, styloglossus, and palatoglossus (Fig. 1). The genioglossus attaches to the root and the front of the tongue, and pulls it towards the chin. It thus can pull downwards at the tip, forwards at the root, and the various angles in between. It also pulls centrally. The styloglossus is like the genioglossus’ inverse. It attaches along the whole body and pulls it backward. For the part of the tongue that is horizontal, especially the front, this will also produce a pull upwards from rotation. It produces a pull outward, and this outward pull, when counteracted by a transverse unit, creates tension at that specific location. The hyoglossus connects the middle of the tongue to the hyoid bone, and the muscle acts to pull the tongue downwards. Obviously, gravity already does this, so it can function with antagonists—the styloglossus and palatoglossus—to produce tension or potential energy. The palatoglossus is then a relatively small muscle attaching to the back-middle of the tongue, and pulls upward (and slightly backwards).

The muscular behavior of speech can be understood as being governed by 2 dimensions of hierarchy: central-to-peripheral and coarse-to-fine. Coarse-to-fine hierarchy occurs with both the face and jaw muscles, but

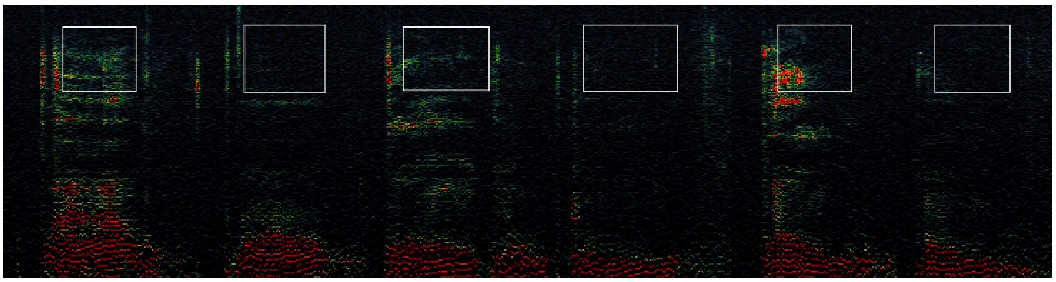

Figure 2: Front Tongue Quadrant Length. Longer front quadrant lengths produce more of a response, over the vowels, in the frequency range of ~9,000 Hz to ~14,000 Hz. The white boxes are approximately the vowels at ~8,600 Hz to ~15,000 Hz. Frequencies shown are 1,800 Hz to 18,000 Hz, on a logarithmic scale, with a window size of 1024 samples.

that will be discussed later. For the tongue, the hierarchy won't interact much with the later topics, so they'll be discussed here. These hierarchies are with regards to planning and temporal variance: coarser and more central elements must be planned before finer and more peripheral elements. They are also more likely to be kept constant over longer stretches of behavior (or multiple syllables) while the finer and more peipheral elements are varied. The most central element of our concern is the skull, and from there the central-to-peripheral order is the mandible, the hyoid, and then the tongue. The coarse-to-fine hierarchy of the tongue starts with the total sum of all the intrinsic muscles, followed by 4 quadrants, 3 subsections of the front two quadrants, and if applicable, smaller elements such as muscle units. There are 2 quadrants for the back part of the tongue, and two quadrants for the front part of the tongue, which will be called the front quadrant and middle quadrant. These quadrants are where lingual prosody occurs, named so since their changes produce a change to the speech act. The exact dividing line between the quadrants isn't clear, but more or less 4 equal parts is a first approximation.

The more invariant central and coarse elements are not arbitrary, but instead are important for contributing meaning to clauses, through a collection of initial parameters. All of these parameters are based on assigning discrete values. The muscles can be relaxed or tautened, but have various fixed points where they come to rest at, and these positions and the neighborhoods around them are considered a value. The prosodic parameters include the default value for the face and jaw bundles and groups (§3.1), as well as the initial front quadrant and middle quadrant values.[2] The values of the middle quadrant come from the change in surface tension in this region. The values of the front quadrant come from the length of the quadrant, which probably also produces increased surface tension with increased length. The middle and front quadrants can have 4 different values, meaning four regions of surface tension or muscle length which change the meaning of the speech act. (The semantics will be discussed in Appendix A.) The tonal area effect by the front tongue length is between ~9000 and ~14,000 Hz (Figure 2). The tension of the middle quadrant shows up between ~5400 and ~9000 Hz (Figure 3).

Its possible to give a rough model of how to produce the tension. The first task for the middle quadrant is to stretch the area of the quadrant. This is done with the styloglossus at the quadrants 4 corners. But the styloglossus pulls wide and back. The front corners need to be pulled forward without negating the wide pull from the styloglossus. This can be achieved with a genioglossus unit at this location, with its pull inward counteracted by the longitudinal muscles. This produces a larger region, and then increasing the tautness of the intrinsic muscles (with transverse and longitudinal forces canceling each other) leads to the tension.

[2]The back two quadrants also receive values, but they are not included since there significance were noticed too late for this study.

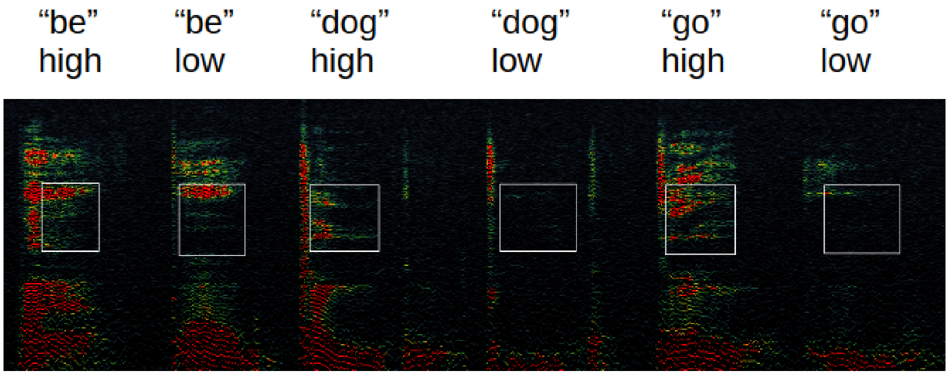

Figure 3: Middle Quadrant Surface Tension. More surface tension in the middle quadrant corresponds to more of a response, over the vowel, from ~5,400 HZ to ~9,000 Hz The white boxes are approximately the vowels at ~4,800 Hz to ~8,900 Hz. Frequencies shown are 1,800 Hz to 18,000 Hz, on a logarithmic scale, with a window size of 1024 samples.

For the front quadrant, lengthening occurs by using a differential of the superior and inferior longitudinal muscles to produce a force up or down to counteract the genioglossus and styloglossus pulls in the opposite directions. By alternating, elongation can occur. While relaxing a muscle doesn't directly produce a force, it will often convert potential energy into kinetic energy of antagonistic muscles. Thus, one can relax the inferior muscle leading to a force upward, and counteract it with a pull from the genioglossus. This will produce pulls which will help the inferior muscle lengthen and not just be relaxed. Then the superior muscle can be lengthened, producing a pull downward which can be counteracted by a styloglossus unit and so on. The inward or outward pull from the extrinsic muscles can be canceled out by an intrinsic muscle.

Neurological Aspects. The syllable needs to be planned out in advance, in that phoneme production can vary based on neighboring phonemes and other aspects of articulation. Larger units than this are likely planned too, but not in their entirety. For instance, tonal coordination suggests that one has to at least avoid certain sequences, though one doesn't have to determine all the tones from the start. Within the syllable, there is a sequence of phonemes, but there is minimal variation from other articulatory properties. The changes that do occur on occasion play a role in language, such as stress, tonal direction, or encoding information about unarticulated arguments. To finish up the physiology of the tongue, we discuss phonemes and their relation to other articulation features.

Most aspects of speech articulation uses one side or the other of the face, jaw, or tongue, so that behaviors are unilateral. Each muscle occurs on both the left and right side of the body. One side of the face is used for prosody, while the other side is used for most of the phonemes. While speech uses both sides of the tongue for phonemes, the vowel is required to be on the opposite side as the prosody. Often, the initial consonant also occurs on the non-prosodic side while the final consonant occurs on the prosodic side.[3] More complicated cases like consonant clusters and diphthongs haven't been looked into. How much of this unilateral activation is obligatory as opposed to preferential also isn't clear.

The motor neurons and thus muscle activations of speech are shared with mastication (Jürgens (1998)), with speculation that all the motor neurons stem from mastication (Lund and Kolta (2006), Hiiemae and Palmer (2003)). The tongue movements for the phonemes simultaneously activate muscles of the face and jaw, and vice versa. It's likely that the face-tongue units are a subpart of behaviors for cleaning the cheeks and gums. This behavior has a raising element (probably to limit food particles going to the back of the mouth), a downwards bending element, a sidewards slanting of the tongue for side locations, plus

[3]Since the face muscles of prosody are also used for phonemes, there may appear to be a concern in the consistency of the claims made here, but prosody's grammatical functions are likely detected over the vowel.

a sweeping motion. The hypothesis here is that these behavioral units of the cleaning movements are also behavioral units of phoneme production. These behaviors are implemented on the 6 subsections of the front and middle quadrants. There are behavioral units for each of the tongue regions, with the exception that some behaviors aren't applicable at the tip.

The raising behavior will be addressed in §2.2. There is a mechanistic relation between the jaw muscles and the tongue movement. The other cases—the face and lip muscles—don't have a mechanistic relation with the tongue, but are the coordinated activation of distinct mechanisms. The bending behavior occurs with the orbicular oris, the muscle around the lips. On each side, the top and bottom segment of the orbicular oris muscle is divided into 3 regions. Thus, all 6 regions, including the tip, can be bent. The bending is downward, so that the inferior is tauter than the superior longitudinal muscle, when the corresponding region is tautened, and the tongue's upward bend occurs when the corresponding lip region is most relaxed. The sideways sweeping behavior occurs with the muscles described in §2.3, plus some others, such as the levator labii superioris, a muscle lying to the side of the nose. Roughly, it appears that the sweep outward widens a subsection of the tongue, and the sweeping back inward occurs by lengthening and thus narrowing the subsection. The description of this behavior can't be given, but once it is discovered in one region, the same description and dynamics should occur for the other regions too. The full outward sweeping motion also involves slanting one side downwards, achieved with moving the jaw sideways and asymmetry in the height of the jaw on each side. The former is likely the basis for the front-back vowel classification, with the chin moved to its widest corresponding to rounded vowels; the latter is used for the mandibular bundle initial value described in §3.1.

To summarize, the jaw muscles are involved in lifting the tongue's subsections, the lip muscle is coordinated with up and down bending of the tongue, with different regions of the lip corresponding to different subsections of the tongue, and the face muscles around the lips coordinate with a widening or elongating mechanism in each region. In mastication, these are used with a side tilt to produce a sweeping motion. These preliminary observations are hypothesized to be used in phoneme and prosody production.

A useful note on trying to detect these values is the interaction with intention. It sometimes happens that the side one intends to articulate the phonemes on is the side one will articulate the prosody and not the phonemes. Other times, the sides align. The condition for the same side is likely the left brain having control, and the opposite side the right brain having executive control.[4] Regardless, even when the intention side doesn't match the phoneme articulation side, there are means of studying what's occurring. To determine what the face is doing during a phoneme (or the tongue for that matter), one has to take note of the initial position, draw one's attention to the other side of the face, form an intention and perform whatever amount of the phoneme(s) one wishes and pause, and compare the resulting positions with the initial positions. One may also note that one's hands don't need to be touching the same side as one's attention, and the changes felt are sometimes strong enough to grab one's attention despite not aiming it at the non-prosodic side.

1.3 Data Collection

Before implementing the spectrogram, the main way of obtaining information for the grammar was by (1) transcription of tone with a musical instrument and (2) sensory identification of the position of the jaw, face, and their muscles' states. With regards to (1), anybody with relative pitch shouldn't have too much difficulty with picking up the end tone of a syllable, though it's more difficult than transcribing notes from musical instruments. This difficulty shows up on a spectrogram, with musical instruments having flat, even tones, while speech does not. The initial tone can be more difficult to figure out. If not using recordings, a methodology for this is to produce false starts so that the initial tone is basically all that's produced. For

[4]One can develop the skill of feeling the difference, by e.g. becoming accustomed to which side of the visual field is stronger. The left eye's field will be stronger when the right brain is in control, and the right's eye field will be stronger for the left brain. One can also learn to control which side is stronger.

recorded audio, the initial tone can be picked up by repeated listening to a slowed down recording. Individual skills may vary and some may be able to pick up the initial tone with more challenging circumstances. The significance of the initial and final tone will be discussed below.

With regards to (2), there is much more to say, especially since the acoustic correlates are not specific enough to use for verification, and the skill of auditory identification is only developed in so far as one can reliably reproduce data. By putting one's fingers at various locations and having a general anatomical and physiological model, one can associate certain sensations or changes in pressure with different muscular behaviors. Around the lips, one can place the fingers, and with the descriptions in §2, begin to pick up on the changes that occur. For other articulation cases, one can place one's fingers atop the bottom of the jaw, and sense its slip under the finger; likewise for under the chin. And then on the face, beneath the outer eye and in front of the ear is another location of sensation.

The challenging aspect of detecting these changes in pressure is the limitations of attention. These face muscles sometimes work as a bundle, so that they all change, and sometimes individually, so that a change in one place doesn't correspond to a change elsewhere. Since the sensations are so subtle, they don't grab one's attention and instead one has to direct one's attention, and this directing can't be towards multiple places at once. Thus, detecting a change in one place leaves open the question of whether there was change elsewhere. One technique for addressing this is to make notes of starting positions, and then to compare the resulting position. However, some muscle movements are subtle and small and difficult to tell whether a change has occurred or not.

There isn't much to be done about such limitations but to repeat experiments over and over again. One helpful technique is to use a separation of intention and attention. One can produce the intention to articulate e.g. with the cheek muscles, and then draw one's attention to the lower lip. Separating intention and attention is helpful in detecting when multiple muscles do not work in sync but work independently. The intention at the end can be verified, but still there is the issue of other locations that weren't attended to. Another issue is detecting the smallest unit of change and detecting the size of change more generally. These issues can't be given straightforward solutions. But large numbers of trials all being consistent with one hypothesis and all other hypotheses having at least some trial being inconsistent with that alternative hypothesis leads to strong evidence. This was the standard used to generate the grammar.

Similar issues with detecting the tongue behavior could be mentioned, but since most of these issues have been eliminated or become irrelevant, they are omitted.

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In this section, we discussed the nature of the spectrograms and perceptual cues. The tongue, being the most peripheral element of speech, plays a prominent role in speech in the same way that the hands play a dominant role in manipulating objects. The tongue has both coarse and fine properties. The tone effects the tongue as a whole and from there it's finer use stems from a division into 4 quadrants, with the front two further divided into three subsections. The quadrants are utilized for prosody while the subsections are important for phonemes. The relation between these subsections and the articulation that's to come was discussed. In the next section, we will discuss the muscles of the face and jaw, as well as tones. All of these have effects on a spectrogram and have perceptual correlates—both auditory and tactile. But moving forward, the physiology of the articulation becomes a secondary concern, as the focus shifts to values that are implemented in the grammar.

2 Articulation

The grammar is built upon 3 main articulation values, which will be called the tonal value, mandibular bundle (MB) value, and the labial bundle (LB) value. The tone corresponds to the fundamental frequency.

The MB and LB are parts of prosody, and involve multiple muscles centered around the jaw and mouth, respectively. Each value is derived from breaking up a continuous signal into discrete groupings. The tone breaks up a continuous frequency range into tonal regions, while the MB and LB's values will be based on discrete regions of muscular length. A fourth value, the direction value, modifies the other 3 articulators. Further, each value type is involved in a system of modular arithmetic, with tonal, MB, and LB values using mod 2, mod 3, and mod 4 arithmetic systems, respectively. The 3 values together are used to articulate the grammar, in the sense that their changes correspond to changes in syntactic structure. For each value type, some notes about articulation and underlying structure will be given, followed by formalizing the value, and introducing any computations that the values are involved in. Lastly, we discuss the basis of the direction value and how it changes the computational properties of the main articulators.

2.1 Tone

There are 3 possible patterns for the tone within English grammar: an increasing, decreasing or flat tone. The pattern is solely based on the starting and ending tone; in between these two tones can be all kinds of fluctuations. It isn't uncommon to see a tonal spike within a syllable, most likely a sudden increase or decrease followed by rapid corrections.[5] But the most common case is for a steady rise or fall or a lack of movement. A flat, increasing, or decreasing tone are properties of the tonal value. The tonal value is assigned to syllables and is used to perform computations between them. When articulating, the tone must be planned before a syllable starts, and especially before the syllable's vowel.[6] So, we first speculate about the physiological correlates for the tone, and then describe the tonal value, the difference computation, and a comparative computation used in the grammar and for semantic purposes.

Currently, the belief is that the physical basis for the tone is the larynx (Olson (1967)), or the changes in the length and tension of the vocal folds (Seikel et al. (2025)). On the other hand, experimentation suggests that the intrinsic muscles of the tongue varies with the change in tone, with the muscles pulled tauter corresponding to higher tones. The most plausible explanation is that the tongue tautness forms a behavioral unit with the muscles of the trachea, so that they both change synchronously. One can experiment and demonstrate that the extrinsic muscles as well as the jaw position are irrelevant to the tone. The surface area of the tongue also doesn't matter, so that it's not the tension nor length of the intrinsic muscles that is relevant. Additionally, tautness or relaxed muscles can occur in various regions and produce the same tonal effect. Since it's the overall tautness, there are multiple realizations of a tone's lingual behavior. Thus, from a control or planning perspective, only a general tonal region is initially needed, and then fine tuning can occur to accommodate other articulatory properties, including phonemes.

A quick note about tonal values may be helpful before describing them. A western 12-tone scale is assumed and was sufficient for analysis. The distance between neighboring tones is considered a half tone, and the distance between every other tone is a whole tone. This convention finds favor with some general patterns we find. In particular, the difference computation suggests that a half tone and whole tone difference are treated differently, and dynamic changes which are or are not integer multiples of whole tones get different treatments, or so current empirical evidence suggests.

A tone may be represented as a vector $\begin{bmatrix} b & k & d & l & c \end{bmatrix}$, or in the form $\int_{-cd\frac{l}{2}}^{dl-cd\frac{l}{2}} 2b-k+t$, where b is the basic tone, k is the key and restricted to 0 or 1, d is the direction of the dynamic change (1, 0 for no change, -1), l is the length of the dynamic change in half-tones, c is whether the length is a non-zero integer multiple of a whole tone or not (values 1 or 0), and t is a variable, such that it's substitution with the integral sign's

[5] This is mainly seen with less samples than will be used here. The correction occurs so fast, it's likely that there are mechanisms to stabilize a tone, similar to how the eyes stabilize an image, for instance keeping an image still or tracking an object (see e.g. Gallistel (1980), Goldberg et al. (2012)).

[6] It's not clear whether tone is applicable over consonants with more sonority, but it's not applicable over stops and fricatives.

limits produce the end tones.[7] When the dynamic change is a whole tone length, the basic tone is taken as the central tone, and when the dynamic change has a half tone part, the basic tone is taken as the initial tone. Thus, to give an example. Let the 110Hz A-note be arbitrarily assigned the value 12, and suppose a syllable goes from G (~98Hz) to A. We have a positive direction ($\begin{bmatrix} ? & ? & 1 & ? & ? \end{bmatrix}$), a multiple of a whole tone ($\begin{bmatrix} ? & ? & 1 & ? & 1 \end{bmatrix}$), a length of 2 (half tones) ($\begin{bmatrix} ? & ? & 1 & 2 & 1 \end{bmatrix}$, and the middle note is $11 = 2 \cdot 6 - 1$. So this would have the notations $\int_{-1}^{+1} 11 + t$ and $\begin{bmatrix} 6 & 1 & 1 & 2 & 1 \end{bmatrix}$.

The difference between two tones can be computed from the vector. But first, there is the question of where the conclusions for the differences come from. The conclusions come from testing the grammar. There were tonal value sequences which were impossible for a given clause or basic sentence (§3.2.1). That it's a difference in values and not absolute comes from getting different results from different starting points. So then, attempting different changes allowed an inference to see what the differences were. For instance, once one knows whether a tone can be repeated or not, one can then test various cases (key changes, dynamic lengths, and so on) in the known contexts to find equivalence classes. By changing the different variables of the vector, and looking for internal consistency across experiments, the vector elements and the difference function were the conclusion.

The difference of two tonal values T_1 and T_2, given that $T_1 = \begin{bmatrix} b_1 & k_1 & d_1 & l_1 & c_1 \end{bmatrix}$ and $T_2 = \begin{bmatrix} b_2 & k_2 & d_2 & l_2 & c_2 \end{bmatrix}$, is

$$\operatorname{diff}(T_2, T_1) = \left\lceil \frac{(2b_2 - k_2) - (2b_1 - k_1)}{2} \right\rceil + c_2 \cdot c_1 \cdot |d_2| \cdot |d_1| \frac{2 - |d_2 + d_1|}{2} + |c_2 \frac{l_2}{2} - c_1 \frac{l_1}{2}| \pmod 2$$

In words, the sum may be stated as the following rules:

- There is a difference of $n \pmod 2$ if the tone changes by $2n$ or $2n-1$, in half tones ($\left\lceil \frac{(2b_2 - k_2) - (2b_1 - k_1)}{2} \right\rceil$).
- If both lengths are whole tones ($c_2 \cdot c_1$), then there is a difference of 1 if the direction changes ($\frac{2 - |d_2 + d_1|}{2}$, and $|d_2| \cdot |d_1|$ to avoid non-directions).
- There is a difference of $n \pmod 2$ if the whole tone length changes by n ($|\frac{l_2}{2} - \frac{l_1}{2}|$), where non-whole tone lengths are set to 0 ($c_i \frac{l_i}{2}$).

Note that because of the ceiling function term in the sum, the difference function is not commutative, so that it's not always true that $\operatorname{diff}(T_2, T_1) = \operatorname{diff}(T_1, T_2)$. All else being the same, it's not commutative when there's a key change. Second, since a direction change can be a distance of 1 from one another, but both are a direction of 1 from a tonal value with no dynamic change, then the triangle equality—or the equality condition of the triangle inequality–isn't satisfied, so that it's not always true that $\operatorname{diff}(T_3, T_2) + \operatorname{diff}(T_2, T_1) = \operatorname{diff}(T_3, T_1)$. These are useful facts for testing the grammar.

For the comparison of end tones, we have two functions $L(T_1, T_2)$ and $R(T_1, T_2)$. The basic idea of the comparison is that a supporting element is compared to a base element. For instance, a noun is compared to a verb within the same clause, or the verb of an embedded clause is compared to the verb of the embedding clause. This gives information about the supporting or embedded element. When this lesser element has a greater tone, the comparison outputs a 1, and outputs a -1 when it has a lesser tone. The tricky part is that this is handled differently when the secondary element is to the left or right of the primary element. When it's to the left (the function L), its final tone gets compared to the primary element's initial tone, and when it's to the right (the function R), its initial tone gets compared to the primary element's final tone. The primary element is listed first among the functions' inputs.

[7]The latter notation should not be mistaken for an integral. In particular, the changing tones don't have to be linear. However, the idea of going from the bottom tone to the top continuously is meant to be conveyed.

The initial tone of a tonal value $T = \begin{bmatrix} b & k & d & l & c \end{bmatrix}$ is

$$\text{start}(T) = (2b-k) - \frac{l}{2} \cdot c \cdot d$$

and the final tone is

$$\text{final}(T) = (2b-k) + l \cdot d - \frac{l}{2} c \cdot d$$

Thus, we have

$$L(T_1, T_2) = \begin{cases} 1 & \text{if final}(T_2) > \text{start}(T_1) \\ -1 & \text{if final}(T_2) < \text{start}(T_1) \end{cases}$$

$$R(T_1, T_2) = \begin{cases} 1 & \text{if start}(T_2) > \text{final}(T_1) \\ -1 & \text{if start}(T_2) < \text{final}(T_1) \end{cases}$$

~ ~ ~

The MB and LB share some general patterns which aren't applicable to tones. Being more central, the muscles themselves suffice for analysis rather than finer divisions. The bundles are split into four regions — south, south-east, east, and north-east,[8] with each region corresponding to at least one muscle. They are called bundles since all 4 regions work in sync. If not otherwise indicated, the effects described refer to the effects of higher values and shorter, tauter muscles. It appears that the auditory and semantic effects don't occur if the length of the muscle doesn't change, and so length is the property emphasized. The positive direction of the MB and LB values correspond to shorter muscles.

2.2 Mandibular Bundle

The mandibular bundle value is determined by its 4 regions. The jaw is held in position by various muscles, and a change to each of the muscles produces distinct changes to its position, such as rotations or translations, or changes its position relative to the hyoid bone. Looking ahead, the four regions and the muscles for each region will be specified, follow by a discussion of the muscles' effects on the tongue. Then the spectral effects of the mandibular bundle will be given. Finally, the MB values for the grammars are formalized.

The muscles are displayed in figure 4. The south region brings the chin and the combination of the hyoid bone and mouth floor closer to one another. If the other muscles of the bundle are longer, it'll lower the chin down towards the hyoid; if the other muscles are shorter, it'll raise the hyoid. The main muscle that does this is the mylohyoid muscle, although it probably has the help of other superior hyoid muscles. The south-east region raises the side base of the jaw, between the chin and the corner or angle, or just the jaw more generally. There can also be a pull backwards towards an overbite alignment. This results from the temporalis muscle. The east region has the sensation of the skin being pulled back toward the ear at the mandible's ramus, i.e. the jaw's side where it is a narrow, vertical strip. This is produced by the pterygoid muscles, both the lateral and medial pterygoid muscle. Finally, the north-east region puffs out the outside of the cheek, below the outer side of the eye. This stems from the shortening of the masseter muscle.

The change at each region demonstrates the effects of changes to more central elements in the central-to-peripheral hierarchy. At any given time, the various extrinsic tongue muscles are all pulling against one

[8] Since prosody may be articulated on either the left or right side, so east may be replaced by west.

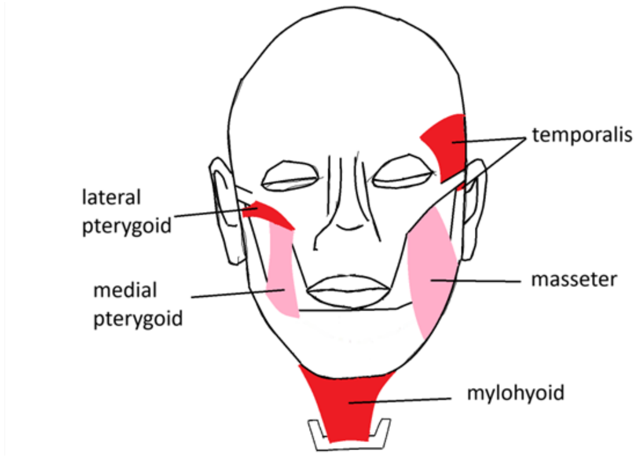


Figure 4: Mandibular Bundle Muscles.
Each muscle occurs on both sides of the face. Face outline traced from medicalartlibrary.com

another to produce the stable location of the tongue; the forces cancel out, but one has potential energy: the forces that would apply if one of the other forces is removed or modified in some way. The south region pulls up the hyoid bone, so that the hyoglossus applies less force, and the potential energy of the styloglossus and palatoglossus muscles apply, and the back mid part of the tongue is raised. (One can use one's molars to feel the pressure of the tongue raised.) The south-east region has two possible effects. The temporalis is a large muscle with many fibers, and some connect straight downward, while others have more of a backwards pull. The pull upward slackens the vertical genioglossus muscles, so that the styloglossus potential energy is released near the tip of the tongue. The pull backwards effects the genioglossus units that are nearly horizontal, and so the styloglossus potential energy is activated for pulling the tongue back, but also for pulling it upwards at the most back part of the top surface. The east and northeast effects are based on rotations upwards. These rotations will slacken the diagonal genioglossus muscles most, thus the styloglossus lifts the tongue somewhere in the center. The lateral pterygoid, for the east region, rotates the jaw around a point closer towards where it meets the skull, and the masseter, for the north-east region, rotates the jaw around a point closer towards the corner of the jaw. Thus, the north-east rotation is smaller while the east rotation is a larger circular path. The result is the east region's diagonal effect being closer to a horizontal effect and the north-east region's diagonal effect being closer to a vertical effect. The medial pterygoid has the lateral ptergoid's rotation plus a translation upwards, making its diagonal effect even more vertical than the lateral pterygoid's effect. If the subsections of the tongue are named 1-6, with 1 being the tip's subsection and 6 being the back most subsection, then we have the mylohyoid/south raising 5, the temporalis/south-east raising 6 and 1, the lateral pterygoid/east raising 3, the medial pterygoid/east raising 2, and the masseter/north-east raising 4.

On the spectrogram, higher MB values produce a larger response in the harmonic region for consonants with less sonority, especially stops, as shown in Figure 5. It likely has an effect on vowels too, but it is less visual. For instance, lowering the gain tends to make the lower MB valued examples disappear in the harmonic region first, but given natural volume variation in the voice, it is not a strong visual effect.

The mandibular bundle values encode information with mod 3 arithmetic. Changes in values occur at any of the 4 regions, with the total changes at each region being summed mod 3. It can be represented as a vector $\begin{bmatrix} b & d & l & c \end{bmatrix}$, where b is the basic value, d is the direction of dynamic change (1, 0, or -1), l is the dynamic change's length, and c is 1 when d is a non-zero even number and 0 otherwise. This corresponds to $\int_{-cd\frac{l}{2}}^{dl-cd\frac{l}{2}} b+t$. The basic value is the sum of the values of each region, at the center tone—between the final and intial tone—if c=1, and at the initial tone if c=0. An MB value may be translated into an integer mod 3,

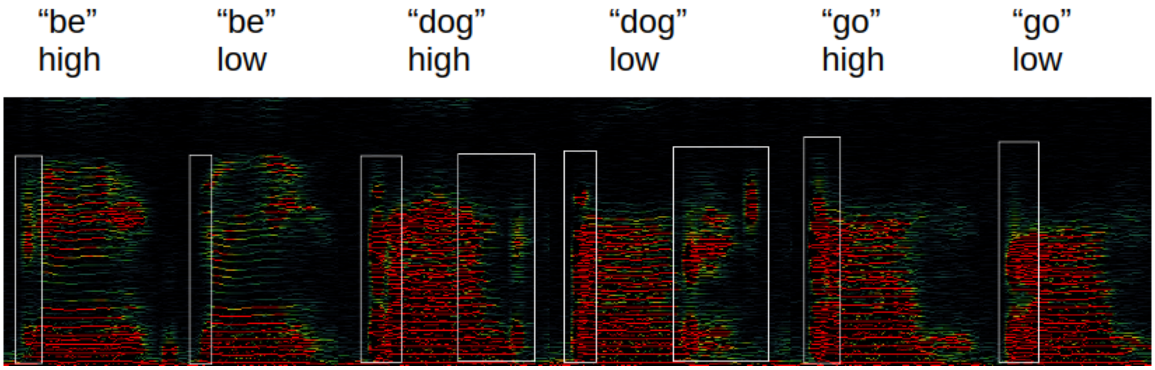


Figure 5: Mandibular Bundle Spectrogram. Higher MB values produce more of a response of the consonants in the harmonic range. The white boxes are approximately the consonants. Frequencies shown are 50 Hz to 3,500 Hz, with a linear scale and a window size of 4096 samples.

given that some initial sum of regional values b' is $0 \pmod 3$ by the formula

$$\mathrm{int}(\begin{bmatrix} b & d & l & c \end{bmatrix}) = (b + c \cdot d\frac{l}{2}) - b' \pmod 3$$

Likewise, if $\begin{bmatrix} b & d & l & c \end{bmatrix}$ is known to be $0 \pmod 3$, then the above formula can be set equal to zero, and then solved for b'. In words, the formula can be translated as:

- The 4 region sum by itself is the arithmetic difference from the $0 \pmod 3$ value.
- If the dynamic length is an even value (c), then add a positive direction (d) half-length ($\frac{l}{2}$), and subtract a negative direction (d) half-length ($\frac{l}{2}$).

Odd dynamic lengths, i.e. when $c = 0$ and $l \neq 0$, are Q values. They will be used at the level of the sentence. They have the property that $q = -1 \leftrightarrow l \equiv 1 \pmod 4$ and $q = 1 \leftrightarrow l \equiv 3 \pmod 4$.

2.3 Labial Bundle

The labial bundle also consists of 4 regions, and each region has 2 muscles, and there's a greater possible range of values than the MB. As a whole, the effect of the labial bundle is to tighten or loosen the skin around the lips. The muscles are illustrated in figure 6.

The south effect is the shortening of the skin between the center of the lip and chin, sometimes with the lip also lowered. The muscles for this effect are the mentalis and the depressor labii inferioris. The mentalis is unique in shortening and becoming more taut for lower rather than higher values. This allows the lower lip to move outward, providing more room for the other muscles to lengthen into. The south-east effect occurs with the shortening of the skin below the corner of the mouth. The muscles used here are the depressor anguli oris and the platysma muscle. The east or west effect draws the mouth corner outwards, with values accomplished by the risorius and buccinator muscles. The risorius and buccinator lie within the cheek, with the risorius being towards the outer surface and the buccinator towards the inner surface. Finally the north-east effect stems from the minor and major zygomaticus muscles. For this region, the side and corner of the upper lip are raised outward. Depending on the configuration of other muscles, this effect can produce a bulge in the upper cheek below the eye, as well.

In typical speech, a given value can be formed by various combinations of shortening either of the muscles in each region, and different ways are used with different syllables. Thus, when one puts one's

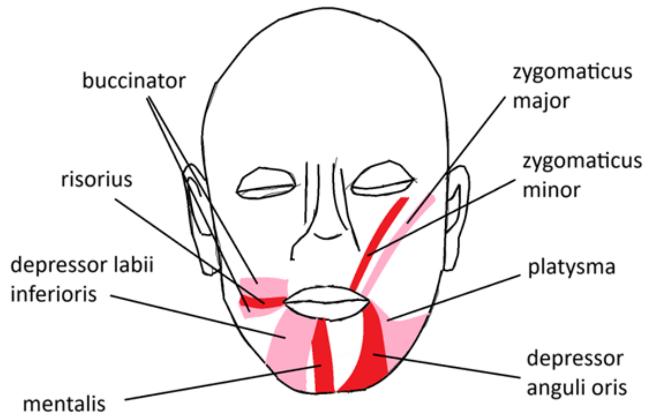


Figure 6: Labial Bundle Muscles. Each muscle occurs on both sides of the face. Face outline traced from medicalartlibrary.com

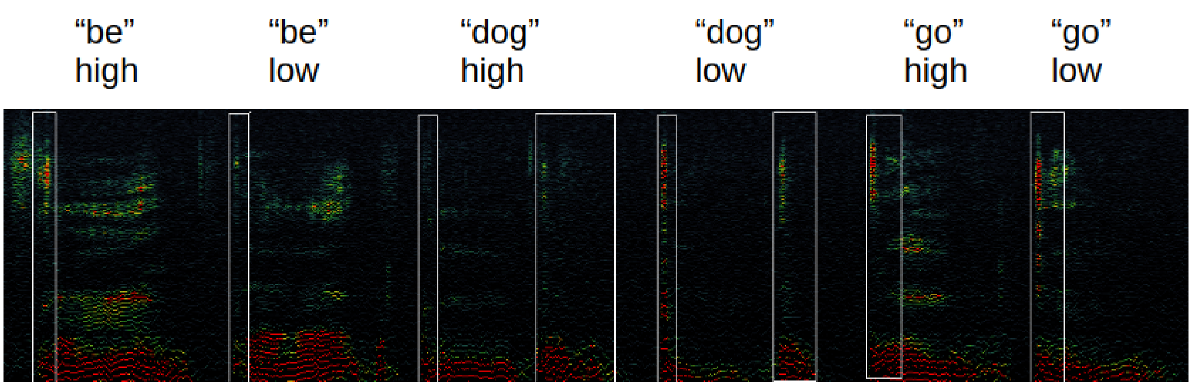


Figure 7: Labial Bundle Spectrogram. Higher LB values produce more of a response for vowels within the prosodic region, relative to the consonant response in the same region. The white boxes are approximately the consonants. Frequencies shown are 1,800 Hz to 18,000 Hz, on a logarithmic scale, with a window size of 1024 samples.

hand on a given region during speech, one will feel movement. This can be diminished by using a slow pace and factual tone, for the sake of obtaining cleaner data to work with. For higher values, the labial bundle shows up on the spectrogram as a stronger response in the prosodic range for vowels, relative to consonants (Figure 7).

A LB value is a vector $\begin{bmatrix} b & d & l & c \end{bmatrix}$, where b is the basic value, d is the dynamic direction (1, 0, or -1), l is the dynamic length, and c is 1 when the dynamic length is even and non-zero, and 0 otherwise, so that the vector corresponds to $\int_{cd\frac{l}{2}}^{dl-cd\frac{l}{2}} b+t$. The basic value is the sum of the values of each muscle, at the center tone—between the final and intial tone—if c=1, and at the initial tone if c=0. The LB value's mod 4 integer, given $b' \equiv 0 \pmod 4$, is

$$(b+c\cdot d\frac{l}{2})-b' \equiv 0 \pmod 4$$

In words this means that

- The 8 muscle sum by itself is the arithmetic difference from the $0 \pmod 4$ value.
- If the dynamic length is an even value (c), then add a positive direction (d) half-length ($\frac{l}{2}$), and subtract a negative direction (d) half-length ($\frac{l}{2}$).

~ ~ ~

When l is odd, the dynamic part is called a Q value. In particular, $q = -1 \leftrightarrow l \equiv 1 \pmod 4$ and $q = 1 \leftrightarrow l \equiv 3 \pmod 4$. These Q values are significant for preserving clausal information when there are unarticulated arguments along with the tonal comparison. They are also used for interrogative pronouns and questions.

2.4 Directional Element

Phoneme articulation is dynamic, not static, and the dynamics can vary from case to case, producing among other things, accents. Different preparatory work for a phoneme can occur before or after its onset, as well as the onset of other phonemes within a syllable. An important variation in English grammar is whether one goes into the vowel or out of the vowel, meaning, respectively, whether the vowel set up starts after its onset or before the onset of the syllable. For instance, the vowel /i/ involves tautening of the platysma muscle on whatever side it's articulated on. If the onset occurs before the onset of e.g. /si/, then the platysma will shorten; but when it's activated after the onset, then it produces tension, but minimal shortening. This going into and out of a vowel for particular constituents is the directional element or direction value. The directional element for a clause has a value of 1 when the verb's main syllable's vowel is gone out of, and -1 when it's gone into. When the value is -1, all of the articulation features get inverse values. For mod 3 and mod 4 systems, it amounts to multiplying the values by -1. But for tones, $-1 \equiv 1 \pmod 2$, so instead, the various functions which take tones as input have the opposite value: differences of 0 become differences of 1 and vice versa, and L and R comparisons become multiplied by -1.

3 Applications

The intonation and prosody are used for various purposes. In the previous section, the focus was on presenting their main use with the grammar, and in this section emphasis is placed on other properties and uses. The face and jaw muscles have coarser properties, applicable at longer temporal lengths, which effect the particular speech act one performs. On the intonation side, clauses take inputs and tones are a major part of distinguishing the inputs, which was the initial source of inferring the tonal properties. This is given a more extended discussion since the case system is not standard analysis of English grammar. The grammar gives few constraints on the tones and so the sentence's melody, the L and R functions, and the dynamic change direction all are used to communicate information about aspect and nouns.

3.1 Prosody

The MB and LB are part of prosody since their initial settings change the speech act. Here we discuss some of its properties. There are constraints that apply to the initial value. Further, these bundles play a role in two other well known phenomenon of speech, namely modal sentences and stress. Modal sentences require certain properties of the bundles in place, and will be a constraint on valid clauses. Finally, the LB's muscles can be divided into two groups, and each region has a muscle that belongs to one of the two groups. Each of these topics will get a brief treatment.

Initial Value. The bundles' regions can vary in their value, but the initial, sentence level value requires that they are all the same. As a hypothesis the MB has 5 possible values for each region, and the LB has 7 possible values for each region. This is a hypothesis about changes to speech acts and not about the muscle's physical capacities, which can show a greater range. In the case of the labial bundle, there can be multiple ways to realize a value in a given region. The semantic effects can be difficult to describe, but one easy case to point out is the difference between angry and sad speech, as seen in Kim (2014). Angry speech has a low initial MB value (probably 2, for values 1-5) while sad speech has a high initial MB value (probably 5). It may be that these initial values are constrained to a coarser level of value distinction than for the grammar.

Stress. Treating the bundle values as unified also occurs with stress. By playing around, it can be shown that various aspects that correlate with stress are not necessary, such as loudness and length. But what is obligatory on a stressed syllable is dynamic change of the labial bundle and mandibular bundle. For stress, the change occurs for every region, in contrast to the value changes of the grammar, where it can be region specific. In particular a stressed syllable moves each region of both bundles 1 unit above (below) the encoding value to 1 unit below (above) the encoding value. This is if there is only one level of stress. For n levels of stress, the dynamic movement occurs from n units above (below) to n units below (above). Knowing this is helpful in transcription: various movements may be from a stressed syllable rather than from other parts of the grammar, and forcing an articulation without any stressed constituents can be helpful for figuring out the grammar.

Prosodic Modals. The LB and MB values are based on the muscle length. But a given LB or MB region may be pulled tauter than the other regions without shortening, and thus producing tension. This increased tension at a given region is a prosodic modal. When it occurs for the muscles of some LB region, it is a LB modal, and when it occurs for the muscles of some MB region, it is an MB modal. This is a preliminary definition for an LB modal and MB modal: if some LB (MB) region k has more muscle tension than all the other regions of the bundle, by some threshold, we call that LB (MB) value a k LB (MB) modal value. We call them prosodic modals since they are obligatory on clauses with modal verbs. In particular, LB modals are obligatory on clauses with the verbs "can," "shall," and "will," and MB modals are obligatory on clauses with the verbs "may," "might," "must" and "ought." Note that the converse doesn't hold. Prosodic modals can occur with any verb, and are not limited to use on clauses with modal verbs.

The shortening is prevented by various antagonist muscles. For instance, many of the labial bundle muscles are antagonized by the orbicularis oris, though this may not be sufficient on its own; the mylohyoid is antagonized by the inferior hyoid muscles; the masseter is likely antagonized by the digastric muscle.

LB Groups. Each region for the labial bundle has 2 muscles, and each region has a muscle belonging to one of two groups, which we call the risorius and buccinator groups. The buccinator group consists of the depressor labii inferioris, platysma, buccinator, and zygomatic major muscles, while the risorius group consists of the mentalis, depressor anguli oris, risorius, and zygomaticus minor muscles. As with the prosodic modals, the key property for these groups is tension, but it's for all regions rather than a single region. The tensing of one of these two groups also changes the speech act, and its semantics is related to positive and negative, approval and disapproval, and so on as discussed in appendix A. The spectral effect of the muscle groups' tension is to change the width of the higher, prosodic range formants, such as those correlated with surface tension (Figure 8). The risorius group produces wide formants over vowels while the buccinator group produces narrow formants over vowels.

3.2 Intonation

3.2.1 Basic Clause Syntax

The articulation values for a clause varies depending on the cases, or theta roles, for the verb's arguments. There are 4 possible cases in English: Ergative, Dative, Oblique, and Stative. The verb "make" demonstrates in (2) the 4 possible cases in English. The oblique case is the topic, focus, element acted upon, or which undergoes change. The stative case is a secondary descriptor for the oblique element, and is the unique case

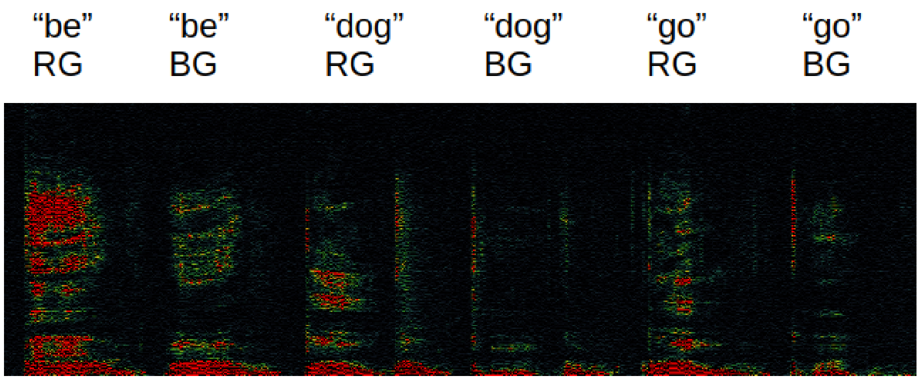


Figure 8: Labial Bundle Groups Spectrogram. The risorius group (RG) produces a wider frequency/formant response in the prosodic region, while the buccinator group (BG) produces a narrower frequency/formant response. Frequencies shown are 1,800 Hz to 18,000 Hz, on a logarithmic scale, with a window size of 1024 samples.

which doesn't reference a new entity. The dative case is a receiver of the oblique entity's state, with receiving understood broadly and includes perceiver, knower, benefactor, owner, target/goal, and accomplisher. Ergative is a causal element which produces change.

(2) a. The enthusiast makes a good teammate. (Obl/Stat)
b. The parents made the enthusiast a good teammate. (Erg/Obl/Stat)
c. The parents made the enthusiast a cake. (Erg/Dat/Obl)

In terms of structure, these cases correspond to a nominal. While most of the grammar can be built up recursively to define unique structures, this is not so for these arguments of the verb. Instead, the difference between them stems from their role in the clause rather than an internal property. Thus, these roles reference their place in a structure and are like "numerator" and "denominator" rather than "number," "teacher" and "student" rather than "person," "antecedent" and "consequent" rather than "proposition," or like "contract" and "homework" rather than "document." So our treatment will be influenced by logical formula. Everything leading up to the clause is analogous to defining the terms of the language; the clause is then analogous to a formula. The atomic formula are defined by terms and relations, and so the leap taken here is that these cases stem from relations.[9] These relations are arbitrarily posited based on the semantics that best fits across all cases. In particular, the Oblique case stems from a relation **T** (for topic) held between a nominal and a verb. The stative case, since it modifies the oblique case will be taken as stemming from a relation **B** (for beacon) between two nominals. Dative will be from a relation **C** (for confirmer) between a noun and a verb. Finally, ergative results from a relation **S** (for source) which is also assumed to hold between a nominal and a verb phrase.

A clause's prosody and intonation is determined by its arguments' cases. A different collection of cases produces a different articulation. Verbs like "make" can be used to detect the articulation variation, and then the intonation and prosody can be used to infer these cases for other verbs; from there, one can start to characterize their semantics to some degree. Each clause receives some relation(s) as an input, with English having 6 possible inputs. A fundamental property of the grammar is the preservation of the relation information, even as arguments or terms of the relations are not articulated. The cases are realized with a unique linear order in the clause, having the general order, from left to right or earlier to later time of Ergative, Dative, Oblique, and Stative. The 6 possible relations or relation compositions are shown in

[9] Another possibility is that some of these cases stem from functions. The semantic insight currently isn't strong enough to distinguish these possibilities.

(3), with the cases and an example for each. These relations can be given definitions from the 4 basic relations and conjunction. Let v be a verb phrase, and n_1, n_2 and n_3 nominals. The relations of the grammar are defined as $\mathbf{T}(v,n_1)$=$\mathbf{T}(v,n_1)$, $\mathbf{TB}(v,n_1,n_2)$:= $\mathbf{T}(v,n_1)\wedge$ $\mathbf{B}(n_1,n_2)$, $\mathbf{CT}(v,n_1,n_2)$:=$\mathbf{C}(v,n_1)\wedge\mathbf{T}(v,n_2)$, and $\mathbf{SR}(v,n_1,\ldots,n_k)$:=$\mathbf{S}(v,n_1)\wedge\mathbf{R}(v,n_2,\ldots,n_k)$, for any relation **R**. In terms of a grammar rule, the relation will be put on the left, and the resulting nominals on the right, e.g. CL(**STB**) $\rightarrow$ Erg VP Obl Stat.

(3) a. **T** [Obl] (The dog sits)

b. **TB** [Obl Stat] (The dog is happy)

c. **CT** [Dat Obl] (The priest had a sister)

d. **ST** [Erg Obl](The child splashed the water)

e. **STB** [Erg Obl Stat] (The sailor made the knot strong)

f. **SCT** [Erg Dat Obl] (The sailor gave the dog a treat)

The tonal rules for the basic clause are as follows. The verb gets the clause's tone. The arguments are then given a tone, based on the difference with its neighbor, going outward from the verb. The difference function takes the right most constituent as its first input. The difference function is 1 for all the arguments except for oblique arguments, which have a value of 0. The number of difference calculations in a clause is equal to the number of arguments. Thus the tonal difference is 0 in the **T** basic clause, the tonal differences are 0 and 1 for the **TB** clause, and 1 and 0 for the **CT** clause. There are also rules for the direction value. Recall that 1 denotes that the vowel is gone into and -1 that the vowel is gone out of. When the direction value is positive, the verbs and cases get assigned the values 1 for oblique and ergative, and -1 for the dative, stative, and the verb phrase. When the direction value is negative (-1), then the value is just the previous value multiplied by -1, so that the values switch. The MB and LB values also vary according to the clausal relations, but these will be given in the grammar.

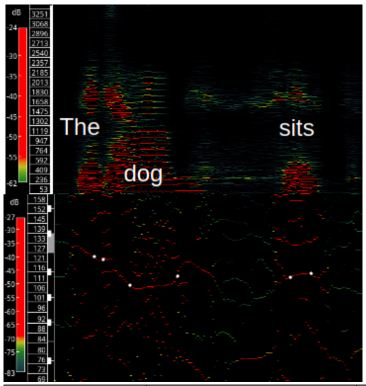
A demonstration of the tones is provided in (4)-(9), with a spectrogram plus a transcription. A close up of the notes, as well as the harmonic region, is shown.[10] The approximate initial and final note are given, and marked in white, and the note and cents measured are provided in the table. Four of the cases, (4), (5), (8), and (9), have a negative direction value, so that the oblique arguments have a difference of 1, while the other arguments have a difference of 0. The transcription is based on auditory judgments, and leave open questions about when different tones are detected as opposed to the same tone. Among the issues are how many cents before a new tone is considered in place, if it's even relative, and how can the initial and final tone be detected by some objective criteria.

~ ~ ~

This paper started with the claim that (1) has its tones constrained with respect to one another. How does a grammar with articulatory properties work? For instance, when we have NP $\rightarrow$ Det N, NP is never articulated. What happens with the articulation properties at this part of the grammatical structure? Each such node in the grammatical structure can be assigned an articulatory value, based on structural evidence. We can use existing information about syntactic structure to infer what values to assign to higher level syntactic elements. The above example is too complicated, so consider the variations in (10). We have complete

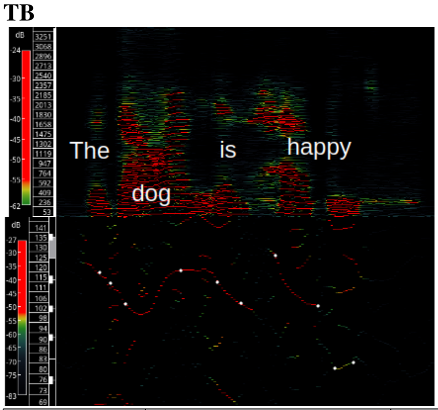
[10] The note region used a window size of 8,192 samples and the harmonic region used a window size of 4,096 samples.

(4) **T**



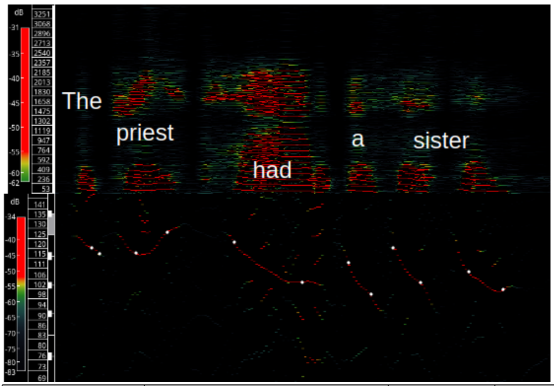
	Obl		VP_f
	Det	N	V_f
	The	dog	sits
Initial Tone	B2+27c	a2+30c	A♯2-36c
Final Tone	B2-7c	A♯2-8c	A♯2+2c
Note	B	A⌒A♯	A♯
Compute	diff(A♯, A⌒A♯)=1		

(5) **TB**



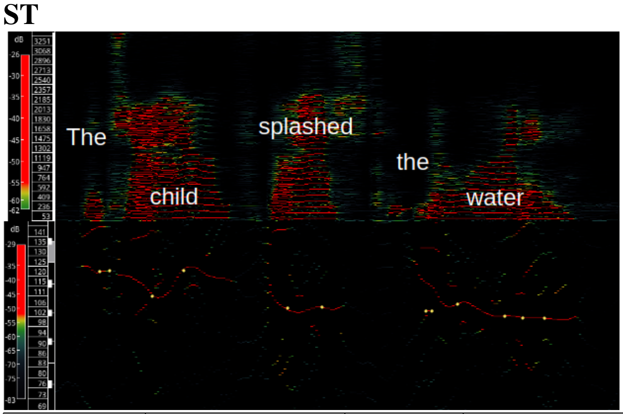
	Obl		VP_f	Stat	
	Det	N	V_f	Adj	
	The	dog	is	hap	-py
Initial Tone	A♯2+33c	G♯2+16c	A♯2+7c	C3-25c	E2-8c
Final Tone	A♯2-31c	B2-22c	A2-4c	G♯+34c	E2+39c
Note	A♯	G♯⌒B	A♯⌒A	C⌒G♯	E
Compute	diff(A♯⌒A, G♯⌒B)=1			diff(C⌒G♯, A♯⌒A)=0	

(6) **CT**



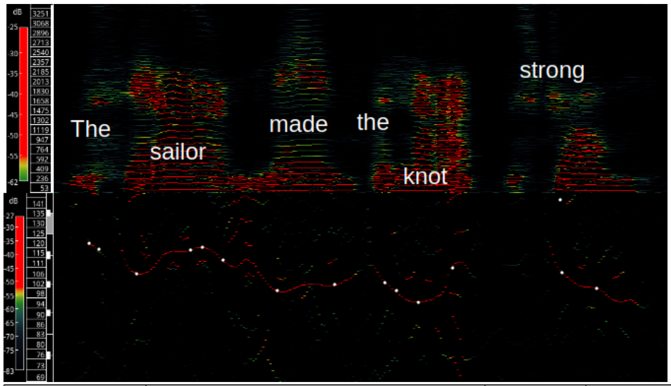
	Dat		VP_f	Obl		
	Det	N	V_f	Det	N	
	The	priest	had	a	sist	-er
Initial Tone	A♯2+18c	A♯+29c	B+4c	A2+47c	B-18c	A2+15c
Final Tone	A♯+4c	C3-20c	G♯+30c	G♯2-14c	G♯-19c	G♯2-30c
Note	A♯	A♯⌢C	B⌢G♯	G♯	B⌢G♯	A
Compute	diff(B⌢G♯, A♯⌢C)=1			diff(B⌢G♯, B⌢G♯)=0		

(7) **ST**



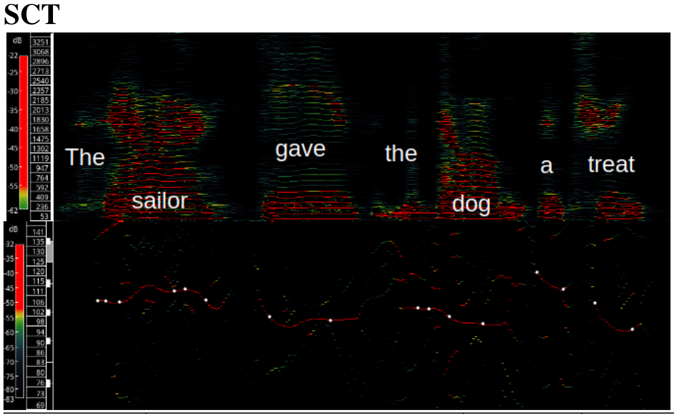
	Erg		VP_f	Obl		
	Det	N	V_f	Det	N	
	The	child	splashed	the	wat	-er
Initial Tone	B2-5c	A2+26c	G♯+40c	G♯2+21c	A2-39c	G♯2-32c
Final Tone	B2-3c	B2+0c	G♯2+42c	G♯2+21c	G♯2-15c	G♯2-34c
Note	B	A⌢B	A	G♯	A⌢G♯	G♯
Compute	diff(A, A⌢B)=1			diff(A⌢G♯, A)=0		

(8) **STB**



	Erg			VP_f	Obl		Stat
	Det	N		V_f	Det	N	Adj
	The	sail	-or	made	the	knot	strong
Initial Tone	B2+12c	A2-4c	B2-27c	G♯-37c	G♯2-3c	G2-12c	A2+18c
Final Tone	A♯2+44c	B2-49c	A♯2+11c	G♯2-1c	G♯2-22c	A2+27c	G♯2-10c
Note	B	A⌢B	B⌢A♯	G♯	G♯	G⌢A	A⌢G♯
Compute	diff(G♯, A⌢B)=0				diff(G⌢A, G♯)=1		diff(A⌢G♯, G⌢A)=0

(9) **SCT**



	Erg			VP_f	Dat		Obl	
	Det	N		V_f	Det	N	Det	N
	The	sail	-or	gave	the	dog	a	treat
Initial Tone	A2-1c	A2-17c	A♯2-24c	G♯2+30c	G♯2+36c	G♯+16c	B2-13c	A2-22c
Final Tone	A2-8c	A♯2-33c	A2+1c	G♯2-37c	G♯2+31c	G2+32c	A♯2-29c	G2-21c
Note	A	A⌢A♯	A♯⌢A	G♯	G♯	G♯⌢G	B⌢A♯	A⌢G
Compute	diff(G♯, A⌢A♯)=0				diff(G♯⌢G, G♯)=0		diff(A⌢G, G♯⌢G)=1	

freedom to make "doctor," "cured," and "patient" whatever tonal values we like. So suppose we notice that "cured" and "happily" have a tonal difference of 1. If the result is robust, then we need to worry about whether this is coincidental and really due to one of the other constituents. Since the difference function doesn't satisfy the triangle equality, it's possible to mix and match differences, so that given $\mathrm{diff}(a,b) = 1$, we can have both the case $\mathrm{diff}(b,c) = 1$ and $\mathrm{diff}(a,c) = 1$, and the case $\mathrm{diff}(b,c) = 1$ and $\mathrm{diff}(a,c) = 0$. We can show this isn't coincidental by setting "patient" and "doctor" tonal values whose difference with "happily" match with and don't match with a value of 1, and so show their differences are independent of the adverb. From the normal interpretation that the adverb modifies the clause/sentence, we can conclude that the clause's values are the same as those of the clause's verb. Thus, the articulation values of non-pronounced syntax can be seen as a way of making the values accessible to other elements.

(10) a. Happily, the doctor cured the patient.
b. The doctor cured the patient happily.

Working with the (a) example, let us give "cured" the tone A, and "happily" the tone $\mathrm{A}^\frown\mathrm{B}$. The stressed syllable of the multisyllabic words will get this tone so /hæp/ gets the tone. The tonal difference between "cured" and "patient" is 0, and the tonal difference between "cured" and "doctor" is 1. For "patient" we can choose the tones $\mathrm{G}^\frown\mathrm{A}$ and $\mathrm{A}^\frown\mathrm{G}$. The difference between G♯ and A♯ is 1, and so the two cases give a difference between "patient" and "happily" of $\mathrm{diff}(\mathrm{G}^\frown\mathrm{A}, \mathrm{A}^\frown\mathrm{B}) = 1$ and $\mathrm{diff}(\mathrm{A}^\frown\mathrm{G}, \mathrm{A}^\frown\mathrm{B}) = 0$, based on having the same and different directions, respectively. By carefully producing these values, we can show that the tone of "happily" is independent of "patient." A similar process can show the independence of "doctor." On the other hand, a native speaker of English will find it challenging to articulate a tonal difference of 0 between "cured" and "happily" with the above structure and a positive direction value. And if one does articulate such a difference, one will perceive a change in structure or meaning upon hearing the articulation.

The data collected, shown in (11)-(14), wasn't as controlled as hoped, but one can see a differentiation for "doctor" and "patient," but not for "cured," though the results have some issues. First, it'd be nice to show that both of the constituents don't have to have the same difference with the adverb, but that's not shown in the data. Second, the cases with the adverb on the right have a negative direction value, which is unfortunate and confusing since the values are switched around. The values haven't been multiplied by the direction value or otherwise adjusted. Regardless, one can see that the difference between the verb and adverb is invariant, but this is not so for the nouns and adverbs.

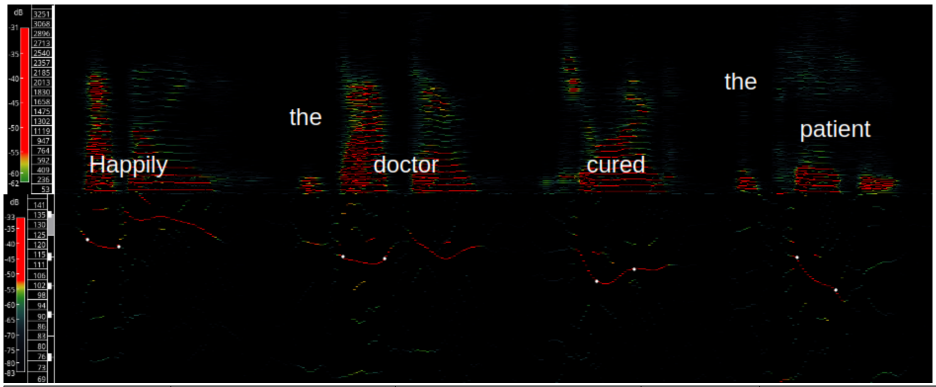
3.2.2 Non-Syntactic Cases

Besides intonation's role in the grammar, there are some semantic effects that one can notice. The following are attempts at spelling out some of these cases.

Dynamic Direction Difference. Each tonal value has a directional component which can take the values 1, 0, or -1. When the last two elements and the first two elements of a clause have non-zero directional components, then they encode information. Thus, this information puts a further limit on the possible tonal combinations, though it isn't a grammatical constraint. When the directional components are both non-zero and don't match, we say the two tonal values are alternating, and when the directional components are non-zero and match, we say the tonal values are non-alternating. This case hasn't been studied extensively for cases like embedded clauses and commands and so on, but only tested on basic clauses restricted to noun phrases and verb phrases.

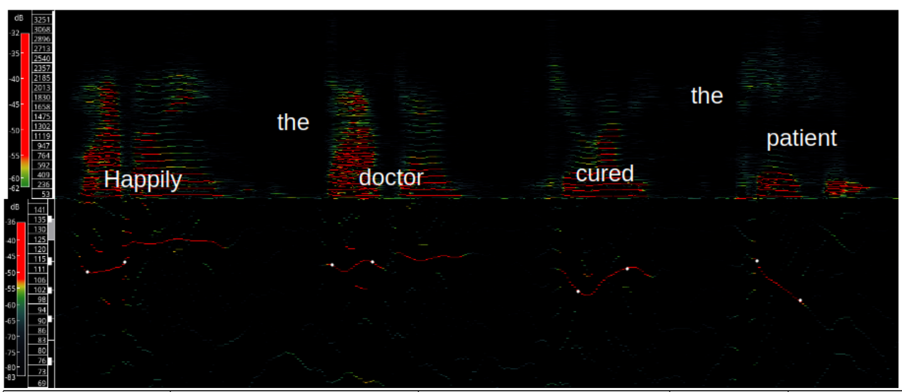
Between the last two elements of a clause, if the tonal values are alternating, then this indicates whether the speaker considers the sentence to be complete within itself, or will be followed up by another sentence. This is based on intention at the time of speaking. It may be that the speaker later decides that there is an

(11)



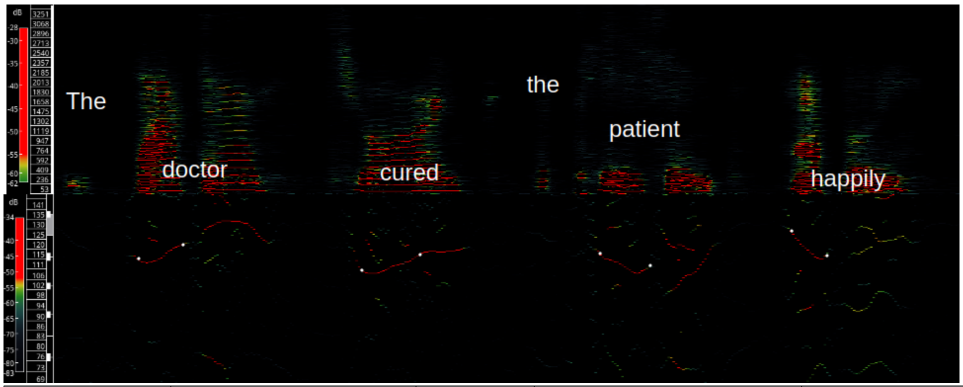
	Adv			Dat			VP_f	Obl		
	Adv			Det	N		VP_f	Det	N	
	hap	-pi	-ly	the	doc	-tor	cured	the	pa	-tient
Initial Tone	B2+22c				A♯2+2c		G♯2+35c		A♯2+1c	
Final	B2-11c				A♯2-15c		A2+25c		G♯2-7c	
Note	B				A♯		G♯⁀A		A♯⁀G♯	
Compute	diff(G♯⁀A, B)=1			diff(G♯⁀A, A♯=1				diff(A♯⁀G♯, G♯⁀A)=0		
Test with Adv				diff(A♯, B)=0				diff(A♯⁀G♯, B)=0		

(12)



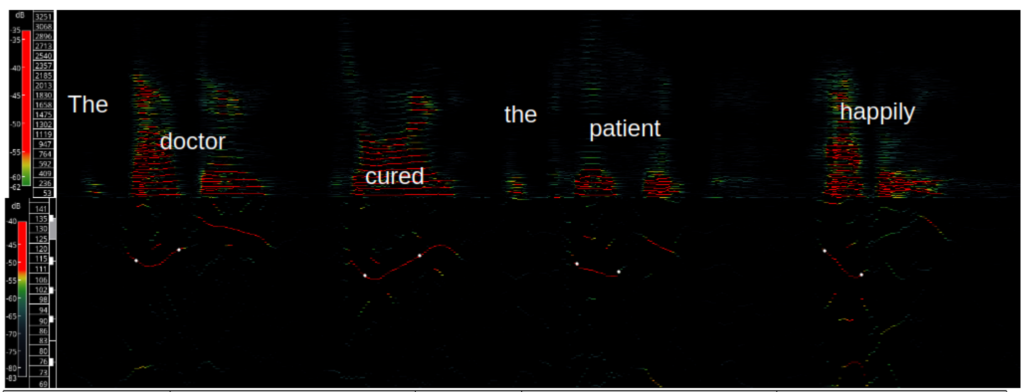
	Adv			Dat			VP_f	Obl		
	Adv			Det	N		VP_f	Det	N	
	hap	-pi	-ly	the	doc	-tor	cured	the	pa	-tient
Initial Tone	A2+37c				A♯2-10c		G♯2+10c		A♯2+13c	
Final	A♯2-27c				A♯2+11c		A2-32c		G2+6c	
Note	A⁀A♯				A♯		G♯⁀A♯		A♯⁀G	
Compute	diff(G♯⁀A♯, A⁀A♯=1			diff(G♯⁀A♯, A♯)=1				diff(A♯⁀G, G♯⁀A♯)=0		
Test with Adv				diff(A♯,A⁀A♯)=1				diff(A♯⁀G, A⁀A♯)=1		

(13)



	Dat			VP_f	Obl			Adv		
	Det	N		VP_f	Det	N		Adv		
	the	doc	-tor	cured	the	pa	-tient	hap	-pi	-ly
Initial Tone		A♯2+3c		A2-1c		A♯2+21c		C3+1c		
Final Tone		B2-16c		A♯2+26c		A2+30c		A♯2+23c		
Note		A♯⌒B		A⌒A♯		A♯⌒A		C⌒A♯		
Compute	diff(A♯⌒B, A⌒A♯)=0				diff(A♯⌒A, A⌒A♯)=1			diff(C⌒A♯, A⌒A♯)=0		
Test with Adv	diff(C⌒A♯, A♯⌒B)=0				diff(C⌒A♯,A♯⌒A)=0					

(14)



	Dat			VP_f	Obl			Adv		
	Det	N		VP_f	Det	N		Adv		
	the	doc	-tor	cured	the	pa	-tient	hap	-pi	-ly
Initial Tone		A♯2+8c		A+7c		A♯-35c		B+2c		
Final Tone		B+5c		A♯+23c		A♯+12c		A+2c		
Note		A♯⌒B		A⌒A♯		A♯⌒A		B⌒A		
Compute	diff(A⌒A♯, A♯⌒B)=0				diff(A♯⌒A, A⌒A♯)=1			diff(B⌒A,A⌒A♯)=0		
Test with Adv	diff(B⌒A, A♯⌒B)=1				diff(B⌒A, A♯⌒A)=1					

addendum, amendment, follow up, or other editing, but such cases don't bare as empirical evidence against this. When the tonal values are non-alternating, this indicates that the person is not done speaking or the current topic is not finished. Again, the caveat that it's the intention at the time of speaking applies; one could later decide that follow up isn't worthwhile and let the topic fade away with time.

The semantics for the first two elements only applies if there's more than two arguments, so that the semantics of the last two elements takes precedent. Between the first two elements of a clause, if there are more than two, alternating tonal values indicates whether the implicit or explicit end of the sentence stems from an efferent or afferent signal. (See Appendix A for more on the implicit or explicit end.) An efferent signal is one that arises from the central nervous system or internally, and an afferent signal is one that arises from an external source or the peripheral nervous system. Alternation indicates the afferent case and non-alternation indicates the efferent case. Efferent cases can produce performative sentences, where we "do things with words," as discussed by Austin (1962). Further investigation is needed to determine whether a 3^{rd} party efferent signal can be encoded by non-alternation.

Final Tone Melody. There are some general patterns of melody that can be noted. The melody of the sentence is the sequence of the syllable's final tones, sensitive only to whole tones, so that the "same tone" means the tone $\pm$ a half tone. Further, there can be groupings which allow melodic properties to start over, and each such grouping will be called a run. This has mainly been studied with single syllable words, so various caveats may exist for more complicated cases. Besides possible complications with more complex, multisyllablic lexical elements, there is also a secondary rule for the final syllable. With these caveats out of the way, the general idea of the melodic categories and then their proper specification can be given, followed by an attempt at their semantics.

There are 4 general categories of melodies: flat, flat monotonic, step monotonic, and ditonic. A flat melody occurs when all the tones are invariant. A flat monotonic melody occurs when within a run, none of the tones are the same and every run rises or every run falls. A step monotonic melody is multiple groups of flat melodies, with the changes between groups being monotonic for each run. Each group has a central tone, and a change from this tone produces a new group, with the change always increasing or always decreasing, unless returning to an extremum for a new run. (How the central tone is determined isn't clear.) For a ditonic melody, each run consists of tonal changes and has one directional change; in other words, it reaches a maxima if it's initially rising, and then falls, or it reaches a minima if it's initially falling, and the minima is followed by a rising melody.

More formally, we can take the collection of syllables as a sequence of temporal regions, which can be denoted $[t_1, t_2, \ldots, t_n]$. Then, we can have subsequences, something like $[t_1, \ldots, t_{i_1}]$, $[t_{i_1+1}, \ldots, t_{i_2}]$, $\ldots$, $[t_{i_{k-1}+1}, \ldots, t_{i_k}]$. Then, for each temporal region, the final-tone function will output the final tone for that temporal region's syllable. These temporal sequences will be given a proper definition in the grammar. The important point here is that each of the subsequences is a run if

$$\text{final-tone}(t_{i_j+1}) \neq \text{final-tone}(t_{i_j}),$$

and

$$\text{final-tone}(t_{i_j+1}) \in \{\max(t_{i_{j-1}+1}, \ldots, t_{i_j}), \min(t_{i_{j-1}+1}, \ldots, t_{i_j})\},$$

where $i_0 = 0$ and $i_k = n$, and the tones are measured in half-tone values. It may be the case that the extremum value condition can be relaxed, but this is a first approximation.

We then can say a tone for t_{i+1} rises if

$$\text{final-tone}(t_{i+1}) > \text{final-tone}(t_i) + 1,$$

it falls if

$$\text{final-tone}(t_{i+1}) < \text{final-tone}(t_i) - 1,$$

and is flat if

$$\text{final-tone}(t_i) - 1 \leq \text{final-tone}(t_{i+1}) \leq \text{final-tone}(t_i) + 1$$

Thus, the rising melody occurs when all non-initial tones of a run are rising, and similarly for falling and flat melodies. A flat monotonic melody is when each run is a rising melody or each run is a falling melody. The melody is step monotonic when for each run or subsequence, the non-initial tones are a mixture of flat and rising, or a mixture of flat and falling. The melody is ditonic when, for each run, there is an m such that the non-initial tones before t_m are rising (falling), and the tones after t_m are falling (rising).

Sentences can be produced that don't fit any of these categories, but such cases don't seem to have any inherit semantics. An exception to these melodic patterns occurs with the last syllable. For a flat melody, a tonal increase or decrease of more than a half tone (from the central tone) can have some meaning or other, for instance, to indicate resolution or reduction in (adrenal) arousal. If a tonic melody has regular intervals, e.g. increasing by two tones $\pm\frac{1}{2}$, then the final tone can "fall" by using 1 tonal difference or less (in the same direction of change) and "rise" by a tonal difference of 3 or more. The semantics of this appears to mimic body language, with increases indicating higher arousal (as when starting an action, or if a mistake is produced), and decreases indicating lower arousal (as when one completes an action, or becomes disappointed that an anticipated, preferred action won't occur or can't be done).

When the directional difference encodes an afferent signal, the semantics of the melody is related to the distance from the event talked about, or the event encoded by the lexical items—especially the verb. The distance is relative to the temporal size of the event: short events have shorter distances away while longer events have longer temporal distances away. This might also be called the temporal proxemics, as it gives a general interval of times within some distance to the event. The order from shortest to longest time is ditonic, strict monotonic, step monotonic, and flat. When the directional difference encodes an efferent signal, the semantics of the melody is related to certainty or commitment, with the same ditonic to flat order as before corresponding to the certain/committed to uncertain/uncommitted order.

End Tone Comparisons. For each noun in a clause, an end tone is compared to an end tone of the verb to specify a property of that noun. How the noun and verb are compared differs based on whether the noun is the subject or not. The L function is used for the subject, and the R function is used for object cases. In particular, where N is a noun's tonal value and V a verb's tonal value, we have $L(V,N)$ and $R(V,N)$.

The semantics of this is that when the relevant function outputs a 1, the noun is a new element being introduced, and when the function outputs a -1, it is an existing element. With the definite article, this amounts to the element having already been apart of and recognized in the situation, or to put it in model theoretic terms, it is a constant of a signature for some language associated with a model and formal system(s). For instance, when asking "Where did we park the car?" in a parking lot full of cars, "car" has a -1 value since driving to and from the location likely involves the car as a constant in that representation. With the indefinite article, there is a second order structure containing a set which the noun belongs to. For instance, when requesting one from a multiple, e.g. "Can you pass me a pen?" referring to a pile of pens, then the pile is a second order element—a set of pens—which exists within the situation. "Do you have a pen?" on the other hand, can be used when one doesn't imply an existing set of pens within the environment, and so the comparison would be a value of 1. Definite and indefinite singular count nouns aren't all the cases, but this gives the general idea.

This sometimes interacts with a verb's meaning as can be seen with the verb "made" in (15) and (16). The (a) cases have the meaning bring into existence while the (b) cases have the meaning bring about a new property for an existing element. The (a) cases require that the comparison is 1 and the (b) cases require that the comparison is -1 for the oblique argument.

(15) a. The clown made a balloon.
b. The clown made a balloon pop.

(16) a. The sculptor made them out of clay.
b. The mentor made them who they are today.

~ ~ ~

This concludes the non-grammar part of the paper. In sections §1 and §2, we provided information about the physical realizability of the grammar and its empirical detection. In this section, we provided some prerequisites for some of the grammar rules, such as prosodic modals and the inputs to clauses, but mainly gave information that is complementary to the grammar. The focus now will shift to a detailed grammar of a fragment of English. The grammar provides constraints for articulation, namely constraining the articulation values to satisfying modular formulas. Each such collection of constraints allow for multiple possible articulations of the given syntactic structure. As much as possible the emphasis will be on justifying the syntactic choices made while describing the articulation.

4 Towards an English Grammar

4.1 Framework

We take Chomsky (1965) as our general framework for approaching a grammar, and especially the symbolism, but modified in a few ways. The grammar will represent more than just the grammatical categories, including the tonal and other articulatory properties, or at least their constraints. This means that rules like S $\rightarrow$ N V N will be replaced with vectors, and with labels in front to help orient the reader to its interpretation as in (17). Along with describing the grammar as a rewriting system, with one symbol being replaced by other symbols, we also take the inverse as a system of defining larger and larger structures. These are not logically distinct, and it's just more of an effort to align the presentation with the structures of a formal language. Everything is defined recursively, so that all "non-terminal" strings are guaranteed to have at least one satisfactory case.

(17)
$$\begin{array}{lcccccc} & S & & N & V & N \\ \text{T} & t_0 & & a_1 & a_2 & a_3 \\ \text{MB} & h_0 & & h_1 & h_2 & h_3 \\ \text{LB} & c_0 & \Rightarrow & c_1 & c_2 & c_3 \\ \text{D} & \theta_0 & & \theta_1 & \theta_2 & \theta_3 \\ Q & q_0 & & q_1 & q_2 & q_3 \\ S & s_0 & & s_1 & s_2 & s_3 \end{array}$$

Formally, we have nested sequences of temporal points, and then functions from these to their grammatical values. For instance, we would have something like

$$T = [t_1, t_2],\ \text{val}(t_1) = \text{Det},\ \text{val}(t_2) = \text{N},\ \text{and val}(T) = \text{NP}$$

This can be composed further to e.g.

$$T' = [[t_1, t_2], [t_3]],\ \text{val}([t_1, t_2]) = \text{NP},\ \text{val}([t_3]) = \text{VP}_\text{f},\ \text{and val}(T') = \text{CL}_\text{FT}$$

where CL_FT indicates a finite total or basic clause. The predicates are more legible when presented in English, so that 'val$(T) = \text{NP}$' will be presented as 'T is an NP.' Such structures can be defined as follows.

Definition 1 A *temporal sequence* $[t_1,\dots,t_n] = [t]_{i=1}^n$ is a sequence of disjoint temporal regions such that order is preserved. That is, $\forall i, t_i$ is a temporal region such that $\forall j, t_i \cap t_j = \emptyset$, and start($t_{i+1}$)<end($t_i$).[11]

Definition 2 $T = [t_1,\dots,t_n]$ is a *temporal sequence tree* (abbreviated tst) if T is a temporal sequence or if $\forall i, 1 \leq i \leq n, t_i$ is a temporal sequence tree, such that the earliest time of t_{i+1} is later than the latest time of t_i.

We interpret $X \to Y_1 Y_2 \dots Y_n$ to mean 'if t_1 is Y_1, t_2 is Y_2, $\dots$, and t_n is Y_n, then $[t_1,\dots,t_n]$ is X.' Often times, the grammatical information in the if-statement doesn't suffice, and so the conditional must be understood to contain all the information of the vector.

Language is known to satisfy the context free rule (Chomsky (1965)), so the grammar has the general rule:

If $A \to \mathscr{U}$, then $\mathscr{V}\ A\ \mathscr{X} \to \mathscr{V}\ \mathscr{U}\ \mathscr{X}$

where A is a vector of the grammar, and $\mathscr{U}, \mathscr{V}$, and $\mathscr{X}$ are (possibly empty) strings of vectors of the grammar.

There will be 6 main properties besides the syntactic category that this grammar will encode. They are the 3 articulation values, the tonal value (T), the mandibular bundle value (MB), the labial bundle value (LB); as well as tracking and modification information: the direction value (D), the Q value (Q), and the source value (S). The source is a tracking mechanism for tonal comparisons and specifies how the Q value is to be articulated. The direction value tracks a variation in production that effects the other values.

The intended interpretation of the grammatical rules and the values within the vectors are as follows.

Tone

- Every rule has a tone on both the left and right side of the arrow, usually denoted a_0, and will be called the default tone.
- Let $\mathrm{C}(x,y) = \frac{1-x}{2} + xy$. Note that this keeps 1 and 0 invariant for $x = 1$ and converts 1 and 0 to 0 and 1 when $x = -1$. The notation δi, when the vector has the direction value θ, for a tone a_k to the left of the default tone in the string of vectors, is to be interpreted as $\mathrm{diff}(a_{k+1}, a_k) = \mathrm{C}(\theta, i)$. For a tone a_m to the right of the default tone, it is to be interpreted as $\mathrm{diff}(a_m, a_{m-1}) = \mathrm{C}(\theta, i)$.
- The notation $a|\phi$ is to be interpreted as a tone that satisfies the condition ϕ.
- A question mark (?) indicates no rule was transcribed or no rule is known. It may be seen as a marker of the transcription and rules being incomplete.

Mandibular and Labial Bundle

- A value in the form $x + y$, when the vector has the direction value θ, is to be interpreted as $x + \theta y$.
- A natural number value n is to be interpreted as a free choice of any value that is $n \pmod 3$ for an MB value and $n \pmod 4$ for an LB value.
- The $0 \pmod 3$ and $0 \pmod 4$ values are determined at the level of the sentence (and not below), and in particular the values in the sentence's vector are $0 \pmod 3$ and $0 \pmod 4$.
- Unless specified by a Q value, an MB or LB value's dynamic change length is 0 or an even number.

[11] These should technically be space-time constraints, but we ignore the spatial part for this simple presentation.

Q Values

- Given a direction value θ, a Q value q, and a S value $L(x)$ in a vector, the tonal value a in the vector is interpreted as $a|L(x,a) = \theta q$. Likewise, given a direction value θ, a Q value q, and a S value $R(x)$ in a vector, the tonal value a in the vector is interpreted as $a|R(x,a) = \theta q$.
- If the S value is λ and the direction value is θ in a vector, then the Q value q is interpreted such that the LB value in the vector has a dynamic change of length $2+\theta q \pmod 4$.
- If the source is m and the direction value is θ in a vector, then the Q value is interpreted such that the MB value has a dynamic change of length $2+\theta q \pmod 4$.
- For both Q and S values, the empty value, for which there is no effect on articulation (from the grammar), is represented as $\cdot$.
- A Q value $[q_1,\ldots,q_n]$ with the source value $[s_1,\ldots,s_n]$ in a vector is interpreted as the Q value q_i with S value s_i, for each i, $1 \le i \le n$.

4.2 Atomic Cases

The grammar is defined over two second order structures, a second order structure of lexical items and a second order structure of grammatical categories.[12] The grammar maps an initial set of tst's to a second set of tst's. The initial set of tst's have a value of a vector, and the vector's syntactic row is a lexical element. The secondary set of tst's have a value of a vector, whose syntactic part is a syntactic category. The treatment of syntactic categories is standard or at least common, except for the use of nominals. Clauses are the main type of the grammar, in the sense that there is a lot of effort that goes into preserving all the information about a clause. Nominals, on the other hand, are introduced for simplification. Anywhere in a clause where a noun phrase can be, also a preposition, a prepositional phrase, or a clause can also be, as the underlined elements in (18)-(20) give examples of. A similar situation occurs with phrases that modify a noun or clause.

(18) a. <u>In</u> goes the flour.
b. The leg gave <u>out</u>.

(19) a. <u>In the kitchen</u> is an oven.
b. The baby stared <u>at the mirror</u>.

(20) a. <u>Playing the game</u> felt satisfying.
b. The team said <u>they won the tournament</u>.

The sets of lexical items and their corresponding lexical categories are as follows. The symbols used in the grammar are written in parentheses. Each lexical element (considered here) is in one of the sets noun (N), verb (V), preposition (P), adjective (Adj), adverb (Adv), determiner (Det), and interrogative pronoun (IPN). It's also useful to split V into the sets non-modal verbs V_{NM}, labial modal verbs V_{LM}, and mandibular modal verbs V_{MM}. Further, there are the sets finite verb (V_f) and non-finite verb (V_n) with the properties that for each element x in V_{NM}, there is a function f such that $f(x) \in V_f$, and a function g such that $g(x) \in V_n$. This is a way of stating that "to be" is a verb, and has the finite forms "is" and "are" and the non-finite form "been," produced from "to be" by functions. In a similar vein, we have the sets V_{lm} and V_{mm} to output the modal forms, so that "shall" and "should" are understood as the same verb. Corresponding to these sets are

[12] The second order aspect of the grammatical categories won't be too relevant to this fragment of the grammar. See the notes section below.

the grammatical categories Noun (N), Preposition (P), Adjective (Adj), Adverb (Adv), Determiner (Det), and Interrogative Pronoun (IPN), as well as the verb categories, finite Verb (V_f), non-finite Verb (V_n), labial modal Verb (V_{lm}), mandibular modal Verb (V_{mm}), and Y Verb (V_Y), with Y to be explained below . This treatment, with the sets and categories appearing redundant is justified by how atomic cases are presented in the grammar; if one could find a means to remove the set condition from the atomic cases, then just the categories could be used.

The grammar has seven phrases: noun phrases (NP), interrogative pronoun phrases (IPP), and five verb phrases: finite verb phrases (VP_f), non-finite verb phrases (VP_n), labial modal verb phrases (VP_{lm}), mandibular modal verb phrases (VP_{mm}), and Y verb phrases (VP_Y). The rest of the grammatical categories will be presented as needed.

4.2.1 Nouns and Nominals

Syntax. It's uncontroversial that a noun by itself and a noun with a determiner is a noun phrase. But consider when there are modifiers as part of the noun phrase and some possible structural issues. A preferred rule to allow adjective modifiers might be N $\rightarrow$ Adj N, since this applies for both the NP $\rightarrow$ N and NP $\rightarrow$ Det N cases. But the issue is that applying the rule twice suggests a modifier of both the noun and an initial adjective, and it also predicts articulation changes that the empirical evidence doesn't bear out. If one intends to have an adjective modify an adjective and noun together, then the adjective and noun get treated like a name, rather than an adjective modifying a noun. [13] And when one has the repetition of two adjectives at two different levels, the meaning is that the second adjective modifies the first. For instance "real small eye" would have "real" modify "small," but this doesn't correspond to it modifying the noun. Thus, the evidence suggest there is no recursion, so N $\rightarrow$ Adj N must be applied only once. The solution to this is modeled on that of mathematical logic: we distinguish between atomic cases and more general cases. Here we can use lexical items in sets as the atomic case. The same is applicable for when an adjective or adverb modifies an adjective. More generally, we will use this as a uniform style of introducing basic grammatical categories.

Nouns are not only modified by adjectives, but also prepositional phrases, other noun phrases, and clauses. Thus, as with a verb's arguments, a nominal will be used to handle all these cases. There is a preference in English for left or right use of these modifiers, but these preferences aren't restrictions. Violating these left and right preferences still gives full interpretation, and so they are not part of the grammar. The nominal types are argument ($\mathrm{Nom_{Arg}}$), noun modifier (NMod), and clause modifier (CLMod) nominal. Most of the rules for one nominal type apply for the others as well, so to make the grammar more succinct, we introduce variables to range over multiple nominal types. Noun modifier and clause modifier are modifier nominals (Mod) and all are nominals in general (Nom). Thus, any rule of the form Mod $\rightarrow \mathscr{U}$ is to be interpreted as 'If $t_1, \ldots, t_n$ are respectively $\mathscr{U}$, then $[t_1, \ldots, t_n]$ is a noun modifier or clause modifier nominal,' and likewise Nom $\rightarrow \mathscr{U}$ stands for 'If $t_1, \ldots, t_n$ are respectively $\mathscr{U}$, then $[t_1, \ldots, t_n]$ is an argument nominal or a noun modifier nominal or a clause modifier nominal.'

To determine the minimal number of rules, we consider where adjectives, noun phrases, and so on can go, restricted to the cases of arguments, noun modifiers, and clause modifiers. Adjectives may be in any of these places and the same is so for noun phrases, prepositional phrases, and prepositions by themselves. Further, preposition phrases allow more general cases than just a noun phrase argument, as in (21). Thus, this will also be a nominal, and its various cases suggest that it's an argument nominal. Another case which might not seem obvious is $\mathrm{Nom_{Arg}} \rightarrow$ NP Mod. This can be seen in sentences like those of (22). What was expected was not the dog, but the dog being happy; what was tried was not the midfielder but the midfielder hanging back, and thus those are the arguments in the clause, and the articulation is consistent with this.

[13] A modifying adjective gets a particular MB value. When one intends a second adjective to modify an initial adjective plus a noun, the initial adjective doesn't get the MB value of a modifying adjective, and instead becomes like "blue" in "blue jay."

This is restricted to a verb's arguments, and doesn't apply to a noun's or clause's modifiers. The remaining nominal rule is the use of adverbs only for clausal modifier nominals.

(21)
a. The doors were painted in red. (adjective)
b. The building looked just like as in the photo. (prepositional phrase)
c. As the team reached the finals, their bones ached. (finite total clause)
d. The dog was wanted for chasing the squirrel. (non-finite total clause)

(22)
a. The dog happy was what we expected.
b. The coach gave the midfielder hanging back a try.
c. The parent sent the child packing.
d. The butterfly had the grasshopper given a scare

The other phrase besides noun phrases, is the interrogative pronoun phrase, which may be an interrogative pronoun alone, or an interrogative pronoun with a noun.

Articulation. Some of the articulatory rules are just copying values downwards and producing invariant values, while others are specific to these constructions or part of more general patterns, and these are worth describing in words. Whenever one is passing down to multiple elements, there is usually a tonal difference of 1 between them, but see 4.7 regarding the determiner. The MB is utilized with modifiers, and has a value of $1 \pmod 3$ for left modifiers and $2 \pmod 3$ for right modifiers. They are written in the form $0+n$ so that the direction value effects it. LB changes come from prepositional phrases and the Nom $\rightarrow$ NP NMod rule. In these cases, there is a $c+2$ value for the preposition and the modifier nominal, where c is the initial LB value.

Besides this, the LB is also involved in an obligatory Q value for all interrogative pronouns (and pronouns more generally), but the treatment is complicated by certain types of questions. When one asks about an argument, the resulting clause types become peculiar; to mark that these distinct clause types will be used, a different Q value is passed down to the first element, which is often an interrogative pronoun. To allow this to override the pronoun's default value, the maximum of the Q value and the interrogative pronoun's Q value is written into the grammar, since the value passed down is 1. The tracking S value is given values for modifiers, but it's not clear if they will be applicable. Two Q and S values, e.g. $[a,b]$, are used for nominals on the right side of a rule. This is due to the capacity to pass values to either a clause's initial constituent or its verb phrase. All of the relevant passing of values here will be passing a tonal comparison value to the verb phrase.

Lexical Items

(26)
$$\begin{array}{lccl} & \text{Det} & & x \\ \text{T} & a & & a \\ \text{MB} & h & & h \\ \text{LB} & c & \Rightarrow & c \\ \text{D} & \theta & & \theta \\ Q & q & & q \\ S & s & & s, \text{ where } x \in Det \end{array}$$

(27)
$$\begin{array}{lccl} & \text{N} & & x \\ \text{T} & a & & a \\ \text{MB} & h & & h \\ \text{LB} & c & \Rightarrow & c \\ \text{D} & \theta & & \theta \\ Q & q & & q \\ S & s & & s, \text{ where } x \in N \end{array}$$

$$(28)\quad \text{a.}\ \begin{array}{llcll}
 & \text{IPN} & & x & \\
\textsc{t} & a & & a & \\
\textsc{mb} & h & & h & \\
\textsc{lb} & c & \Rightarrow & c & \\
\textsc{d} & \theta & & \theta & \\
Q & q & & \max(q,0) & \text{if } s=\lambda, \\
S & s & & s, & \text{where } x \in IPN
\end{array}
\qquad
\text{b.}\ \begin{array}{llcll}
 & \text{IPN} & & x & \\
\textsc{t} & a & & a & \\
\textsc{mb} & h & & h & \\
\textsc{lb} & c & \Rightarrow & c & \\
\textsc{d} & \theta & & \theta & \\
Q & q & & [q,-1] & \text{if } s\neq\lambda, \\
S & s & & [s,\lambda], & \text{where } x \in IPN
\end{array}$$

$$(29)\quad \begin{array}{llcl}
 & \text{P} & & x \\
\textsc{t} & a & & a \\
\textsc{mb} & h & & h \\
\textsc{lb} & c & \Rightarrow & c \\
\textsc{d} & \theta & & \theta \\
Q & q & & q \\
S & s & & s, \text{ where } x \in Prep
\end{array}
\qquad
(30)\quad \begin{array}{llcl}
 & \text{Adj} & & x \\
\textsc{t} & a & & a \\
\textsc{mb} & h & & h \\
\textsc{lb} & 1c & \Rightarrow & c \\
\textsc{d} & \theta & & \theta \\
Q & q & & q \\
S & s & & s, \text{ where } x \in Adj
\end{array}$$

$$(31)\quad \begin{array}{llcl}
 & \text{Adv} & & x \\
\textsc{t} & a & & a \\
\textsc{mb} & h & & h \\
\textsc{lb} & c & \Rightarrow & c \\
\textsc{d} & \theta & & \theta \\
Q & q & & q \\
S & s & & s, \text{ where } x \in Adv
\end{array}$$

Phrases

$$(32)\quad \begin{array}{llcl}
 & \text{NP} & & \text{N} \\
\textsc{t} & a & & a \\
\textsc{mb} & h & & h \\
\textsc{lb} & c & \Rightarrow & c \\
\textsc{d} & \theta & & \theta \\
Q & q & & q \\
S & s & & s
\end{array}
\qquad
(33)\quad \begin{array}{llccc}
 & \text{NP} & & \text{Det} & \text{N} \\
\textsc{t} & a & & ? & a \\
\textsc{mb} & h & & 0+2 & h \\
\textsc{lb} & c & \Rightarrow & c & c \\
\textsc{d} & \theta & & \theta & \theta \\
Q & q & & \cdot & q \\
S & s & & L(a) & s
\end{array}$$

$$(34)\quad \begin{array}{llcl}
 & \text{IPP} & & \text{IPN} \\
\textsc{t} & a & & a \\
\textsc{mb} & h & & h \\
\textsc{lb} & c & \Rightarrow & c \\
\textsc{d} & \theta & & \theta \\
Q & q & & q \\
S & s & & s
\end{array}
\qquad
(35)\quad \begin{array}{llccc}
 & \text{IPP} & & \text{IPN} & \text{N} \\
\textsc{t} & a & & a & \delta 1 \\
\textsc{mb} & h & & h & h \\
\textsc{lb} & c & \Rightarrow & c & c \\
\textsc{d} & \theta & & \theta & \theta \\
Q & q & & q & \cdot \\
S & s & & s & L(a)
\end{array}$$

Nominals

$$(36)\quad \begin{array}{lccc} & \text{Nom} & & \text{NP} \\ \text{T} & a & & a \\ \text{MB} & h & & h \\ \text{LB} & c & \Rightarrow & c \\ \text{D} & \theta & & \theta \\ Q & q & & q \\ S & s & & s \end{array}$$

$$(37)\quad \begin{array}{lccc} & \text{Nom} & & \text{Adj} \\ \text{T} & a & & a \\ \text{MB} & h & & h \\ \text{LB} & c & \Rightarrow & c \\ \text{D} & \theta & & \theta \\ Q & q & & q \\ S & s & & s \end{array}$$

$$(38)\quad \begin{array}{lcccc} & \text{Nom} & & \text{P} & \text{Nom}_{\text{Arg}} \\ \text{T} & a & & \delta 1 & a \\ \text{MB} & h & & h & h \\ \text{LB} & c & \Rightarrow & c+2 & c \\ \text{D} & \theta & & \theta & \theta \\ Q & q & & \cdot & q \\ S & s & & L(a) & s \end{array}$$

$$(39)\quad \begin{array}{lccc} & \text{Nom} & & \text{P} \\ \text{T} & a & & a \\ \text{MB} & h & & h \\ \text{LB} & c & \Rightarrow & c \\ \text{D} & \theta & & \theta \\ Q & q & & q \\ S & s & & s \end{array}$$

$$(40)\quad \begin{array}{lccc} & \text{Nom}_{\text{Arg}} & & \text{IPP} \\ \text{T} & a & & a \\ \text{MB} & h & & h \\ \text{LB} & c & \Rightarrow & c \\ \text{D} & \theta & & \theta \\ Q & q & & q \\ S & s & & s \end{array}$$

$$(41)\quad \begin{array}{lcccc} & \text{Nom}_{\text{Arg}} & & \text{NP} & \text{NMod} \\ \text{T} & a & & a & ? \\ \text{MB} & h & & h & h \\ \text{LB} & c & \Rightarrow & c & c+2 \\ \text{D} & \theta & & \theta & \theta \\ Q & q & & q & [\cdot,\cdot] \\ S & s & & s & [\cdot,R(a)] \end{array}$$

$$(42)\quad \begin{array}{lccc} & \text{CLMod} & & \text{Adv} \\ \text{T} & a & & a \\ \text{MB} & h & & h \\ \text{LB} & c & \Rightarrow & c \\ \text{D} & \theta & & \theta \\ Q & q & & q \\ S & s & & s \end{array}$$

Modifiers

$$(43)\quad \begin{array}{lccccl} & \text{N} & & x & \text{NMod} & \\ \text{T} & a & & a & \delta 1 & \\ \text{MB} & h & & h & 0+2 & \\ \text{LB} & c & \Rightarrow & c & c & \\ \text{D} & \theta & & \theta & \theta & \\ Q & q & & q & [\cdot,\cdot] & \\ S & s & & s & [\cdot,R(a)], & \text{where } x \in N \end{array}$$

$$(44)\quad \begin{array}{lccccl} & \text{N} & & \text{NMod} & x & \\ \text{T} & a & & \delta 1 & a & \\ \text{MB} & h & & 0+1 & h & \\ \text{LB} & c & \Rightarrow & c & c & \\ \text{D} & \theta & & \theta & \theta & \\ Q & q & & [\cdot,\cdot] & q & \\ S & s & & [\cdot,L(a)] & s, & \text{where } x \in N \end{array}$$

$$(45)\quad \begin{array}{cc} & \text{N} \\ \text{T} & a \\ \text{MB} & h \\ \text{LB} & c \\ \text{D} & \theta \\ Q & q \\ S & s \end{array} \Rightarrow \begin{array}{cccc} \text{NMod} & x & \text{NMod} & \\ \delta 1 & a & \delta 1 & \\ 0+1 & h & 0+2 & \\ c & c & c & \\ \theta & \theta & \theta & \\ [\cdot,\cdot] & q & [\cdot,\cdot] & \\ [\cdot,L(a)] & s & [\cdot,R(a)], & \text{where } x \in N \end{array}$$

$$(46)\quad \begin{array}{cc} & \text{Adj} \\ \text{T} & a \\ \text{MB} & h \\ \text{LB} & c \\ \text{D} & \theta \\ Q & q \\ S & s \end{array} \Rightarrow \begin{array}{cc} \text{Adj} & x \\ \delta 1 & a \\ 0+1 & h \\ c & c \\ \theta & \theta \\ \cdot & q \\ L(a) & s \end{array} \qquad (47)\quad \begin{array}{cc} & \text{Adj} \\ \text{T} & a \\ \text{MB} & h \\ \text{LB} & c \\ \text{D} & \theta \\ Q & q \\ S & s \end{array} \Rightarrow \begin{array}{cc} \text{Adv} & x \\ \delta 1 & a \\ 0+1 & h \\ c & c \\ \theta & \theta \\ \cdot & q \\ L(a) & s \end{array}$$

~ ~ ~

For the determiner, we have a possessive construction. This will be defined over the string of a noun phrase, and the only interesting thing to mention is that the [s] ending applies to the last element of the noun phrase, even if this is a modifier, so that "the dog in the kitchen" becomes "the-dog-in-the-kitchen's bowl" and not "*the-dog's-in-the-kitchen bowl." The [s]-ending is marked with a Q value, articulated with the MB.

(48) Possessive rule: If

$$\begin{array}{cc} & \text{NP} \\ \text{T} & a_0 \\ \text{MB} & h_0 \\ \text{LB} & c_0 \\ \text{D} & \theta_0 \\ Q & s_0 \\ S & q_0 \end{array} \Rightarrow \mathscr{A}_1 \Rightarrow \ldots \Rightarrow \mathscr{A}_k \Rightarrow \begin{array}{ccc} x_1 & \ldots & x_n \\ a_1 & \ldots & a_n \\ h_1 & \ldots & h_n \\ c_1 & \ldots & c_n, \\ \theta & \ldots & \theta \\ q_1 & \ldots & q_n \\ s_1 & \ldots & s_n \end{array}$$

then

$$\begin{array}{cc} & \text{Det} \\ \text{T} & a_0 \\ \text{MB} & h_0 \\ \text{LB} & c_0 \\ \text{D} & \theta_0 \\ Q & q_0 \\ S & s_0 \end{array} \Rightarrow \begin{array}{cccc} x_1 & \ldots & x_{n-1} & \sigma(x_n) \\ a_1 & \ldots & a_{n-1} & a_n \\ h_1 & \ldots & h_{n-1} & h_n \\ c_1 & \ldots & c_{n-1} & c_n^q \\ \theta_1 & \ldots & \theta_{n-1} & \theta_n \\ q_1 & \ldots & q_{n-1} & [q_n,-1] \\ s_1 & \ldots & s_{n-1} & [s_n,m] \end{array} \quad \text{or} \quad \begin{array}{c} \text{Det} \\ a_0 \\ h_0 \\ c_0 \\ \theta_0 \\ q_0 \\ s_0 \end{array} \Rightarrow \begin{array}{ccccc} x_1 & \ldots & x_{n-1} & \sigma(x_n)_1 & \sigma(x_n)_2 \\ a_1 & \ldots & a_{n-1} & a_n & \delta 1 \\ h_1 & \ldots & h_{n-1} & h_n & h_n \\ c_1 & \ldots & c_{n-1} & c_n & c_n \\ \theta_1 & \ldots & \theta_{n-1} & \theta_n & \theta_n \\ q_1 & \ldots & q_{n-1} & q_n & -1 \\ s_1 & \ldots & s_{n-1} & s_n & m \end{array}$$

where each $\mathscr{A}_i$ is a string of grammar vectors, each generated from the last by the context free rule, each x_i is a syllable, generated by the context free rule with the lexicon, and $\sigma(x_n)$ outputs a modified syllable (with /s/ or /z/ at the end) or outputs two syllables, the original one ($\sigma(a)_1$) and the second /ɪz/ ($\sigma(a)_2$).

4.2.2 Verbs

Syntax. There are 5 verb phrases, finite (VP_f), non-finite (VP_n), labial modal (VP_{lm}), mandibular modal (VP_{mm}), and Y (VP_Y). The verb can have adverb modifiers akin to modifiers of nouns. But unlike for nouns, the adverbs are the unique modifier of the verb, and so a nominal isn't implemented. Recall that there are input sets (V_{NM}, V_{LM}, and V_{MM}) and functions which go to particular output sets, V_f, V_n, V_{lm}, and V_{mm}. These will be used for the uniform presentation with how nouns were presented.

To built verb phrases, the next issue to address is repeated verbs. While it's common to assume that all such verbs are part of the same phrase (Chomsky (1965), DeMarco (2022)), the articulation suggests that at least the final verb is part of a clause, which is then embedded in a nominal, and so is outside the main verb phrase. But nothing rules out there being multiple verbs in the embedded clause, it just appears to be uncommon in speech. So, cases like "should have been playing" involve 2 verb phrases and are ambiguous, with the most common separation being the VP "should have been" and the verb phrase "playing," but could also be e.g. "should" and "have been playing." In a multiverb construction, all of the verbs but the first must be a non-finite verb, and unlike with modifiers can be applied over and over. Thus, the recursive rule $VP_n \rightarrow V_n\ VP_n$ is used.

The final issue to address is the Y verb phrase, which is justified in the following way. First, it is known that there are limitations to forming questions, so that embedded finite verbs cannot be used to form a question (49). Second, the semantic and articulatory evidence points to only the finite verb and the subject switching places in a question. In particular, the articulation doesn't "move," so that e.g. the $1 \pmod 3$ MB value, which normally applies for the verb, applies to the NP, and the 0 or $2 \pmod 3$ MB value, which normally applies for the subject, applies to the finite verb. And on the semantics side, (50)-(52) suggest that only the finite verb lexical element ($x \in V_{NM}$) moves. In particular, the intuition is that the (a) and (b) cases have the same interpretation for the adverb, suggesting that the adverb doesn't follow the verb but stays in place. These semantic intuitions are not clear, distinct, and evident, but are part of the justification for using variables. A variable approach allows for a simple means of handling all of the evidence. The embedding clause, but not any embedded clause, has verb movement because only the embedding clause has a variable; the articulation doesn't move because the variable is only for the syntax; and the adverbs don't move because the variable occurs where the lexical element would normally occur.

(49) Statement: The dog said the squirrel is getting a raise.

a. *What the dog said is the squirrel getting?
b. *What is the dog said the squirrel getting?
c. *What is the squirrel the dog said getting?
d. What did the dog say the squirrel is getting?

(50) a. Easily, the child is over 4 feet.
b. Easily, is the child over 4 feet?

(51) a. The child easily is over 4 feet.
b. Is easily the child over 4 feet?

(52) a. The child is easily over 4 feet.
b. Is the child easily over 4 feet?

Articulation. The labial bundle value has some freedom: it can be some inputted value x, or it can be $x+2 \pmod 4$. The $x+2 \pmod 4$ case comes across as modifying aspect, indicating repetition such as habitual aspect. It is applicable to both the verb and any adverb modifier, so the rules applies at the phrase level. It further appears to be applicable to any verb, including in longer sequences, and multiple times, though it's hard to keep track of what would be communicated in such cases. Besides this, the adverbs get $1 \pmod 3$ on the left and $2 \pmod 3$ on the right for the MB value, and the S value is speculated and made uniform with similar cases, though there were no empirical cases of it coming up.

Lexical Items

(53) Y is a variable of the grammar, as are $X_{\text{Erg}}, X_{\text{Obl}}$, and X_{Dat}.

(54) $$\begin{array}{llcl} & \mathrm{V_f} & & f(x) \\ \mathrm{T} & a & & a \\ \mathrm{MB} & h & & h \\ \mathrm{LB} & c & \Rightarrow & c \\ \mathrm{D} & \theta & & \theta \\ Q & q & & q \\ S & s & & s, \end{array} \quad \text{where } x \in V_{NM} \text{ and } f(x) \in V_f$$

(55) $$\begin{array}{llcl} & \mathrm{V_{lm}} & & f(x) \\ \mathrm{T} & a & & a \\ \mathrm{MB} & h & & h \\ \mathrm{LB} & c & \Rightarrow & c \\ \mathrm{D} & \theta & & \theta \\ Q & q & & q \\ S & s & & s, \end{array} \quad \text{where } x \in V_{LM} \text{ and } f(x) \in V_{lm}$$

(56) $$\begin{array}{llcl} & \mathrm{V_{mm}} & & f(x) \\ \mathrm{T} & a & & a \\ \mathrm{MB} & h & & h \\ \mathrm{LB} & c & \Rightarrow & c \\ \mathrm{D} & \theta & & \theta \\ Q & q & & q \\ S & s & & s, \end{array} \quad \text{where } x \in V_{MM} \text{ and } f(x) \in V_{mm}$$

(57) $$\begin{array}{llcl} & \mathrm{V_n} & & f(x) \\ \mathrm{T} & a & & a \\ \mathrm{MB} & h & & h \\ \mathrm{LB} & c & \Rightarrow & c \\ \mathrm{D} & \theta & & \theta \\ Q & q & & q \\ S & s & & s, \end{array} \quad \text{where } x \in V_{NM} \text{ and } f(x) \in V_n$$

Modifiers

Let u be a variable which may stand for Y, f, lm, mm, or n.

(58) If

$$\begin{array}{llcl} & \mathrm{V}_u & & z \\ \mathrm{T} & a & & a \\ \mathrm{MB} & h & & h \\ \mathrm{LB} & c & \Rightarrow & c \\ \mathrm{D} & \theta & & \theta \\ Q & q & & q \\ S & s & & s \end{array}$$

then

$$\begin{array}{llccl} & \mathrm{V}_u & & \mathrm{Adv} & z \\ \mathrm{T} & a & & ? & a \\ \mathrm{MB} & h & & 1 & h \\ \mathrm{LB} & c & \Rightarrow & c & c \\ \mathrm{D} & \theta & & \theta & \theta \\ Q & q & & \cdot & q \\ S & s & & L(a) & s \end{array}$$

(59) If

$$\begin{array}{llcl} & \mathrm{V}_u & & z \\ \textsc{t} & a & & a \\ \textsc{mb} & h & & h \\ \textsc{lb} & c & \Rightarrow & c \\ \textsc{d} & \theta & & \theta \\ \textit{\textsc{q}} & q & & q \\ \textit{\textsc{s}} & s & & s \end{array}$$

then

$$\begin{array}{llcll} & \mathrm{V}_u & & z & \mathrm{Adv} \\ \textsc{t} & a & & a & ? \\ \textsc{mb} & h & & h & 2 \\ \textsc{lb} & c & \Rightarrow & c & c \\ \textsc{d} & \theta & & \theta & \theta \\ \textit{\textsc{q}} & q & & q & \cdot \\ \textit{\textsc{s}} & s & & s & R(a) \end{array}$$

(60) If

$$\begin{array}{llcl} & \mathrm{V}_u & & z \\ \textsc{t} & a & & a \\ \textsc{mb} & h & & h \\ \textsc{lb} & c & \Rightarrow & c \\ \textsc{d} & \theta & & \theta \\ \textit{\textsc{q}} & q & & q \\ \textit{\textsc{s}} & s & & s \end{array}$$

then

$$\begin{array}{llclll} & \mathrm{V}_u & & \mathrm{Adv} & z & \mathrm{Adv} \\ \textsc{t} & a & & ? & a & ? \\ \textsc{mb} & h & & 1 & h & 2 \\ \textsc{lb} & c & \Rightarrow & c & c & c \\ \textsc{d} & \theta & & \theta & \theta & \theta \\ \textit{\textsc{q}} & q & & \cdot & q & \cdot \\ \textit{\textsc{s}} & s & & L(a) & s & R(a) \end{array}$$

Phrases

$$(61)\quad \begin{array}{llcl} & \mathrm{VP}_u & & \mathrm{V}_u \\ \textsc{t} & a & & a \\ \textsc{mb} & h & & h \\ \textsc{lb} & c & \Rightarrow & c+2n \\ \textsc{d} & \theta & & \theta \\ \textit{\textsc{q}} & q & & q \\ \textit{\textsc{s}} & s & & s, \quad \text{where } n \in \mathbb{N} \end{array}$$

$$(62)\quad \begin{array}{llcll} & \mathrm{VP}_u & & \mathrm{V}_u & \mathrm{VP}_\mathrm{n} \\ \textsc{t} & a & & a & ? \\ \textsc{mb} & h & & h & h \\ \textsc{lb} & c & \Rightarrow & c+2n & c \\ \textsc{d} & \theta & & \theta & \theta \\ \textit{\textsc{q}} & q & & q & \cdot \\ \textit{\textsc{s}} & s & & s & L(a), \text{ where } n \in \mathbb{N} \end{array}$$

4.3 Declarative Clauses

Syntax. Within the grammar are 6 clause types defined from nominals and verb phrases. There are main clauses ($\mathrm{CL_M}$), finite total clauses ($\mathrm{CL_{FT}}$), finite partial clauses ($\mathrm{CL_{FP}}$), non-finite total clauses ($\mathrm{CL_{NFT}}$), non-finite partial clauses ($\mathrm{CL_{NFP}}$), and non-finite degenerate clauses ($\mathrm{CL_{NFD}}$). Besides the main clause, clauses are categorized by whether the verb is finite or non-finite, and how many arguments are not articulated. The finite total clause will be the base clause, and all other clauses will be defined by transformations, or conditional rules. Main clauses contain a variable and indirectly a second variable in the verb phrase. For non-finite clauses, the subject is always absent. From the total clause cases as bases, we have partial clauses with one argument less and degenerate clauses with two arguments less than the total case. Degenerate clauses stem from questions and will be addressed in the next section. (63) underlines an example of each clause type, and gives a corresponding basic clause.

(63) a. $\underline{\text{The dog is in the kitchen}}$. ($\mathrm{CL_M}$)

b. The ref said $\underline{\text{we were cheating}}$. ($\mathrm{CL_{FT}}$)

c. The child, we often see on the playground, is seated a few rows ahead of us. (CL_{FP})
(We often see the child on the playground.)

d. The cat is heating up the seat. (CL_{NFT})
(The cat heats up the seat)

e. The man was given an answer. (CL_{NFP})
(Someone gave the man the answer.)

f. The Wolves are the team to beat. (CL_{NFP})
(Anyone beats the Wolves.)

g. What was the woman made? (CL_{NFD})
(Someone made the woman what/a gift.)

As mentioned in (§3.2.1), clauses take an input. The rules for specific inputs will only be given for finite total clauses, and only the subset of those cases with a finite verb. There will be a separation of finite verb and modal verbs cases, so that non-finite clauses are based only on the former. From the finite total case, we can use transformations to specify the other clauses, but the transformations will be based on the number of arguments. This could technically be simplified, but there are so few cases that the extra level of abstraction strikes one as unnecessary.

A note with regard to interpretation is that $t_1,\ldots,t_n$ can't be used in both sides of the conditional, since we want the conditional to apply at different times and if there were any distinct values, then the shared temporal indexes would lead to a space-time inconsistency. The statement 'If $X \to \mathscr{U}$, then $Z \to \mathscr{V}$' is to be interpreted as

> If $t_1,\ldots,t_n$ are respectively $\mathscr{U}$ implies $[t_1,\ldots,t_n]$ is X, then $r_1,\ldots,r_k$ are respectively $\mathscr{V}$ implies $[r_1,\ldots,r_k]$ is Z.

The treatment of clauses is based on semantic and articulatory evidence. The semantic evidence is the intuition that has been a theme throughout the history of generative grammar regarding the semantic relation between a given noun and verb despite various placements relative to one another, such as with fronting. The articulatory evidence is of two kinds. As mentioned before, the articulation of multi-verb sequences suggests the final verb is external to the main verb phrase. Thus, the grouping is consistent with the verb being part of a clause. Second, the Q values that occur with some verbs is also evidence of the treatment here. The articulation of Q values in derived clauses can vary, and the variation matches our intuitions of what the corresponding basic clause would be.

Articulation. For the finite total clause, the tone gets copied to the verb, and the tonal difference is 1 for all but the oblique case, which has a difference of 0. The MB value is carried over for oblique and ergative. For the verb phrase, the MB is a relative value, being one more than the value for the clause. The Dative and Stative nominals get a $2 \pmod 3$ MB value. The LB value is +1 for the last argument whenever there exists a dative or stative case, and is carried over otherwise. We also denote the S values for the tonal comparison semantics discussed in §3.2.1. The direction value is independent of any embedding element, and so the right and left side have different values.

For derived clauses, the tonal, MB, and LB values are retained for the articulated constituents. The direction value continues to be independent of any embedding element, except for the main clause, which takes the sentence's directional element. The tonal difference for the non-articulated arguments get realized as Q values that match the finite total case's tonal difference. This requires converting 1 and 0 to 1 and -1, respectively, for which the formula $2i-1$ is used, and for readability is represented by the variable j. An exception is for a ditransitive argument structure, in which case it encodes whether the unarticulated argument is the first or second argument to the right of the verb. If the right most argument is not articulated, it's a -1 Q value, and if it's the argument to the right of the verb, the Q value is 1. All of these Q values get realized on the verb. The first non-articulated argument gets realized by a tonal comparison for the verb, with the corresponding S value denoted by the variable s. The second non-articulated argument gets realized by a LB value with an odd dynamic change length.

When Q values are passed to a clause, they can effect the verb or the first constituent. Thus, the Q and S values of the clause are presented as a vector. The first element of the vector goes to the first constituent of the clause and the second element of the vector goes to the verb phrase. This is true of all clause types. Another complication with the Q values is that some of them are not only effected by the direction value of its vector but also the direction value for the clause. Thus, in "water is filling the cup," both the direction value of the whole sentence and of "filling the cup" effect the embedded clauses Q values. Since effects on the Q value by the vector's direction value is specified in the grammar's interpretation rules, only the direction value of the embedding entity is denoted.

Finite Total Clauses

(64)
$$\begin{array}{cccccc}
 & \mathrm{CL}(\mathbf{T})_{\mathrm{FT}} & & \mathrm{Obl} & \mathrm{VP_f} \\
\mathrm{T} & a & & \delta 0 & a \\
\mathrm{MB} & h & & h & h+1 \\
\mathrm{LB} & c & \Rightarrow & c & c \\
\mathrm{D} & \theta_0 & & \theta_1 & \theta_1 \\
Q & [u,\cdot] & & [\cdot,\cdot_1,u] & \cdot \\
S & [v,s] & & [\cdot,L(a),v] & s
\end{array}$$

(65)
$$\begin{array}{cccccc}
 & \mathrm{CL}(\mathbf{TB})_{\mathrm{FT}} & & \mathrm{Obl} & \mathrm{VP_f} & \mathrm{Stat} \\
\mathrm{T} & a & & \delta 0 & a & \delta 1 \\
\mathrm{MB} & h & & h & h+1 & 0+2 \\
\mathrm{LB} & c & \Rightarrow & c & c & c+1 \\
\mathrm{D} & \theta_0 & & \theta_1 & \theta_1 & \theta_1 \\
Q & [u,\cdot] & & [u,\cdot] & \cdot & [\cdot,\cdot] \\
S & [v,s] & & [v,L(a)] & s & [\cdot,R(a)]
\end{array}$$

(66)
$$\begin{array}{cccccc}
 & \mathrm{CL}(\mathbf{CT})_{\mathrm{FT}} & & \mathrm{Dat} & \mathrm{VP_f} & \mathrm{Obl} \\
\mathrm{T} & a & & \delta 1 & a & \delta 0 \\
\mathrm{MB} & h & & 0+2 & h+1 & h \\
\mathrm{LB} & c & \Rightarrow & c & c & c+1 \\
\mathrm{D} & \theta_0 & & \theta_1 & \theta_1 & \theta_1 \\
Q & [u.\cdot] & & [u,\cdot] & \cdot & [\cdot,\cdot] \\
S & [v,s] & & [v,L(a)] & s & [\cdot,R(a)]
\end{array}$$

(67)
$$\begin{array}{cccccc}
 & \mathrm{CL}(\mathbf{ST})_{\mathrm{FT}} & & \mathrm{Erg} & \mathrm{VP_f} & \mathrm{Obl} \\
\mathrm{T} & a & & \delta 1 & a & \delta 1 \\
\mathrm{MB} & h & & h & h+1 & h \\
\mathrm{LB} & c & \Rightarrow & c & c & c+1 \\
\mathrm{D} & \theta_0 & & \theta_1 & \theta_1 & \theta_1 \\
Q & [u,\cdot] & & [u,\cdot] & \cdot & [\cdot,\cdot] \\
S & [v,s] & & [v,L(a)] & s & [\cdot,R(a)]
\end{array}$$

$$(68)\quad \begin{array}{llclllll}
 & \mathrm{CL}(\mathbf{STB})_{\mathrm{FT}} & & \mathrm{Erg} & \mathrm{VP_f} & \mathrm{Obl} & \mathrm{Stat} \\
\textsc{t} & a & & \delta 1 & a & \delta 0 & \delta 1 \\
\textsc{mb} & h & & h & h+1 & h & 0+2 \\
\textsc{lb} & c & \Rightarrow & c & c & c & c+1 \\
\textsc{d} & \theta_0 & & \theta_1 & \theta_1 & \theta_1 & \theta_1 \\
Q & [u,\cdot] & & [u,\cdot] & \cdot & [\cdot,\cdot] & [\cdot,\cdot] \\
S & [v,s] & & [v,L(a)] & s & [\cdot,R(a)] & [\cdot,R(a)]
\end{array}$$

$$(69)\quad \begin{array}{llclllll}
 & \mathrm{CL}(\mathbf{SCT})_{\mathrm{FT}} & & \mathrm{Erg} & \mathrm{VP_f} & \mathrm{Dat} & \mathrm{Obl} \\
\textsc{t} & a & & \delta 1 & a & \delta 1 & \delta 1 \\
\textsc{mb} & h & & h & h+1 & 0+2 & h \\
\textsc{lb} & c & \Rightarrow & c & c & c & c+1 \\
\textsc{d} & \theta_0 & & \theta_1 & \theta_1 & \theta_1 & \theta_1 \\
Q & [u,\cdot] & & [u,\cdot] & \cdot & [\cdot,\cdot] & [\cdot,\cdot] \\
S & [v.s] & & [v,L(a)] & s & [\cdot,R(a)] & [\cdot,R(a)]
\end{array}$$

Let k be one of the regions south, south-east, north, or north-east, and Ω a variable for the clausal relations.

(70) If $\forall i, c_i$ is a k prosodic modal value and

$$\begin{array}{llclll}
 & \mathrm{CL}(\Omega)_{\mathrm{FT}} & & A_1 & \mathrm{VP_f} & \dots & A_{n-1} \\
\textsc{t} & a_0 & & a_1 & a_2 & \dots & a_n \\
\textsc{mb} & h_0 & & h_1 & h_2 & \dots & h_m \\
\textsc{lb} & c_0 & \Rightarrow & c_1 & c_2 & \dots & c_n \\
\textsc{d} & \theta_0 & & \theta_1 & \theta_1 & \dots & \theta_1 \\
Q & q_0 & & q_1 & q_2 & \dots & q_n \\
S & s_0 & & s_1 & s_2 & \dots & s_n,
\end{array}$$

then

$$\begin{array}{llclll}
 & \mathrm{CL}(\Omega)_{\mathrm{FT}} & & A_1 & \mathrm{VP_{lm}} & \dots & A_{n-1} \\
\textsc{t} & a_0 & & a_1 & a_2 & \dots & a_n \\
\textsc{mb} & h_0 & & h_1 & h_2 & \dots & h_n \\
\textsc{lb} & c_0 & \Rightarrow & c_1 & c_2 & \dots & c_n \\
\textsc{d} & \theta_0 & & \theta_1 & \theta_1 & \dots & \theta_1 \\
Q & q_0 & & q_1 & q_2 & \dots & q_n \\
S & s_0 & & s_1 & s_2 & \dots & s_n,
\end{array}$$

(71) If $\forall i, h_i$ is a k prosodic modal value and

$$\begin{array}{llclll}
 & \mathrm{CL}(\Omega)_{\mathrm{FT}} & & A_1 & \mathrm{VP_f} & \dots & A_{n-1} \\
\textsc{t} & a_0 & & a_1 & a_2 & \dots & a_n \\
\textsc{mb} & h_0 & & h_1 & h_2 & \dots & h_m \\
\textsc{lb} & c_0 & \Rightarrow & c_1 & c_2 & \dots & c_n \\
\textsc{d} & \theta_0 & & \theta_1 & \theta_1 & \dots & \theta_1 \\
Q & q_0 & & q_1 & q_2 & \dots & q_n \\
S & s_0 & & s_1 & s_2 & \dots & s_n,
\end{array}$$

then

$$\begin{array}{ccccccc}
 & \mathrm{CL}(\Omega)_{\mathrm{FT}} & & A_1 & \mathrm{VP}_{\mathrm{mm}} & \dots & A_{n-1} \\
\text{T} & a_0 & & a_1 & a_2 & \dots & a_n \\
\text{MB} & h_0 & & h_1 & h_2 & \dots & h_m \\
\text{LB} & c_0 & \Rightarrow & c_1 & c_2 & \dots & c_n \\
\text{D} & \theta_0 & & \theta_1 & \theta_1 & \dots & \theta_1 \\
Q & q_0 & & q_1 & q_2 & \dots & q_n \\
S & s_0 & & s_1 & s_2 & \dots & s_n,
\end{array}$$

Main

Let u be a variable, so that VP_u may denote $\mathrm{VP}_{\mathrm{f}}, \mathrm{VP}_{\mathrm{lm}}$, or $\mathrm{VP}_{\mathrm{mm}}$, and Ω a variable for the clausal relations.

(72) If

$$\begin{array}{ccccc}
 & \mathrm{CL}(\Omega)_{\mathrm{FT}} & & A_1 & \mathrm{VP}_u \\
\text{T} & a_0 & & a_1 & a_2 \\
\text{MB} & h_0 & & h_1 & h_2 \\
\text{LB} & c_0 & \Rightarrow & c_1 & c_2 \\
\text{D} & \theta_0 & & \theta_1 & \theta_1 \\
Q & q_0 & & q_1 & q_2 \\
S & s_0 & & s_1 & s_2
\end{array}$$

then

$$\begin{array}{cccc}
\mathrm{CL}(\Omega)_{\mathrm{M}} & & X_{A_1} & \mathrm{VP}_Y \\
a_0 & & a_1 & a_2 \\
h_0 & & h_1 & h_2 \\
c_0 & \Rightarrow & c_1 & c_2 \\
\theta_1 & & \theta_1 & \theta_1 \\
q_0 & & q_1 & q_2 \\
s_0 & & s_1 & s_2
\end{array}$$

(73) If

$$\begin{array}{cccccc}
 & \mathrm{CL}(\Omega)_{\mathrm{FT}} & & A_1 & \mathrm{VP}_u & A_3 \\
\text{T} & a_0 & & a_1 & a_2 & a_3 \\
\text{MB} & h_0 & & h_1 & h_2 & h_3 \\
\text{LB} & c_0 & \Rightarrow & c_1 & c_2 & c_3 \\
\text{D} & \theta_0 & & \theta_1 & \theta_1 & \theta_1 \\
Q & q_0 & & q_1 & q_2 & 1_3 \\
S & s_0 & & s_1 & s_2 & s_3
\end{array}$$

then

$$\begin{array}{ccccc}
\mathrm{CL}(\Omega)_{\mathrm{M}} & & X_{A_1} & \mathrm{VP}_Y & A_3 \\
a_0 & & a_1 & a_2 & a_3 \\
h_0 & & h_1 & h_2 & h_3 \\
c_0 & \Rightarrow & c_1 & c_2 & c_3 \\
\theta_1 & & \theta_1 & \theta_1 & \theta_1 \\
q_0 & & q_1 & q_2 & q_3 \\
s_0 & & s_1 & s_2 & s_3
\end{array}$$

(74) If

$$\begin{array}{ccccccc}
 & \mathrm{CL}(\Omega)_{\mathrm{FT}} & & A_1 & \mathrm{VP}_u & A_3 & A_4 \\
\text{T} & a_0 & & a_1 & a_2 & a_3 & a_4 \\
\text{MB} & h_0 & & h_1 & h_2 & h_3 & h_4 \\
\text{LB} & c_0 & \Rightarrow & c_1 & c_2 & c_3 & c_4 \\
\text{D} & \theta_0 & & \theta_1 & \theta_1 & \theta_1 & \theta_1 \\
Q & q_0 & & q_1 & q_2 & q_3 & q_4 \\
S & s_0 & & s_1 & s_2 & s_3 & s_4
\end{array}$$

then

$$\begin{array}{cccccc}
\mathrm{CL}(\Omega)_{\mathrm{M}} & & X_{A_1} & \mathrm{VP}_Y & A_3 & A_4 \\
a_0 & & a_1 & a_2 & a_3 & a_4 \\
h_0 & & h_1 & h_2 & h_3 & h_4 \\
c_0 & \Rightarrow & c_1 & c_2 & c_3 & c_4 \\
\theta_1 & & \theta_1 & \theta_1 & \theta_1 & \theta_1 \\
q_0 & & q_1 & q_2 & q_3 & q_4 \\
s_0 & & s_1 & s_2 & s_3 & s_4
\end{array}$$

Non-Finite Total

Let $j_1 = 2i_1 - 1$, φ some comparison S value, and let Ω be a variable for the clausal relations.

(75) If

$$\begin{array}{lccccc}
 & \mathrm{CL}(\Omega)_{\mathrm{FT}} & & A_1 & \mathrm{VP_f} \\
\mathrm{T} & a_0 & & \delta i & a_0 \\
\mathrm{MB} & h_0 & & h_1 & h_2 \\
\mathrm{LB} & c_0 & \Rightarrow & c_1 & c_2 \\
\mathrm{D} & \theta_0 & & \theta_1 & \theta_1 \\
Q & [u,\cdot] & & [u,\cdot] & \cdot \\
S & [v,s] & & [v,L(a_0)] & s
\end{array}$$

then

$$\begin{array}{ccc}
\mathrm{CL}(\Omega)_{\mathrm{NFT}} & & \mathrm{VP_n} \\
a_0 & & a_0 \\
h_0 & & h_2 \\
c_0 & \Rightarrow & c_2 \\
\theta_0 & & \theta_1 \\
[u,\cdot] & & [u,\theta_0 j_1] \\
[v,s] & & [v,s]
\end{array}$$

(76) If

$$\begin{array}{lcccccc}
 & \mathrm{CL}(\Omega)_{\mathrm{FT}} & & A_1 & \mathrm{VP_f} & A_3 \\
\mathrm{T} & a_0 & & \delta i_1 & a_0 & \delta i_2 \\
\mathrm{MB} & h_0 & & h_1 & h_2 & h_3 \\
\mathrm{LB} & c_0 & \Rightarrow & c_1 & c_2 & c_3 \\
\mathrm{D} & \theta_0 & & \theta_1 & \theta_1 & \theta_1 \\
Q & [u,\cdot] & & [u,\cdot] & \cdot & [\cdot,\cdot] \\
S & [v.s] & & [v,L(a_0)] & s & [\cdot,R(a_0)]
\end{array}$$

then

$$\begin{array}{cccc}
\mathrm{CL}(\Omega)_{\mathrm{NFT}} & & \mathrm{VP_n} & A_3 \\
a_0 & & a_0 & \delta i_2 \\
h_0 & & h_2 & h_3 \\
c_0 & \Rightarrow & c_2 & c_3 \\
\theta_0 & & \theta_1 & \theta_1 \\
[u,\cdot] & & [u,\theta_0 j_1] & [\cdot,\cdot] \\
[v,s] & & [v,s] & [\cdot,R(a_0)]
\end{array}$$

(77) If

$$\begin{array}{lccccccc}
 & \mathrm{CL}(\Omega)_{\mathrm{FT}} & & A_1 & \mathrm{VP_f} & A_3 & A_4 \\
\mathrm{T} & a_0 & & \delta i_1 & a_0 & \delta i_3 & \delta i_4 \\
\mathrm{MB} & h_0 & & h_1 & h_2 & h_3 & h_4 \\
\mathrm{LB} & c_0 & \Rightarrow & c_1 & c_2 & c_3 & c_4 \\
\mathrm{D} & \theta_0 & & \theta_1 & \theta_1 & \theta_1 & \theta_1 \\
Q & [u,\cdot] & & [u,\cdot] & \cdot & [\cdot,\cdot] & [\cdot,\cdot] \\
S & [v.s] & & [v,L(a_0)] & s & [\cdot,R(a_0)] & [\cdot,R(a_0)]
\end{array}$$

then

$$\begin{array}{ccccc}
\mathrm{CL}(\Omega)_{\mathrm{NFT}} & & \mathrm{VP_n} & A_3 & A_4 \\
a_0 & & a_0 & \delta i_3 & \delta i_4 \\
h_0 & & h_2 & h_3 & h_4 \\
c_0 & \Rightarrow & c_2 & c_3 & c_4 \\
\theta_0 & & \theta_1 & \theta_1 & \theta_1 \\
[u,\cdot] & & [u,\theta_0 j_1] & [\cdot,\cdot] & [\cdot,\cdot] \\
[v,s] & & [v,s] & [\cdot,R(a_0)] & [\cdot,R(a_0)]
\end{array}$$

Non-Finite Partial

Let $j_3 = 2i_3 - 1$. and let Ω be a variable for the clausal relations.

(78) If

$$\begin{array}{cccccc}
 & \mathrm{CL}(\Omega)_{\mathrm{NFT}} & & \mathrm{VP_n} & A_3 \\
\mathrm{T} & a_0 & & a_0 & \delta i_3 \\
\mathrm{MB} & h_0 & & h_2 & h_3 \\
\mathrm{LB} & c_0 & \Rightarrow & c_2 & c_3 \\
\mathrm{D} & \theta_0 & & \theta_1 & \theta_1 \\
Q & [u,\cdot] & & [u,\theta_0 j_1] & [\cdot,\cdot] \\
S & [v,s] & & [v,s] & [\cdot,R(a_0)]
\end{array}$$

then

$$\begin{array}{ccc}
\mathrm{CL}(\Omega)_{\mathrm{NFP}} & & \mathrm{VP_n} \\
a_0 & & a_0 \\
h_0 & & h_2 \\
c_0 & \Rightarrow & c_2 \\
\theta_0 & & \theta_1 \\
[u,\cdot] & & [u,\theta_0 j_1,\theta_0 j_2] \\
[v,s] & & [v,s,\lambda]
\end{array}$$

(79) If

$$\begin{array}{cccccc}
 & \mathrm{CL}(\Omega)_{\mathrm{NFT}} & & \mathrm{VP_n} & A_3 & A_4 \\
\mathrm{T} & a_0 & & a_0 & \delta i_3 & \delta i_4 \\
\mathrm{MB} & h_0 & & h_2 & h_3 & h_4 \\
\mathrm{LB} & c_0 & \Rightarrow & c_2 & c_3 & c_4 \\
\mathrm{D} & \theta_0 & & \theta_1 & \theta_1 & \theta_1 \\
Q & [u,\cdot] & & [u,\theta_0 j_1] & [\cdot,\cdot] & [\cdot,\cdot] \\
S & [v,s] & & [v,s] & [\cdot,R(a_0)] & [\cdot,R(a_0)]
\end{array}$$

then

$$\begin{array}{cccc}
\mathrm{CL}(\Omega)_{\mathrm{NFP}} & & \mathrm{VP_n} & A_4 \\
a_0 & & \delta_{i_2=v} & a_4 \\
h_0 & & h_2 & h_4 \\
c_0 & \Rightarrow & c_2 & c_4 \\
\theta_0 & & \theta_1 & \theta_1 \\
[u,\cdot] & & [u,\theta_0 j_1,\theta_0(-1)] & [\cdot,\cdot] \\
[v,s] & & [v,s,\lambda] & [\cdot,R(a_0)]
\end{array}$$

and

$$\begin{array}{cccc}
\mathrm{CL}(\Omega)_{\mathrm{NFP}} & & \mathrm{VP_n} & A_3 \\
a_0 & & a_0 & \delta i_3 \\
h_0 & & h_2 & h_3 \\
c_0 & \Rightarrow & c_2 & c_3 \\
\theta_0 & & \theta_1 & \theta_1 \\
[u,\cdot] & & [u,\theta_0 j_1,\theta_0 1] & [\cdot,\cdot] \\
[v,s] & & [v,s,\lambda] & [\cdot,R(a)]
\end{array}$$

Finite Partial

Let $j_1 = (2i_1 - 1)$ and $j_3 = (2i_3 - 1)$, let u be a variable so that VP_u may denote $\mathrm{VP_f}, \mathrm{VP_{lm}}$, or $\mathrm{VP_{mm}}$, and let Ω be a variable for the clausal relations.

(80) If

$$
\begin{array}{cccccc}
 & \mathrm{CL}(\Omega)_{\mathrm{FT}} & & A_1 & \mathrm{VP}_u \\
\mathrm{T} & a_0 & & \delta i_1 & a_0 \\
\mathrm{MB} & h_0 & & h_1 & h_2 \\
\mathrm{LB} & c_0 & \Rightarrow & c_1 & c_2 \\
\mathrm{D} & \theta_0 & & \theta_1 & \theta_1 \\
Q & [u,\cdot] & & [u,\cdot] & \cdot \\
S & [v,s] & & [v,L(a)] & s
\end{array}
$$

then

$$
\begin{array}{ccc}
\mathrm{CL}(\Omega)_{\mathrm{FP}} & & \mathrm{VP}_u \\
a_0 & & a_0 \\
h_0 & & h_2 \\
c_0 & \Rightarrow & c_2 \\
\theta_0 & & \theta_1 \\
[u,\cdot] & & [u,\theta_0 j_1] \\
[v,s] & & [v,s]
\end{array}
$$

(81) If

$$
\begin{array}{ccccccc}
 & \mathrm{CL}(\Omega)_{\mathrm{FT}} & & A_1 & \mathrm{VP}_u & A_3 \\
\mathrm{T} & a_0 & & \delta i_1 & a_0 & \delta i_3 \\
\mathrm{MB} & h_0 & & h_1 & h_2 & h_3 \\
\mathrm{LB} & c_0 & \Rightarrow & c_1 & c_2 & c_3 \\
\mathrm{D} & \theta_0 & & \theta_1 & \theta_1 & \theta_1 \\
Q & [u,\cdot] & & [u,\cdot] & \cdot & [\cdot,\cdot] \\
S & [v,s] & & [v,L(a)] & s & [\cdot,R(a)]
\end{array}
$$

then

$$
\begin{array}{cccc}
\mathrm{CL}(\Omega)_{\mathrm{FP}} & & \mathrm{VP}_u & A_3 \\
a_0 & & a_0 & \delta i_3 \\
h_0 & & h_2 & h_3 \\
c_0 & \Rightarrow & c_2 & c_3 \\
\theta_0 & & \theta_1 & \theta_1 \\
[u,\cdot] & & [u,\theta_0 j_1] & [\cdot,\cdot] \\
[v,s] & & [v,s] & [\cdot,R(a_0)]
\end{array}
$$

and

$$
\begin{array}{cccc}
\mathrm{CL}_{\mathrm{FP}} & & A_1 & \mathrm{VP}_u \\
a_0 & & \delta i_1 & a_0 \\
h_0 & & h_1 & h_2 \\
c_0 & \Rightarrow & c_1 & c_2 \\
\theta_0 & & \theta_1 & \theta_1 \\
[u,\cdot] & & [u,\cdot] & \theta_0 j_3 \\
[v.s] & & [v,L(a_0)] & s
\end{array}
$$

(82) If

$$
\begin{array}{cccccccc}
 & \mathrm{CL}(\Omega)_{\mathrm{FT}} & & A_1 & \mathrm{VP}_u & A_3 & A_4 \\
\mathrm{T} & a_0 & & \delta i_1 & a_0 & \delta i_3 & \delta i_4 \\
\mathrm{MB} & h_0 & & h_1 & h_2 & h_3 & h_4 \\
\mathrm{LB} & c_0 & \Rightarrow & c_1 & c_2 & c_3 & c_4 \\
\mathrm{D} & \theta_0 & & \theta_1 & \theta_1 & \theta_1 & \theta_1 \\
Q & [u,\cdot] & & [u,\cdot] & \cdot & [\cdot,\cdot] & [\cdot,\cdot] \\
S & [v.s] & & [v,L(a)] & s & [\cdot,R(a)] & [\cdot,R(a)]
\end{array}
$$

then

$$
\begin{array}{ccccc}
\mathrm{CL}(\Omega)_{\mathrm{FP}} & & \mathrm{VP}_u & A_3 & A_4 \\
a_0 & & a_0 & \delta i_3 & \delta i_4 \\
h_0 & & h_2 & h_3 & h_4 \\
c_0 & \Rightarrow & c_2 & c_3 & c_4 \\
\theta_0 & & \theta_1 & \theta_1 & \theta_1 \\
[u,\cdot] & & [u,\theta_0 j_1] & [\cdot,\cdot] & [\cdot,\cdot] \\
[v,s] & & [v,s] & [\cdot,R(a_0)] & [\cdot,R(a_0)]
\end{array}
$$

and

$$\begin{array}{ccccc} \mathrm{CL}(\Omega)_{\mathrm{FP}} & & A_1 & \mathrm{VP}_u & A_4 \\ a_0 & & \delta i_1 & a_0 & \delta i_4 \\ h_0 & & h_1 & h_2 & h_4 \\ c_0 & \Rightarrow & c_1 & c_2 & c_4 \\ \theta_0 & & \theta_1 & \theta_1 & \theta_1 \\ [u,\cdot] & & [u,\cdot] & \theta_0(-1) & [\cdot,\cdot] \\ [v,s] & & [v,L(a_0)] & s & [\cdot,R(a_0)] \end{array}$$

and

$$\begin{array}{ccccc} \mathrm{CL}(\Omega)_{\mathrm{FP}} & & A_1 & \mathrm{VP}_u & A_3 \\ a_0 & & \delta i_1 & a_0 & \delta i_3 \\ h_0 & & h_1 & h_2 & h_3 \\ c_0 & \Rightarrow & c_1 & c_2 & c_3 \\ \theta_0 & & \theta_1 & \theta_1 & \theta_1 \\ [u,\cdot] & & [u,\cdot] & \theta_0 1 & [\cdot,\cdot] \\ [v,s] & & [v,L(a_0)] & s & [\cdot,R(a_0)] \end{array}$$

4.4 Question Clauses

Syntax. The final set of cases come from questions. There are two types of questions, which have different markings, but all of them take a similar syntactic structure. First, the syntax of the questions, inferred from their tonal and MB values, is that of an **SCT** or **STB** clause for *who, what, which,* and *why*, with the question word being the subject and **CT** or **TB** for *when, where*, and *how*, with the question word occurring as a clause modifier nominal (and the verb occurring with the *X* variable). Yes-no questions use the normal main clause structure. The discussion here will be focused on *who, what,* and *which* questions, which require further clausal and clause-like categories. The analysis will be based on the levels of clausal embedding within the oblique nominal.

When we have *who/what/which* questions with just one clause in the last nominal, we require the non-finite partial clause, as well as a new clause type, the non-finite degenerate clause (examples in (83)). This clause is for when a non-finite partial's argument is asked about. For non-finite degenerate clauses, we have 3 non-articulated arguments, and thus it is restricted to the 2 possible ditransitive cases. Despite this, there are 4 cases, for the degenerate case not only preserves the initial syntax, but also encodes which partial case it is derived from. This is the main reason why all clauses aren't being derived from the finite total base case, but instead are derived from a sensical predecessor, e.g. non-finite partial from non-finite total.

(83) a. What was the doctor given? ($\mathrm{CL}_{\mathrm{NFD}}$)
b. What were you playing? ($\mathrm{CL}_{\mathrm{NFP}}$)
c. Who did Erin tell to be quiet? ($\mathrm{CL}_{\mathrm{NFP}}$)

The analysis becomes more complicated when the oblique nominal has embedded clauses, as in (84). The problem here is not with the most embedded clause. The most embedded clause is one of the clauses discussed already. The problem is with the clauses that embed them. We need a means of allowing e.g. non-finite degenerate clauses in these question cases, but not in a regular clause. The recursion makes this task non-trivial.

(84) a. What did you [say the biologist [had the spider [given __]]]?
(The biologist gave the spider something.)

b. Who did Effy [make the prize [baking a cake [for __]]?
(Someone bakes a cake for someone)

c. What did Kyle [believe [we were to be [shown __]]]?
(Someone showed us what/something.)

To handle these, we will do two things. First, we will need, for handling the recursion of the embedded clauses, to create *q*-clauses, corresponding to the main clauses, where the *q*-clause has one of the nominals allow for the question clauses. For this, we will create a partial nominal, the argument type for the *q*-clause. Partial nominals are posited to deal with questions asked about a subpart of a nominal, such as those in (85). In the *who/what/which* questions, the oblique case will actually be a partial oblique nominal. Note that the preposition's articulation for partial nominals is different than when they standalone in a nominal.

(85) a. What was the dog in __ ?
(the dog was in the kitchen)

b. Who did everyone give __ playing a game a try?
(everyone game John playing a game a try)

c. What had everyone given Mary playing __ a try?
(Everyone had given Mary playing a game a try)

d. What is acceptable __ in the kitchen?
(The dog in the kitchen is acceptable.)

A partial nominal (Nom_{P}) has the clause forms of Ergative (Erg_{P}), Dative (Dat_{P}), Oblique (Obl_{P}), and Stative (Stat_{P}). It should be clear that a partial nominal is *not* a nominal, and likewise *q*-clauses are not clauses. And continuing the pattern of introducing any dependencies before they are utilized, we begin with some partial nominal cases. The partial nominal rules stem from the nominal rules Nom $\to$ P NP and $\text{Nom}_{\text{Arg}} \to$ NP NMod. The partial nominal can then be used in a *q*-clause to allow non-finite degenerate clauses without them being allowed in a non-*who, what*, or *which* question clause. The recursion means that the partial nominal can go anywhere. Thus, besides introducing the non-finite degenerate clause, there will be a *q*-clause for each of the clause types that are embedded in argument nominals. These are the finite total, non-finite total, and non-finite partial clauses. There will also be the base case, the main *q*-clause, limited to an **SCT** or **STB** input, with a partial nominal for the last argument.

There is not currently any rule for interrogative pronouns to be in nominals, so these will be given with the partial nominals.

Articulation. As before, the transformations keep the same articulation, but there are new additions for the Q and S values. The non-finite degenerate clause has all its Q and S values on its single constituent. The final arguments information is encoded by the direction value, multiplied by the embedding direction value. Thus, the direction value for non-finite degenerate clauses is not a free choice. The *q*-clauses have the same articulation as the corresponding regular clause. The main *q*-clause has the direction value passed over as with the main clause, but otherwise the direction value is a free choice for the clauses and the *q*-clauses. For the partial nominals, a Q value of 1 is given and paired with an S value of m, meaning that the Q value is realized by the MB. This is always passed to any first element, and so is placed in the left position of the vectors of Q and S values. When an interrogative pronoun is used in a clausal modifier, it gets a -1 Q value realized by the MB.

Let us summarize some of the Q and S articulation rules on their own. First, the Q and S vectors are such that the left most element goes to the first constituent of a clause, and the second element goes to the verb. The first element handles a few sentential cases to be discussed later, marks partial nominals that don't embed any further, and marks possessives and some interrogative pronouns; everything else is multiplied by the direction value of whatever is embedding it and assigned to the clause's verb. In particular, the tonal comparison for the clause is passed to the verb, and is used for the first non-articulated argument. The second non-articulated argument is encoded by the LB. The third non-articulated argument is indicated with the direction value. The values are derived from the tonal difference value of the removed argument, except when removing one of the two arguments to the right of the verb, as described in §4.3. This is so regardless of the clause type and input, though the rules can only apply if the clausal input or relation has enough corresponding arguments.

Nominals and Partial Nominals

(86) $$\begin{array}{cccc} & \mathrm{Nom_{Arg}} & & \mathrm{IPP} \\ \mathrm{T} & a & & a \\ \mathrm{MB} & h & & h \\ \mathrm{LB} & c & \Rightarrow & c \\ \mathrm{D} & \theta & & \theta \\ Q & [u,\cdot] & & [1,\cdot] \\ S & [v,s] & & [m,s] \end{array}$$

(87) $$\begin{array}{cccc} & \mathrm{Mod} & & \mathrm{IPP} \\ \mathrm{T} & a & & a \\ \mathrm{MB} & h & & h \\ \mathrm{LB} & c & \Rightarrow & c \\ \mathrm{D} & \theta & & \theta \\ Q & [u,\cdot] & & [-1,\cdot] \\ S & [v,s] & & [m,s] \end{array}$$

(88) $$\begin{array}{cccc} & \mathrm{Nom_P} & & \mathrm{NMod} \\ \mathrm{T} & a & & a \\ \mathrm{MB} & h & & h \\ \mathrm{LB} & c & \Rightarrow & c+2 \\ \mathrm{D} & \theta & & \theta \\ Q & [u,\cdot] & & [1,\cdot] \\ S & [v,s] & & [m,s] \end{array}$$

(89) $$\begin{array}{cccc} & \mathrm{Nom_P} & & \mathrm{P} \\ \mathrm{T} & a & & a \\ \mathrm{MB} & h & & h \\ \mathrm{LB} & c & \Rightarrow & c+2 \\ \mathrm{D} & \theta & & \theta \\ Q & [u,\cdot] & & [1,\cdot] \\ S & [v,s] & & [m,s] \end{array}$$

Non-Finite Degenerate

Let $j_2 = 2i - 1$ and let Ω be a variable for the clausal relations.

(90) If

$$\begin{array}{ccccc} \mathrm{CL}(\Omega)_{\mathrm{NFP}} & & \mathrm{VP_n} & & A \\ a_0 & & a_0 & & \delta i \\ h_0 & & h_1 & & h_2 \\ c_0 & \Rightarrow & c_1 & & c_2 \\ \theta_0 & & \theta_1 & & \theta_1 \\ [u,\cdot] & & [u,\cdot,\theta_0 n] & & [\cdot,\cdot] \\ [v,s] & & [v,s,\lambda] & & [\cdot,R(a_0)] \end{array}$$

then

$$\begin{array}{ccc} \mathrm{CL}(\Omega)_{\mathrm{NFD}} & & \mathrm{VP_n} \\ a_0 & & a_0 \\ h_0 & & h_1 \\ c_0 & \Rightarrow & c_1 \\ \theta_0 & & j_2\theta_0 \\ [u,\cdot] & & [u,\theta_0 j_1,\theta_0 n] \\ [v,s] & & [v,s,\lambda] \end{array}$$

Main q-Clause

Let u be a variable, so that V_u may denote $\mathrm{V_f}, \mathrm{V_{lm}}$, or $\mathrm{V_{mm}}$, and let Ω be a variable for the clausal relations.

(91) If

$$\begin{array}{llclccc}
 & \mathrm{CL}(\Omega)_{\mathrm{M}} & & X_{Z_1} & \mathrm{VP}_Y & Z_2 & Z_3 \\
\mathrm{T} & a_0 & & a_1 & a_2 & a_3 & a_4 \\
\mathrm{MB} & h_0 & & h_1 & h_2 & h_3 & h_4 \\
\mathrm{LB} & c_0 & \Rightarrow & c_1 & c_2 & c_3 & c_4 \\
\mathrm{D} & \theta_0 & & \theta_1 & \theta_1 & \theta_1 & \theta_1 \\
Q & q_0 & & q_1 & q_2 & q_3 & q_4 \\
S & s_0 & & s_1 & s_2 & s_3 & s_4
\end{array}$$

then

$$\begin{array}{lcclccc}
 & q\text{-}\mathrm{CL}(\Omega)_{\mathrm{M}} & & Z_1 & \mathrm{V}_u & Z_2 & (Z_3)_{\mathrm{P}} \\
\mathrm{T} & a_0 & & a_1 & a_2 & a_3 & a_4 \\
\mathrm{MB} & h_0 & & h_1 & h_2 & h_3 & h_4 \\
\mathrm{LB} & c_0 & \Rightarrow & c_1 & c_2 & c_3 & c_4 \\
\mathrm{D} & \theta_0 & & \theta_1 & \theta_1 & \theta_1 & \theta_1 \\
Q & q_0 & & q_1 & q_2 & q_3 & q_4 \\
S & s_0 & & s_1 & s_2 & s_3 & s_4
\end{array}$$

FT, NFT, and NFP q-Clause

Let v be a variable over FT, NFT, and NFP, and Ω a variable for the clausal relations.

(92) If

$$\begin{array}{llclcc}
 & \mathrm{CL}(\Omega)_v & & Z_1 & \dots & Z_m \\
\mathrm{T} & a_0 & & a_1 & \dots & a_m \\
\mathrm{MB} & h_0 & & h_1 & \dots & h_m \\
\mathrm{LB} & c_0 & \Rightarrow & c_1 & \dots & c_m \\
\mathrm{D} & \theta_0 & & \theta_1 & \dots & \theta_1 \\
Q & q_0 & & q_1 & \dots & q_m \\
S & s_0 & & s_1 & \dots & s_m
\end{array}$$

then

$$\begin{array}{lclcc}
q\text{-}\mathrm{CL}(\Omega)_v & & V_1 & \dots & V_m \\
a_0 & & a_1 & \dots & a_m \\
h_0 & & h_1 & \dots & h_m \\
c_0 & \Rightarrow & c_1 & \dots & c_m \\
\theta_0 & & \theta_1 & \dots & \theta_1 \\
q_0 & & q_1 & \dots & q_m \\
s_0 & & s_1 & \dots & s_m
\end{array}$$

where $\exists i, V_i = (Z_i)_P$ and $\forall j, j \neq i \rightarrow V_j = Z_j$.

4.5 Recursive Cases

So far we have defined a bunch of clauses and *q*-clauses, but there isn't yet any rules to compose the clauses, which is the topic of this section. The recursive cases can be split into 4 categories: declarative arguments, noun modifiers, clause modifiers, and question arguments. For clausal arguments, we have finite total, non-finite, and non-finite partial cases, which can also be after a preposition, shown in (93). For noun modifiers, examples are shown in (94), and (95) shows clausal modifiers. Finally, we have the question cases, which amount to base cases plus the recursion, handled by *q*-clauses, shown in (96).

(93) a. The dog thinks <u>the suitcases mean they'll leave again</u>. (CL_{FT})
b. The moons are <u>twirling the space dust</u>. (CL_{NFT})
c. The monolopy had their laws <u>written to their liking</u>. (CL_{NFP})
d. As <u>the dog went to bed</u>, ... (CL_{FT})
e. The AI is good at <u>playing the game</u>. (CL_{NFT})
f. For <u>given a chance</u>, they'd succeed. (CL_{NFP})

(94) a. The dog, whom <u>the children pet</u>, ...(CL_{FP})
b. The dog, <u>the child said got in the trash</u>, ... (q-CL_{FT})
c. The dog, <u>playing with the child</u>, ... (CL_{NFT})
d. The dog, <u>given a treat</u>, ... (CL_{NFP})

(95) a. <u>Playing fetch</u>, the dog ran wildly. (CL_{NFT})
b. <u>Given a pet</u>, the cat purred gently. (CL_{NFP})
c. <u>Is going to the moon</u>, the monkey prepared all night. (CL_{FP})
d. <u>The child said their parent works</u>, the job sounded bizarre. (q-CL_{FT})
(The child said their parent works the job.)

(96) a. What did Lee say <u>I am</u>? (CL_{FP})
(I am tired.)
b. What did Helen say <u>you are given</u>? (q-CL_{FT})
(Someone gives you what/something.)
c. What did the saxophonist <u>play</u>? (CL_{NFP})
(The saxophonist played a stomp.)
d. What did the operator <u>hear the saxophonist playing</u>? (q-CL_{NFT})
(The operator heard the saxophonist playing a rag.)
e. What is the adult <u>taken</u>? (CL_{NFD})
(Someone took the adult a picture.)
f. What is the adult <u>saved from saying out loud</u>? (q-CL_{NFP})
(Someone saved the adult from saying mean things out loud.)

So all of this needs to be parsed into the smallest number of rules. We can see that non-finite total and non-finite partial clauses are used for all the nominal types, so they can be given rules with the Nom generalization. Finite total clauses are applicable only for argument nominals, while finite partial clauses are applicable for the modifier nominals. On the partial nominal side, we have only a single nominal type and it has the clauses CL_{NFD}, CL_{NFP}, and CL_{FP}, along with the *q*-clauses q-CL_{NFT}, q-CL_{NFP}, q-CL_{FT}.

Besides these, there are a few other loose ends. First, we have cases where finite clauses—both total and partial—can be used with an interrogative pronoun. This is demonstrated in (97). For prepositions, besides the partial nominal cases given in §4.4, there is also the possibility that the preposition's argument is a partial nominal. There is, in *who/what/which* questions, a case where Nom $\rightarrow$ NP NMod can have an argument removed from the clause in the noun modifier nominal, but only for the non-finite total clause. So we will posit the rule $\text{Nom}_{\text{P}} \rightarrow$ NP CL_{NFP}. Finally, and to start the rules given, clausal modifiers need to be attached to clauses and *q*-clauses. This is done with 3 different cases, based on whether the clausal modifier occurs at the beginning, end, or in between the clause's constituents.

(97) a. The cat prances when the family comes home. (CL_{FT})
b. They forgot which car the child's favorite is. (CL_{FP})
c. We don't know what the cat is hissing at. (q-CL_{FT})

Articulation. The articulation of the embedded clauses just passes down values, with two exceptions. First is that the LB value distinguishes between the different clause types, and each of the 3 gets its own LB value. For nominals, non-finite total clauses get a $+1$ from the input LB value, non-finite partial clauses get a $+2$ value, and finite clauses get $+3$, both finite and partial which are in a mutually exclusive distribution. Further, the finite total *q*-clause also gets a $+3$ value. For partial nominals, the corresponding *q*-clauses get the same values, but the (non-*q*) clauses are shifted to a clause with one less argument. As a result, non-finite partial clauses and non-finite total *q*-clauses get the $+1$ value and non-finite degenerate clauses and non-finite partial *q*-clauses get the $+2$ value. Second, the partial nominal is indicated with a Q value of 1 paired with an S value of m, so that the Q value is realized with the MB. This is so for clauses and non-clause-like elements, but is not so for *q*-clauses.

Since the clause modifier's tonal value isn't necessarily given a difference to its neighboring tone, but is instead a difference from the verb's or clause's tone, we indicate this with a condition. The clause modifier also gets a +3 LB value.

Clausal Modifiers

Let v be a variable that ranges over the possible clause types, and Ω a variable for the clausal relations.

(98) If

$$
\begin{array}{lcccccc}
 & \text{CL}(\Omega)_v & & Z_1 & \ldots & Z_n \\
\text{T} & a_0 & & a_1 & \ldots & a_n \\
\text{MB} & h_0 & & h_1 & \ldots & h_n \\
\text{LB} & c_0 & \Rightarrow & c_1 & \ldots & c_n \\
\text{D} & \theta_0 & & \theta_1 & \ldots & \theta_1 \\
Q & q_0 & & q_1 & \ldots & q_n \\
S & s_0 & & s_1 & \ldots & s_n
\end{array}
$$

then

$$
\begin{array}{lccccccc}
 & \text{CL}(\Omega)_v & & \text{CLMod} & Z_1 & \ldots & Z_n \\
\text{T} & a_0 & & a_{n+1}\big|_{\text{diff}(a_0,a_{n+1})=1} & a_1 & \ldots & a_n \\
\text{MB} & h_0 & & h_0 & h_1 & \ldots & h_n \\
\text{LB} & c_0 & \Rightarrow & c_0+3 & c_1 & \ldots & c_n \\
\text{D} & \theta_0 & & \theta_1 & \theta_1 & \ldots & \theta_1 \\
Q & q_0 & & [\cdot,\cdot] & q_1 & \ldots & q_n \\
S & s_0 & & [\cdot,L(a_0)] & s_1 & \ldots & s_n
\end{array}
$$

and

$$\begin{array}{lccccccc}
 & \mathrm{CL}(\Omega)_v & & Z_1 & \dots & Z_n & \mathrm{CLMod} \\
\mathrm{T} & a_0 & & a_1 & \dots & a_n & a_{n+1}|_{\mathrm{diff}(a_{n+1},a_0)=1} \\
\mathrm{MB} & h_0 & & h_1 & \dots & h_n & h_0 \\
\mathrm{LB} & c_0 & \Rightarrow & c_1 & \dots & c_n & c_0+3 \\
\mathrm{D} & \theta_0 & & \theta_1 & \dots & \theta_1 & \theta_1 \\
Q & q_0 & & q_1 & \dots & q_n & [\cdot,\cdot] \\
S & s_0 & & s_1 & \dots & s_n & [\cdot,R(a_0)]
\end{array}$$

and

$$\begin{array}{lcccccccccc}
 & \mathrm{CL}(\Omega)_v & & Z_1 & \dots & Z_i & \mathrm{CLMod} & Z_{i+1} & \dots & Z_n \\
\mathrm{T} & a_0 & & a_1 & \dots & a_i & a_{n+1}|\psi & a_{i+1} & \dots & a_n \\
\mathrm{MB} & h_0 & & h_1 & \dots & h_i & h_0 & h_{i+1} & \dots & h_n \\
\mathrm{LB} & c_0 & \Rightarrow & c_1 & \dots & c_i & c_0+3 & c_{i+1} & \dots & c_n \\
\mathrm{D} & \theta_0 & & \theta_1 & \dots & \theta_i & \theta_1 & \theta_1 & \dots & \theta_1 \\
Q & q_0 & & q_1 & \dots & q_i & [\cdot,\cdot] & q_{i+1} & \dots & q_n \\
S & s_0 & & s_1 & \dots & s_i & [\cdot,\varphi] & s_{i+1} & \dots & s_n
\end{array}$$

for $1 \leq i < n$, where $Z_{i+1} \neq \mathrm{Dat}$, and φ is $L(a_0)$ and ψ is $\mathrm{diff}(a_0,a_{n+1}) = 1$ if the verb phrase is to CLMod's left, and φ is $R(a_0)$ and ψ is $\mathrm{diff}(a_{n+1},a_0) = 1$ if the verb phrase is to CLMod's right.

(99) If

$$\begin{array}{lcccccc}
 & q\text{-}\mathrm{CL}(\Omega)_v & & Z_1 & \dots & Z_n \\
\mathrm{T} & a_0 & & a_1 & \dots & a_n \\
\mathrm{MB} & h_0 & & h_1 & \dots & h_n \\
\mathrm{LB} & c_0 & \Rightarrow & c_1 & \dots & c_n \\
\mathrm{D} & \theta_0 & & \theta_1 & \dots & \theta_1 \\
Q & q_0 & & q_1 & \dots & q_n \\
S & s_0 & & s_1 & \dots & s_n
\end{array}$$

then

$$\begin{array}{lccccccc}
 & q\text{-}\mathrm{CL}(\Omega)_v & & \mathrm{CLMod} & Z_1 & \dots & Z_n \\
\mathrm{T} & a_0 & & a_{n+1}|_{\mathrm{diff}(a_0,a_{n+1})=1} & a_1 & \dots & a_n \\
\mathrm{MB} & h_0 & & h_0 & h_1 & \dots & h_n \\
\mathrm{LB} & c_0 & \Rightarrow & c_0+1 & c_1 & \dots & c_n \\
\mathrm{D} & \theta_0 & & \theta_1 & \theta_1 & \dots & \theta_1 \\
Q & q_0 & & [\cdot,\cdot] & q_1 & \dots & q_n \\
S & s_0 & & [\cdot,L(a_0)] & s_1 & \dots & s_n
\end{array}$$

and

$$\begin{array}{lccccccc}
 & q\text{-}\mathrm{CL}(\Omega)_v & & Z_1 & \dots & Z_n & \mathrm{CLMod} \\
\mathrm{T} & a_0 & & a_1 & \dots & a_n & a_{n+1}|_{\mathrm{diff}(a_{n+1},a_0)=1} \\
\mathrm{MB} & h_0 & & h_1 & \dots & h_n & h_0 \\
\mathrm{LB} & c_0 & \Rightarrow & c_1 & \dots & c_n & c_0 \\
\mathrm{D} & \theta_0 & & \theta_1 & \dots & \theta_1 & \theta_1 \\
Q & q_0 & & q_1 & \dots & q_n & [\cdot,\cdot] \\
S & s_0 & & s_1 & \dots & s_n & [\cdot,R(a_0)]
\end{array}$$

and

$$\begin{array}{lccccccccc}
 & q\text{-CL}(\Omega)_v & & Z_1 & \dots & Z_i & \text{CLMod} & Z_{i+1} & \dots & Z_n \\
\text{T} & a_0 & & a_1 & \dots & a_i & a_{n+1}|\psi & a_{i+1} & \dots & a_n \\
\text{MB} & h_0 & & h_1 & \dots & h_i & h_0 & h_{i+1} & \dots & h_n \\
\text{LB} & c_0 & \Rightarrow & c_1 & \dots & c_i & c_0+1 & c_{i+1} & \dots & c_n \\
\text{D} & \theta_0 & & \theta_1 & \dots & \theta_1 & \theta_1 & \theta_1 & \dots & \theta_1 \\
Q & q_0 & & q_1 & \dots & q_i & [\cdot,\cdot] & q_{i+1} & \dots & q_n \\
S & s_0 & & s_1 & \dots & s_i & [\cdot,\varphi] & s_{i+1} & \dots & s_n
\end{array}$$

for $1 \leq i < n$, where $Z_{i+1} \neq \text{Dat}$, and φ is $L(a_0)$ and ψ is $\text{diff}(a_0, a_{n+1}) = 1$ if the verb phrase is to CLMod's left, and φ is $R(a_0)$ and and ψ is $\text{diff}(a_{n+1}, a_0) = 1$ if the verb phrase is to CLMod's right.

Declarative Clauses

Let v be a variable that ranges over the possible clause types, and Ω a variable for the clausal relations.

(100)
$$\begin{array}{lccc}
 & \text{Nom} & & \text{CL}(\Omega)_{\text{NFT}} \\
\text{T} & a & & a \\
\text{MB} & h & & h \\
\text{LB} & c & \Rightarrow & c+1 \\
\text{D} & \theta & & \theta \\
Q & q & & q \\
S & s & & s
\end{array}$$

(101)
$$\begin{array}{lccc}
 & \text{Nom} & & \text{CL}(\Omega)_{\text{NFP}} \\
\text{T} & a & & a \\
\text{MB} & h & & h \\
\text{LB} & c & \Rightarrow & c+2 \\
\text{D} & \theta & & \theta \\
Q & q & & q \\
S & s & & s
\end{array}$$

(102)
$$\begin{array}{lccc}
 & \text{Nom}_{\text{Arg}} & & \text{CL}(\Omega)_{\text{FT}} \\
\text{T} & a & & a \\
\text{MB} & h & & h \\
\text{LB} & c & \Rightarrow & c+3 \\
\text{D} & \theta & & \theta \\
Q & q & & q \\
S & s & & s
\end{array}$$

(103)
$$\begin{array}{lccc}
 & \text{Mod} & & \text{CL}(\Omega)_{\text{FP}} \\
\text{T} & a & & a \\
\text{MB} & h & & h \\
\text{LB} & c & \Rightarrow & c+3 \\
\text{D} & \theta & & \theta \\
Q & q & & q \\
S & s & & s
\end{array}$$

(104) a.
$$\begin{array}{lcccc}
 & \text{Nom} & & \text{IPP} & \text{CL}(\Omega)_{\text{FP}} \\
\text{T} & a & & ? & a \\
\text{MB} & h & & h & h \\
\text{LB} & c & \Rightarrow & c+3 & c+3 \\
\text{D} & \theta & & \theta & \theta \\
Q & q & & [\cdot,\cdot] & q \\
S & s & & [\cdot,L(a)] & s
\end{array}$$

b.
$$\begin{array}{lcccc}
 & \text{Nom} & & \text{IPP} & q\text{-CL}(\Omega)_{\text{FT}} \\
\text{T} & a & & ? & a \\
\text{MB} & h & & h & h \\
\text{LB} & c & \Rightarrow & c+3 & c+3 \\
\text{D} & \theta & & \theta & \theta \\
Q & q & & [\cdot,\cdot] & q \\
S & s & & [\cdot,L(a)] & s
\end{array}$$

(105)
$$\begin{array}{lcccc}
 & \text{Nom} & & \text{IPP} & \text{CL}(\Omega)_{\text{FT}} \\
\text{T} & a & & ? & a \\
\text{MB} & h & & h & h \\
\text{LB} & c & \Rightarrow & c+3 & c+3 \\
\text{D} & \theta & & \theta & \theta \\
Q & q & & [\cdot,\cdot] & q \\
S & s & & [\cdot,L(a)] & s
\end{array}$$

Question Clauses

Let v be a variable that ranges over the possible q-clause types, and Ω a variable for the clausal relations.

$$(106)\quad \begin{array}{lcccc} & \text{Nom}_{\text{P}} & & \text{P} & \text{Nom}_{\text{P}} \\ \textsc{t} & a & & \delta 1 & a \\ \textsc{mb} & h & & 1 & h \\ \textsc{lb} & c & \Rightarrow & c+2 & c \\ \textsc{d} & \theta & & \theta & \theta \\ \textsc{q} & [u,\cdot] & & \cdot & [u,\cdot] \\ \textsc{s} & [v,s] & & L(a) & [v,s] \end{array}$$

$$(107)\quad \begin{array}{lcccc} & \text{Nom}_{\text{Arg}} & & \text{NP} & \text{CL}(\Omega)_{\text{NFP}} \\ \textsc{t} & a & & a & ? \\ \textsc{mb} & h & & h & h \\ \textsc{lb} & c & \Rightarrow & c & c+2 \\ \textsc{d} & \theta & & \theta & \theta \\ \textsc{q} & q & & u & [1,\cdot] \\ \textsc{s} & s & & v & [m,R(a)] \end{array}$$

$$(108)\quad \text{a.}\ \begin{array}{lccc} & \text{Nom}_{\text{P}} & & \text{CL}(\Omega)_{\text{FP}} \\ \textsc{t} & a & & a \\ \textsc{mb} & h & & h \\ \textsc{lb} & c & \Rightarrow & c+3 \\ \textsc{d} & \theta & & \theta \\ \textsc{q} & [u,\cdot] & & [1,\cdot] \\ \textsc{s} & [v,s] & & [m,s] \end{array} \qquad \text{b.}\ \begin{array}{lccc} & \text{Nom}_{\text{P}} & & q\text{-CL}(\Omega)_{\text{FT}} \\ \textsc{t} & a & & a \\ \textsc{mb} & h & & h \\ \textsc{lb} & c & \Rightarrow & c+3 \\ \textsc{d} & \theta & & \theta \\ \textsc{q} & q & & q \\ \textsc{s} & s & & s \end{array}$$

$$(109)\quad \text{a.}\ \begin{array}{lccc} & \text{Nom}_{\text{P}} & & \text{CL}(\Omega)_{\text{NFP}} \\ \textsc{t} & a & & a \\ \textsc{mb} & h & & h \\ \textsc{lb} & c & \Rightarrow & c+1 \\ \textsc{d} & \theta & & \theta \\ \textsc{q} & [u,\cdot] & & [1,\cdot] \\ \textsc{s} & [v,s] & & [m,s] \end{array} \qquad \text{b.}\ \begin{array}{lccc} & \text{Nom}_{\text{P}} & & q\text{-CL}(\Omega)_{\text{NFT}} \\ \textsc{t} & a & & a \\ \textsc{mb} & h & & h \\ \textsc{lb} & c & \Rightarrow & c+1 \\ \textsc{d} & \theta & & \theta \\ \textsc{q} & q & & q \\ \textsc{s} & s & & s \end{array}$$

$$(110)\quad \text{a.}\ \begin{array}{lccc} & \text{Nom}_{\text{P}} & & \text{CL}(\Omega)_{\text{NFD}} \\ \textsc{t} & a & & a \\ \textsc{mb} & h & & h \\ \textsc{lb} & c & \Rightarrow & c+2 \\ \textsc{d} & \theta & & \theta \\ \textsc{q} & [u,\cdot] & & [1,\cdot] \\ \textsc{s} & [v,s] & & [m,s] \end{array} \qquad \text{b.}\ \begin{array}{lccc} & \text{Nom}_{\text{P}} & & q\text{-CL}(\Omega)_{\text{NFP}} \\ \textsc{t} & a & & a \\ \textsc{mb} & h & & h \\ \textsc{lb} & c & \Rightarrow & c+2 \\ \textsc{d} & \theta & & \theta \\ \textsc{q} & q & & q \\ \textsc{s} & s & & s \end{array}$$

4.6 Sentences

There are 4 sentence types. Two of the sentence types are based on clauses and 2 are based on an argument nominal. The sentences are main sentences (S_{M}), q-sentences (S_q), answer sentences (S_{A}), and behavior sentences (S_{B}). A main sentence is a main clause, with the X variable an argument nominal and the Y variable a finite or modal verb. The q-sentences are the various possible questions, which include either variable assignment for the main clause, plus the main q-clause. The answer sentence is for any direct answer, such as a noun phrase, prepositional phrase, adjective, non-finite clause, or even a finite partial clause, and thus are handled by a clause modifier nominal. Behavior clauses also use an argument nominal, and include imperatives, greetings, and exclamations.

At the level of the sentence, from a production standpoint, one doesn't have passed down or conditional articulation, but instead specific articulation can be thought of as inputs. At this level, the $0 \pmod 3$ MB value and $0 \pmod 4$ LB value is decided. Some of these set values are free to specify semantic values or free more generally, but a few are needed for the sentence to have a valid syntax, or change the speech act. In particular, a Q value is used in this way. All sentence types, besides a main sentence, require a Q value. These values correspond to what has been denoted as u, and the S value denoted v. One may recall that the marking occurs to a temporally early or spatially left constituent. q-sentences use an LB realization of the Q value. *Who/what/which* get a 1 value, while all other cases get a -1 value. Note that when the interrogative pronoun is an adverb, the verb (as the X variable) gets the Q value. The answer and behavior sentences have the Q value realized with the MB. Answer sentences take a -1 value while behavior sentences produce a 1 value.

$$(111)\quad \begin{array}{lccc} & \mathrm{S_M} & & \mathrm{CL}(\Omega)_\mathrm{M} \\ \mathrm{T} & a_0 & & a_0 \\ \mathrm{MB} & h_0 & & h_0 \\ \mathrm{LB} & c_0 & \Rightarrow & c_0 \\ \mathrm{D} & \theta & & \theta \\ Q & \cdot & & \cdot \\ S & \cdot & & \cdot \end{array}$$

where X_z is a z nominal and $Y = v$, for some $v \in V_f \cup V_{lm} \cup V_{mm}$.

$$(112)\quad \text{a.}\quad \begin{array}{lccc} & \mathrm{S}_q & & \mathrm{CL}(\Omega)_\mathrm{M} \\ \mathrm{T} & a_0 & & a_0 \\ \mathrm{MB} & h_0 & & h_0 \\ \mathrm{LB} & c_0 & \Rightarrow & c_0 \\ \mathrm{D} & \theta & & \theta \\ Q & [-1,\cdot] & & [-1,\cdot] \\ S & [\lambda,\cdot] & & [\lambda,\cdot] \end{array}$$

where Y is a z nominal and $X_z = v$, or $Y = v$ and X_z is a z nominal, for some $v \in V_f \cup V_{lm} \cup V_{mm}$.

$$\text{b.}\quad \begin{array}{lccc} & \mathrm{S}_q & & q\text{-}\mathrm{CL}(\Omega)_\mathrm{M} \\ \mathrm{T} & a_0 & & a_0 \\ \mathrm{MB} & h_0 & & h_0 \\ \mathrm{LB} & c_0 & \Rightarrow & c_0 \\ \mathrm{D} & \theta & & \theta \\ Q & [1,\cdot] & & [1,\cdot] \\ S & [\lambda,\cdot] & & [\lambda,\cdot] \end{array}$$

where X_z is a z nominal and $Y = v$, for some $v \in V_f \cup V_{lm} \cup V_{mm}$.

$$(113)\quad \begin{array}{lccc} & \mathrm{S_A} & & \mathrm{CLMod} \\ \mathrm{T} & a_0 & & a_0 \\ \mathrm{MB} & h_0 & & h_0 \\ \mathrm{LB} & c_0 & \Rightarrow & c_0 \\ \mathrm{D} & \cdot & & \cdot \\ Q & [-1,\cdot] & & [-1,\cdot] \\ S & [m,\cdot] & & [m,\cdot] \end{array}$$

$$(114)\quad \begin{array}{lccc} & \mathrm{S_B} & & \mathrm{Nom_{Arg}} \\ \mathrm{T} & a_0 & & a_0 \\ \mathrm{MB} & h_0 & & h_0 \\ \mathrm{LB} & c_0 & \Rightarrow & c_0 \\ \mathrm{D} & \cdot & & \cdot \\ Q & [1,\cdot] & & [1,\cdot] \\ S & [m,\cdot] & & [m,\cdot] \end{array}$$

4.7 Some Notes and Remarks

This concludes this fragment of an English grammar. From lexical elements and sets of lexical elements, the basic categories of the grammar, such as noun, adjective, and verb, are built up. From these preliminary categories, the phrases can be built up. To go from phrases to clauses, one has phrases which, besides their internal make up, have structural properties from the clause, and so the ergative, dative, and so on, case-like descriptors are introduced. Further, since multiple phrases or other constituents are possible within a clause, the non-verb phrases of the clause are placed within a nominal, which allows for these generalizations. An initial finite total clause can be used, with transformations, to produce 5 further clauses, one redundant but for the use of variables which allow for linear order variation. The rest of the clauses vary in the verb being finite or non-finite, and the number of arguments that are not articulated. When questions are asked about particular constituents of a clause, new structures emerge, in particular partial nominals, and q-clauses, which contain a partial nominal. Once all the clauses and q-clauses are defined by previously defined elements, the recursive part of the grammar can begin, in particular rules for allowing clauses and q-clauses within nominals and partial nominals. Finally, the initial rules that must be applied first within a production grammar, the sentence rules, allow for producing main clauses, q-clauses, and nominals with particular articulation markers to make valid sentences.

Such a grammar doesn't contain every possible sentence of English, and some of the expansion will require more of a second order structure for grammatical categories. In particular conjunctions apply over multiple basic categories. And while they could be defined disjunctively, it makes more sense to present with variables constrained to certain sets. The commutative and associative conjunctions, "and" and "or," apply to all of the possible categories. But, the subordinate conjunctions, such as "if" and "when" are applicable over $\mathrm{CL}_{\mathrm{FT}}$, Adj, and Adv, having the rule 'If $A \to x$, then $A \to \mathrm{Conj}_{\mathrm{sub}}\ x$.' The commutative and associative conjunctions have the rule 'if $A \to x_1, \ldots, A \to x_n$, then $A \to x_1\ \ldots\ x_{n-1}\ \mathrm{Conj}_{\mathrm{CA}}\ x_n$, for $n \geq 1$.'

Other aspects of the grammar will be more of the same. For instance, words like "all," "even" and "probably" before a noun phrase likely receive the same articulation as a preposition out front. Pronouns have the same q-value properties as interrogative pronouns, and their case marking can be coordinated to the L and R functions within the source row of the vector to indicate subject or not. While the possessive forms "ours" and "theirs" don't line up with any current rules, they match the form of a gapping rule, which can occur when an adjective is used for a noun phrase, such as in the sentence "They don't want the red car, they want the blue." "Blue" can be marked in a certain way to make this a valid noun phrase. That's not to say that the articulation for the possessive pronouns will match up. Other gapping cases will be more complex conjunction rules that apply for clauses.

A sentence of the grammar is any finite application of the production rules, starting from a sentence rule, which leaves a string of vectors, such that their top row is a lexical element. For any collection of such rules, there will be a set of valid articulations for the lexical elements. A key question is whether parsing is unique: whether there is an algorithm which, given any such articulation, reproduces the temporal sequence tree, and it produces only one such structure. Within a finite total clause, the tonal differences and MB and LB values produce uniqueness for each input. The Q values are used to preserve this uniqueness, if one knows the clause type. The clause type can be inferred from LB values, though there is some overlap with clauses to be concerned about. Showing that parsing is unique is not a trivial task, but it falls short for one particular reason, which may not be applicable in real life circumstances. Nonetheless it appears that certain parts that produce unique parsing are not required, in particular the change of the LB and MB value at the level of the bundle's initial value.

The problem that will occur for unique parsing is that multiple modifiers lead to structural ambiguity. Noun modifiers are marked by an MB value, but it's absolute and not relative to an input, so the value can become redundant and indistinguishable. In natural speech, the level of embedding and modifying is tracked by the LB and MB. In particular, the initial value will be manipulated, so that every region has a +1 to indicate the embedding. Either the LB or MB is used, but not both, for a given embedding level. The direction ia dependent on the main clause's direction value. For the LB, the modification leads to invariance mod 4 but for the MB, the mod 3 value is not invariant, and so appropriate adjustments are needed. Whether this technique for tracking embedding is optional or the the transcribing techniques led to incomplete and improper production isn't clear.

Movement in the opposite direction for the LB initial value appears to be a means of destressing a syllable. In particular, this appears to be used with the determiners so that an otherwise existing $\delta 1$ tonal rule can be dropped, but this needs to be investigated further.

A Semantics of Prosody: Action and Ethics

The semantics for the risorius and buccinator groups and the front tongue quadrant are based on encoding *good* and *bad*. These terms can be understood in a completely formal way, dependent on an understanding of actions. In particular, they may be understood as analogous to the truth values *true* and *false*, and so will be called benevolence values. (Benevolence will be abbreviated bnv.) We start with describing two basic forms of actions which are logical negations of one another. We then move on to describe ethical systems and describe how *good* and *bad* are recursively assigned to states, actions, and objects. Next we address the obvious objections to fixed, universal *good* and *bad* judgments, and distinguish between ethical and moral systems. Lastly, we discuss how various elements of this apply to the semantics of prosody.

Actions

There are two basic kinds of actions: pursuit and prevention. To pursue something is to try to bring it about; to prevent something is to try to stop it from coming about. These actions are used in any domain, including the biological and knowledge domains. For instance, "secrets" are dependent upon knowledge prevention; the good of eating, if the food does not bring any particular pleasurable taste, is to prevent starvation. While there are actions that are based on pursuit of good ends such as pleasures, the *good* focused on here come from preventing *bad* results.

Any action—a pursuit or prevention or more complex form—has dependencies. An action may be seen as a means (a machine or a sequence of machines) to make change, plus the end result of that change. Thus, an action may be considered a mechanism or a sequence of mechanisms (probably a temporal sequence tree), where a mechanism is understood as an ordered pair of a means or change, and an end state. The collection of dependencies are of one of two kinds, which may be demonstrated with an example. To throw an object at a target, the following two conditions must be met. First, the direction of the thrower's hand must be facing a particular direction, usually towards the target. Second, there mustn't be any objects that obstruct the path. The first is existential in nature: there exists a direction the person's hand must face; the second is universal, every object satisfying certain conditions (such as mass) should not be in the path. The quantification differences for these different kinds of conditions stem from pursuit and prevention being used in a recursive manner. The existential conditions are ones that are pursued, while the universal conditions are the ones that are prevented. And this is so for an action understood as a means and an end.

Bnv values are assigned recursively in a way that's naturally related to practical reasoning. For practical reasoning, one doesn't start with an action, but with an end. The classical formulation of practical reasoning is presented in (115) (Fodor (2008)). But, often we know multiple actions for the same ends: if we want or need to eat, we might cook, order out, shop for something ready to eat, request food from others, or some other action, and some of these can be a collection of possible actions, such as multiple ways of cooking. It is known that people and other mammals have an expectation of minimizing costs for actions (Spelke (2022)). Combining this fact with the classical formulation produces something like the rule in (116). But, this is a special case where one is pursuing an end and nothing is preventing *m*.

(115) want/need x
x only if m
therefore, (do) m

(116) want/need x
x if m
$\forall m', x \text{ if } m' \rightarrow \text{work}(m') \geq \text{work}(m)$
therefore m

When one starts with an end and wishes to build an action, the reasoning for pursuit or prevention are the negations of one another. To pursue an end, one needs the existence of a mechanism to do; to prevent an end, one needs every mechanism to not be done. Then, within a given mechanism, as mentioned there are conditions, and again pursuit and prevention are logical negations as to how they handle them. For pursuit, every condition or dependency of the mechanism must be met; for prevention, all one needs is for there to exist a condition that is not met to prevent the mechanism. We can see also that the practical reasoning formulation applies for pursuit, and the prevention analogue would be (117).[14]

(117) $\neg x$
$\exists m, x$ if m
Therefore, $\forall m, x$ if $m \rightarrow \neg m$

A lock on a door might demonstrate the use of prevention. A wall, or multiple such walls, are a means of preventing people from entering a space. An opening, such as a door way, prevents this prevention. But then, there is a mechanism of entering, and so the initial prevention by the wall fails. A door prevents the entering while producing a mechanism to pursue entering (opening the door). But again, the pursuit is available to all. By producing a lock and a key, one can prevent the mechanism of opening the door, but now with a mechanism to pursue the opening which is available only to the key holder.

The general reasoning for pursuit and prevention is summarized in the table below.

	Mechanisms for an end A $(\mu(A))$	Preconditions for a mechanism m (C_m)
pursuit	$\exists m \in \mu(A),\ m$	$\forall x \in C_m,\ x$
prevention	$\forall m \in \mu(A),\ \neg m$	$\exists x \in C_m,\ \neg x$

Ethics and Bnv Values

The general idea is that there is probably ethical systems and even an ethical calculus, analogous to the logical calculi. That is, they are deductive systems such that a desiderata is that they are benevolently consistent, or consistent with respect to bnv values. Here, we want to spell out intuitions about how *good* and *bad* will be assigned from primitive cases. The ontological types that the bnv values will be assigned to are states, mechanisms/actions, and objects. The primitives or axioms will be states from which the deductions will occur from. Two possible candidates of axioms for ethical theories are starving being *bad* and being physically hit by something with a momentum greater than some threshold being *bad*. One can include with the latter squeezing and stress and strains on the body, above some threshold, also being *bad*. It's not clear that any axioms involve something being *good*, and more generally ethics is likely based on needs rather than wants.

Deducing bnv values from some initial collection of states which are assigned bnv values can occur from the following axioms. These are axioms not of any particular ethical theory and which apply across all theories. That is, they are analogous to propositional or predicate calculus axioms, while the starving and being hit axioms are akin to theory specific axioms, such as geometric or natural number axioms. *Modus ponens* would still be used as the single inference rule. The logical connectives apply, but the bnv values don't necessarily have the same rules as the truth values. E.g. 'x and y is *good*' doesn't necessarily imply 'x is *good*' and 'y is *good*.'

A given mechanism or action has some intended end, but normally this is not the only consequence of the action. For instance, if an object is thrown at a target and it hits the target (the intended end), the collision will also produce a sound, and the various movements will produce a slight heat as well. These would also

[14] There is probably a work condition: for all mechanisms lower than some level of work.

be consequences of the action. Let $\sigma(m)$ denote all the consequences of an action.[15] Note that if x is a consequence of a mechanism m, then m is a mechanism of x (118). The general formula for assigning bnv values to the mechanism m, if the bnv values of each consequence is known or unassigned, is that m is *bad* whenever at least one of its consequences are *bad*, and m is *good* whenever every consequence is not *bad*, and at least one consequence is *good* (120). If a *good* action brings about at least one *good* end, and we start with only *bad* states, then we need a means to deduce *good* from *bad*. It's easy to check intuition, that if something is *bad*, then it not being the case is *good* (119). Finally, from mechanisms, objects and further states can also derive bnv values (121). These objects and states are dependencies of the mechanism, or in other words, the states are in C_m, and the object is a term of the state.

(118) $m \in \mu(A) \leftrightarrow A \in \sigma(m)$

(119) If A is *bad*, then $\neg A$ is *good* .[16]

(120) If A is *bad*, and $A \in \sigma(m)$, then m is *bad*.

If A is *good* , and $A \in \sigma(m)$, then $\forall x \in \sigma(m), \neg(x$ is *bad*$) \rightarrow m$ is *good*

(121) If m is *good* , and $\exists a, a \in C_m$, then a is *good* .

If m is *good* , and $\exists a, a \in C_m$, and c is a constant in a, then c is *good* .

If m is *bad*, and $\exists a, a \in C_m$ and $\forall m', m'$ is *good* $\rightarrow a \notin C_{m'}$, then a is *bad*.

If m is *bad*, and $\exists a, a \in C_m$ and $\forall m', m'$ is *good* $\rightarrow a \notin C_{m'}$, and c is a constant in a, then c is *bad*.

Let's return to a lock on a door to demonstrate these rules. We shall use three ethical systems, two containing one *bad* primitive each, and one with both *bad* primitives. The first *bad* primitive is if there is no mechanism for an agent p to get to an object x (122). The second *bad* element is that for everybody else, it's bad if there's a way to get the object x (123). Call the system with just the first *bad* state S1, the system with just the second *bad* state S2, and the system with both S3. In the initial wall-less state of the world, the world is *good* in S1 since there's a mechanism (locomotion) for p to get to x, but also for anybody else, so the world is *bad* in S2 and S3. Constructing walls then is *good* in S2, but bad in S1 and S3. A door is *good* in S1, but *bad* in S2 and S3. Producing a lock, and giving the only key to p then produces a mechanism to be *good* in S2 (locking and preventing any one but p from entering), *good* in S1 (as p has a means of entering), and thus *good* in S3.

(122) $\forall m, [\text{dist}(p,x) < \alpha] \notin \sigma(m)$ m is *bad*, for some constant α.

(123) $\forall q, q \neq p \wedge \exists m, [\text{dist}(q,x) < \alpha] \in \sigma(m)$ is *bad*, where *bad* is in the scope of the quantifier

~ ~ ~

A complication is considerations of time. For instance, the threat of starving or otherwise having an empty belly is temporally varying and goes away after eating a sufficient amount. When one is full, and if taste is irrelevant, then food is no longer good *at that time*, though it will eventually become good again. This type of periodic bnv values is generally treated as just universal across time, or at least the periodic change to value is treated as the default, so that eating requires no explanation while not eating does. Another case complicated by time are the conditions for a mechanism: some conditions are needed throughout an action while others are only needed at a certain time in an action. Only the former get treated with bnv values, while the latter are treated as neutral (if not assigned bnv values for some other reason), though it may be temporarily given a bnv value.

[15] Claiming that such a set exists is not so obvious, there could be an infinite sequence of consequences and so on, but we might claim a restriction of events having energy greater than some threshold or otherwise make restrictions for the set to exist.

[16] The inverse does not hold: the logical negation of something *good* is not necessarily *bad*.

Ethics and Morality

At an intuitive level, the objection to claims of objective conditions for *good* and *bad* is disagreement about what's good, bad, right, and wrong. But such claims normally involve benevolently inconsistent cases. For instance, questions of punishment assume that some *bad* resulted, and so a benevolently consistent system, or an ethical system, isn't applicable, as its desiderata isn't satisfied. Instead, when there are benevolently inconsistent circumstances, one uses a moral system. Moral systems deal with right and wrong, which aren't going to have the same objective properties as good and bad.

A simple case of when we would use a moral system rather than an ethical system is when a child gets a splinter. A *bad* result has already occurred, and the child feels pain, irritation, and has heightened risks of worse outcomes. To diminish these, typically people use actions that cause more pain, but do so temporarily in order to eliminate long term suffering and risk. This is ambiguous and can lead to disagreement about right and wrong, but right and wrong aren't good and bad. Both pains are bad, but its common in this culture to consider the splinter removal to be right, despite the immediate pain, which is *bad*.

More generally, moral systems make decisions which are justified by getting more of something. For instance, the road systems for vehicles is an instance of something like a moral system being used to allow more travel in less time. The speed at which cars move is *bad*, as it produces risks and dangers which are normally, and when not highly coordinated like with the road systems, unacceptable. The result is inconsistency. For instance, stop signs and stop lights produce mechanistic inconsistency: one doesn't drive (or stops) in order to drive somewhere. In the sciences, distrust as a default is justified by more knowledge and more precise knowledge. In everyday life, trust is the default while distrust is earned. But in the sciences, distrust being the default is so accepted and expected that one presumes one has to present an argument for any claim one makes. Moral system will have similar properties, but will have an obvious bnv contradiction in place, or some unavoidable *bad* state.

The Semantics of Prosody

Trying to discover the semantics of a sentence is a difficult task. One starts with a speech act and has to make a hypothesis about how it's different than other speech acts, and then try to attribute this difference to some particular part of that utterance. It is a guess and check process which is utterly superstitious in nature and one can only hope that one doesn't become so attached to such superstitious ideas that one becomes blinded to counterevidence. On top of this, the hypothesis is then attempted to put into a framework—a classificatory process—and often by assigning it a symbol. But this symbol or class only has meaning in so far as it does work in reasoning. In other words, semantics requires developing a theory of the outputs and not just the language, and the lack of robust understanding of those outputs is currently a large hindrance. This will have the same issue, which is why the semantics above were endeavored: although we will give the prosody some interpretation that is wanting, the actions and ethics analysis suggests that such classification won't always be so vacuous, though there is still work to do to go from these interpretations to get to the psychology of actions, practical reasoning, or any reasoning that occurs in a moral system or across multiple moral systems. With that being said, action structure and bnv values are related to the semantics of prosody.

Most of the semantics is based around an end or goal—call it the prosodic end—which may be implicit or explicit, depending on the middle quadrant value. The LB and MB initial values appear to work together to specify some constraints that this end meets, but this is not worked out to a satisfactory extent, and will be left aside. Whether this end is *good* or *bad* is encoded indirectly through two articulation features, the front quadrant length and the labial bundle group. The front quadrant determines whether pursuit or prevention reasoning is used. The risorius and buccinator groups encode whether there is a *good* or *bad* circumstance. A *good* circumstance is when a *good* end is pursued or a *bad* end is prevented. A *bad* circumstance is one where a *bad* end is pursued or a *good* end is prevented. Thus, the prosodic end is *good* when either the front quadrant indicates pursuit and the LB group indicates a *good* circumstance, or when the front quadrant indicates prevention and the LB group encodes a *bad* circumstance, and the prosodic end is *bad* when one of the other two possible combinations is used.

When the buccinator group is more tense than the risorius group, by some threshold, the speech act encodes a *bad* bnv circumstance, and when the risorius group is more tense than the buccinator group by some threshold, a *good* bnv circumstance is encoded. General behavioral cases which make this function clear are asking yes/no questions where it's obvious whether the speaker approves or disapproves of a 'yes' answer. "Did you take an apple?" as a prelude to a potential invitation requires the risorius group value, while as an accusation, it requires a buccinator group value.

The front tongue's semantics indicates two properties. Axiomatic ends like hunger and being hit being *bad* are defined relative to an individual, and the relevant individual is the benefactor. More specifically, the benefactor is the relevant individual, or one of the individuals, that the prosodic end benefits. The longer tongue lengths indicate the speaker is not the benefactor and the shorter two lengths indicate a first person benefactor. Most of the time, the longer tongue lengths indicate a second person benefactor, but there are exceptions including cases in which the semantics indicates a third person moral authority. So this splits the values into two groups, and for each group, the shorter tongue length encodes the prosodic end is being prevented and the longer tongue length encodes that the prosodic end is being pursued. This property can be very subtle, but a sentence like "Did you lock the door(s)?" can sometimes be helpful. The difference between wanting to get into the house versus worry about the house being secure are clear differences of pursuit and prevention.

From practical reasoning, one can produce a partial order for which a least upper bound exists for any two elements. This can be generated by the pursuit rules (124) applied recursively. For prevention, the negation of each mechanism that would lead to the prosodic end becomes an end which the pursuit rules can be applied and so a partial order generated. The states and events encoded by the lexical elements—call it the lexical topic—is related to the prosodic end, and the middle quadrant indicates how so. It's high values imply the lexical topic is greater in such a structure while it's lower values imply the lexical topic is lesser in such an order. The highest value, and only the highest value, is the prosodic end, but from there the specifics aren't clear. A hypothesis is that the lowest value corresponds to the "minimal" elements of irrelevant states which are inconsequential to the prosodic end. These can be used to achieve a distant, factual tone. The second lowest element is related to conditions, but it's not clear if it's conditions lower in the order, or stemming from more than one iterations of (124) or not. The general pattern of language is that functions and there outputs often get conflated, so that skepticism is warranted for the mechanisms being the second highest values. A tentative distinction between the middle two values is that the higher value indicates a controllable condition while the lower value indicates an uncontrollable condition within the relevant space-time, however that may be determined. The more general claim, that it's distance from the actions end,[17] can be demonstrated with answers of questions: a direct answer will have a higher value than an indirect answer, as seen in (125).

(124) a. $(m \in \mu(A) \wedge \forall m', m' \in \mu(A) \rightarrow \text{work}(m) < \text{work}(m')) \rightarrow m < A$.

b. $\forall x, x \in C_m \rightarrow x < m$.

(125) Question: "Feel like going to the park?"
Answer:

a. "Sure."
(direct answer, high middle quadrant value.)

[17]No distance has been defined, but an example would be a discrete distance, where a child/parent in the tree structure is a distance of 1.

b. "I need to eat before I think about anything else."
(indirect answer, low middle quadrant value.)

In summary, the LB group value and front quadrant encode a bnv value for an end. The bnv value derives ultimately from at least one agent being harmed or benefited. The front quadrant encodes whether some such agent, the benefactor, is the speaker (first person) or not. The class that end belongs to is known from the LB and MB initial values. The end itself may not be stated, but a state or event that's related to it is what the lexical items encode. What relation holds between the lexical items and that end is encoded by the middle quadrant value. The relation stems from a partial order structure defined from the end.

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